

Accepted Manuscript

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PII: S0079-6565(15)00029-1

DOI: <http://dx.doi.org/10.1016/j.pnmrs.2015.05.003>

Reference: JPNMRS 1406

To appear in: *Progress in Nuclear Magnetic Resonance Spectroscopy*

Received Date: 3 July 2014

Accepted Date: 20 May 2015



Please cite this article as: Z. Chen, S. Cai, Y. Huang, Y. Lin, High-resolution NMR spectroscopy in inhomogeneous fields, *Progress in Nuclear Magnetic Resonance Spectroscopy* (2015), doi: <http://dx.doi.org/10.1016/j.pnmrs.2015.05.003>

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High-resolution NMR spectroscopy in inhomogeneous fields

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Numbers of pages: 90

Numbers of figures: 38

Keywords: High-resolution, NMR spectroscopy, inhomogeneous magnetic field, iMQCs, nutation echoes, spatial encoding

Abstract

High-resolution NMR spectroscopy, providing information on chemical shifts, J coupling constants, multiplet patterns, and relative peak areas, is a mainstream tool for analysis of molecular structures, conformations, compositions, and dynamics. Generally, a homogeneous magnetic field is a prerequisite for obtaining high-resolution NMR information. Magnetic field inhomogeneity, whether from non-ideal experimental conditions or from intrinsic magnetic susceptibility discontinuities in samples, represents a hurdle for applications of high-resolution NMR. Numerous techniques have been proposed for measuring high-resolution NMR spectra free from the influence of inhomogeneous magnetic fields. Besides developments and improvements in NMR instrumentation, various types of experimental approaches have been established for recovering NMR information in inhomogeneous magnetic fields. Three main types are systematically described in this review. In addition, other high-resolution NMR approaches or data processing methods are also briefly described. All the high-resolution NMR approaches covered in this review have individual advantages and disadvantages in practical applications, and no one technique is applicable to all practical circumstances. Hence, they are complementary for high-resolution NMR applications in inhomogeneous fields. The underlying mechanisms of these approaches are presented, together with analyses of their applicability and efficiency.

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1. Introduction

High-resolution nuclear magnetic resonance (NMR) spectroscopy is a powerful and non-invasive tool for studies of molecular structures, conformations, compositions, and dynamics [1-7]. It enables the detection of detailed molecule-level information, such as chemical shifts, J couplings, and relaxation rates. Chemical shifts, depending on the magnetic environment of a given nucleus (e.g. carbon or hydrogen), reveal subtle differences between the environments of different nuclei, and are used for determinations of molecular compositions [8, 9]. The J couplings, mediated through chemical bonds between atoms, appear as peak splittings (multiplets) in NMR spectra and provide information regarding bond distances, bond angles, and atom connectivity. Thus J couplings are applied to determine molecular structures, monomer sequences, and conformations [10-12]. Spin relaxation times, including longitudinal (T_1) and transverse (T_2) relaxation times, are also one of the fundamental aspects of magnetic resonance. They describe processes in which spin magnetizations return to equilibrium from non-equilibrium states. In NMR experiments, relaxation rates determine line-widths of resonances in the acquisition period, and provide information regarding molecular motions and intramolecular distances [13, 14]. Generally, relaxation rates can be used to study chemical dynamic processes [15, 16]. Owing to the capability of providing multitudinous valuable information, NMR spectroscopy has found wide applications in various fields, from biomolecules to living organisms [17, 18], from chemical assignment to polymer characterization [19-21], and from nanoparticles to macroscopic materials [22-24].

In general, the spectral resolution in a conventional NMR spectrum is directly related to the overall magnetic field homogeneity experienced by the detected sample. A strong and highly homogeneous static magnetic field is a basic requirement for high-resolution NMR. Spatial

variations in magnetic field should be less than 10^{-8} for a liquid high-resolution proton (^1H) NMR spectrum [25-27]. Although magnets used in modern NMR spectrometers have been carefully designed to maintain field homogeneity as high as possible, circumstances still arise in which the spatial and temporal homogeneities of magnetic fields are degraded. For example, experimental samples are subject to intrinsic variations in magnetic susceptibility over the sample volume and between different structural components. Heterogeneous biological tissues in human or animal *in vivo* NMR detection, rock samples for oil exploration, and porous resin beads in combinatorial chemistry all suffer from severe field inhomogeneities and make high-resolution NMR detection challenging [28-30]. Tissues with microscopic air inclusions, such as the lungs of mammals or leaves of plants, and interfaces between tissue and air or bone, are generally recognized to be inaccessible for high-resolution NMR information [31]. In addition, some technology designs such as inside-out or single-sided magnets suffer from intrinsic field inhomogeneities [32]. The resulting inhomogeneous line broadening may be tens or hundreds of times greater than the homogeneous broadening, leading to severe signal overlapping and loss of spectral features, thus precluding spectral analyses based on interpretations of the familiar NMR parameters mentioned above.

To compensate for the effects of inhomogeneous magnetic fields and maximize spectral resolution, researchers have made great efforts in NMR experiment preparation, including careful shimming, sample spinning, rigorous exclusion of paramagnetic or particulate impurities from sample, and the utilization of magnetic susceptibility matched glasses in sample containers. These operations prior to NMR data acquisitions can improve the homogeneity of liquid samples to a certain extent. For some practical circumstances, however, they are not suitable. On the other hand, techniques based on hardware system designs have been proposed for special

applications in inhomogeneous fields. For example, magic-angle spinning (MAS) is as an effective tool for removing line broadening caused by chemical shift anisotropy and magnetic susceptibility discontinuities [33, 34]. For example the Varian Nano-NMR probe is specifically designed for performing MAS on very small ($\leq 40 \mu\text{L}$) samples, and can dramatically improve the resolution and sensitivity of proton signals of polymer support-bound compounds [35]. Blümich and coworkers have designed a magnet array to generate a volume of relatively homogeneous magnetic field external to a portable single-sided NMR magnet with a spectral resolution of 0.25 ppm for ^1H NMR spectra [36, 37]. At the same time, methods such as linear programming (LP) algorithms [38] and stream functions [39] have been proposed for shim coil design. A combined approach of passive (ferromagnetic) and active (electronic) shimming was developed to generate strong and highly homogeneous fields for *in vivo* NMR detection in a high field system where shim coils alone might be insufficient to permit homogenization at the desired spatial scale [40].

Besides development and improvement of hardware techniques for maintaining field homogeneity, experimental techniques have been designed to solve the aforementioned problems of inhomogeneous fields. Three main types of NMR experimental technique are discussed in this review. The first type is based on intermolecular multiple-quantum coherences (iMQCs) originating from long-range dipolar interactions among spins in different molecules [41-63]. A series of iMQC high-resolution pulse sequences have been proposed by Warren's group, Faber's group, and our group. These sequences include intermolecular zero-quantum coherence (iZQC) [41, 43], intermolecular double-quantum coherence (iDQC) [44, 57], and intermolecular single quantum coherence (iSQC) [47, 48] sequences. The second type is based on the use of nutation echo and/or z-rotation pulse techniques to extract chemical shift information in highly inhomogeneous fields [64-72]. A high-resolution *ex situ* NMR technique has been proposed for

the recovery of chemical shift in the presence of field inhomogeneities *via* nutation echoes [65, 71]. Blümich and coworkers developed a unilateral and mobile NMR sensor for high-resolution NMR spectra at low magnetic field strength [70]. This approach can be used in highly inhomogeneous fields, and does not rely on any special property of a given mechanism, such as intra- or inter-molecular interactions. The third type is based upon spatial encoding and phase correction. It is an extension of single-scan two-dimensional (2D) NMR spectroscopy suggested by Frydman and coworkers [73]. Differing from conventional Fourier-transform (FT) NMR, spatially-encoded NMR methods make direct phase compensation for the dephasing caused by field inhomogeneities [74-79]. In this review, we present an overview of mechanisms and practical implementations of these three types of experimental technique. Detailed descriptions are presented in the second to fifth sections. Other high-resolution experimental or data processing methods used in inhomogeneous fields are also described briefly.

2. High-resolution NMR spectroscopy based on iMQCs

2.1. Mechanism for iMQC spectroscopy

In contrast to other spectroscopic techniques, NMR spectroscopy has a comprehensive theoretical framework [80-82], which can be used to derive signal expressions and predict the effects of most pulse sequences. Generally, an NMR experiment has to be adjusted when the theoretical prediction and experimental result are inconsistent [83]. This is the reason why an anomalous experimental phenomenon observed by Warren and coworkers in highly polarized nuclear spin systems [84-86] drew extensive attention. In that experiment, Warren and coworkers applied a simple pulse sequence composed of two radio-frequency (RF) pulses plus two gradient pulses, also referred to as the CRAZED sequence (COSY Revamped with Asymmetric Z-

gradient Echo Detection) to a pure water sample. Conventional NMR theory predicts no observable signals with CRAZED. However, obvious signals were found in the resulting 2D NMR spectrum, and these signals had iMQC characteristics. This abnormal observation generated tremendous interest and controversy in the NMR community.

In subsequent studies, it was shown that iMQC signals are similar to the multiple spin echoes (MSEs) in concentrated solutions first explored by Deville *et al.* in ^3He [87]. Both iMQCs and MSEs originate from distant dipolar interactions of spins. To explain the observation of iMQC signals, two new theoretical methods were proposed. The classical distant dipolar field (DDF) treatment [88-90] adds a non-linear term to the Bloch equations. Alternatively, the quantum-mechanical iMQC treatment [41, 91] represents the observable signals as deriving from multi-spin operators in the equilibrium density matrix. Although the fundamental assumptions of the classical and quantum treatments are different, it has been shown that both lead to exactly the same predictions of signals for simple pulse sequences such as the CRAZED sequence [89, 90, 92]. Due to their attractive properties, iMQCs have been extensively explored. Low-order iMQCs originating from two-spin terms, including iZQCs [83], iDQCs [44] and iSQCs [93], have been analyzed theoretically and experimentally. High-order iMQCs originating from multi-spin terms have also been observed in high magnetic fields. Although high order iMQC signals are observable, most iMQC applications focus on iZQCs, iDQCs and iSQCs due to their intrinsically stronger signal intensities in most common cases.

Apart from their fundamental significance, the unique properties of iMQCs make them applicable in both NMR spectroscopy and MRI. In the past two decades, a variety of spectroscopic techniques based on iZQCs, iDQCs and iSQCs have been established for high-resolution NMR spectroscopy in inhomogeneous fields [41, 45, 53]. All these iMQC

spectroscopic methods are variations on the basic CRAZED sequence, with improvements in acquisition efficiency, spectral dimension and resolution. To show the intrinsic robustness of iMQC signal against field inhomogeneities, we analyse the basic iZQC CRAZED sequence as an example. The quantum iMQC treatment is applied to deduce the signal evolution and spectral pattern in inhomogeneous fields.

Consider a liquid mixture consisting of I (corresponding to the solvent) and S (corresponding to the solute) components, where I is a single spin-1/2 system, and S is an AX spin-1/2 system (comprising spins S_k and S_l spins with a mutual scalar coupling constant J_{kl}). Assume that I is abundant and S is either abundant or dilute, that I and S are spins of the same isotope, that ω_m is the frequency offset of spin m ($m = I, S$) in the rotating frame in the absence of field inhomogeneity, and that $\Delta B_m(\mathbf{r})$ is the inhomogeneous magnetic field deviation of experienced by spin m at position $\mathbf{r}$. For simplification, effects of radiation damping, diffusion, relaxation, and intermolecular NOE are ignored. (In this review, all the spin systems we refer to in theoretical descriptions are such $I+S$ spin-1/2 system except where otherwise noted, and the J coupling is not taken into account in the analysis when it is not the focus.)

For the quantum iMQC treatment, two basic assumptions in conventional NMR theory are abandoned. The dipolar interaction between spins in solution (which is only partially averaged away by diffusion) is now included, and the high-temperature approximation to the spin density operators at equilibrium is removed. For the homonuclear $I+S$ spin system, the spin density operators at thermal equilibrium without the high-temperature approximation is given by [94]

$$\rho_{eq} = 2^{-(N_I+N_S)} \prod_i (1 - \mathfrak{I}I_{iz}) \times \prod_j (1 - \mathfrak{I}S_{jz}), \quad (1)$$

where $\mathfrak{S} = 2 \tanh(\hbar\omega_0 / 2kT)$. N_I and N_S represent the numbers of I and S spins, respectively. The indices i and j run over every spin in the sample; $\hbar$ is Planck's constant, k is the Boltzmann constant, and T is absolute temperature; ω_0 is the Larmor frequency; I_{iz} and S_{jz} stand for the longitudinal components of the I_i and S_j spins, respectively. When the basic iZQC CRAZED sequence represented by $(\pi/2)_y \rightarrow t_1 \rightarrow G \rightarrow (\pi/4)_y \rightarrow t_2$ is applied, where G represents a coherence selection gradient pulse (CSG), the first non-selective $(\pi/2)_y$ pulse rotates the longitudinal components of I and S spins into the xy plane. During the evolution period t_1 , I and S spins evolve under chemical shifts and the CSG. The density operator just before the second RF pulse is

$$\begin{aligned} \rho(z, t_1^-) = & 2^{-(N_I+N_S)} \prod_i \left\{ 1 - \mathfrak{S} [I_{ix} \cos(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z) + I_{iy} \sin(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z)] \right\} \\ & \times \prod_j \left\{ 1 - \mathfrak{S} [S_{jx} \cos(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z) + S_{jy} \sin(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z)] \right\}. \end{aligned} \quad (2)$$

The dephasing angle for a spin at position $\mathbf{r}$ due to the CSG is $\gamma G \delta z$, where γ is the gyromagnetic ratio, z is the component of $\mathbf{r}$ along the z direction, and G and δ are the amplitude and duration of the CSG. I_{ix} , I_{iy} , S_{jx} and S_{jy} stand for the x and y components of I_i and S_j spins, respectively. After the second RF pulse, Eq. (2) becomes

$$\rho(z, t_1^+) = 2^{-(N_I + N_S)} \prod_i \left\{ \begin{aligned} &1 - \Im I_{iy} \sin(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z) \\ &-\frac{\sqrt{2}}{2} \Im [I_{ix} \cos(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z) - I_{iz} \cos(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z)] \end{aligned} \right\} \\ \times \prod_j \left\{ \begin{aligned} &1 - \Im S_{jy} \sin(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z) \\ &-\frac{\sqrt{2}}{2} \Im [S_{jx} \cos(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z) - S_{jz} \cos(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z)] \end{aligned} \right\}. \quad (3)$$

The Hamiltonian for homonuclear dipolar interactions can be approximated as

$$H_{d,eff}^{sec} = \sum_{i=1}^{N_I} \sum_{j=1}^{N_S} D_{ij} 2I_{zi} S_{zj}. \quad (4)$$

The residual intermolecular dipolar coupling constant is given by [95]

$$D_{ij} = \frac{\mu_0}{4\pi} \frac{\gamma^2 \hbar}{2r_{I_i S_j}^3} (1 - 3 \cos^2 \theta_{I_i S_j}), \quad (5)$$

where μ_0 is the magnetic vacuum permeability, $r_{I_i S_j}$ is the distance between the spins I_i and S_j ,

and $\theta_{I_i S_j}$ is the angle between the $I_i - S_j$ internuclear vector and the static magnetic field.

During the detection period t_2 , the spin density operators evolve under chemical shifts and the effective dipolar interaction Hamiltonian. The result is

$$\rho(z, t_1 + t_2) = 2^{-(N_I + N_S)} \times \prod_i \left\{ \begin{aligned} &1 - \frac{\sqrt{2}}{2} \Im \cos(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z) \\ &\times \left[\begin{aligned} &I_{ix} \cos(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \cos(D_{ij} t_2) + 2I_{iy} S_{jz} \cos(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &+ I_{iy} \sin(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{ix} S_{jz} \sin(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &- I_{iz} \end{aligned} \right] \\ &- \Im \sin(\omega_I t_1 + \gamma \Delta B_I(\mathbf{r}) t_1 + \gamma G \delta z) \\ &\times \left[\begin{aligned} &I_{iy} \cos(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{ix} S_{jz} \cos(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &- I_{ix} \sin(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{iy} S_{jz} \sin(\omega_I t_2 + \gamma \Delta B_I(\mathbf{r}) t_2) \sin(D_{ij} t_2) \end{aligned} \right] \end{aligned} \right\} \\ \times \prod_j \left\{ \begin{aligned} &1 - \frac{\sqrt{2}}{2} \Im \cos(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z) \\ &\times \left[\begin{aligned} &S_{jx} \cos(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \cos(D_{ij} t_2) + 2I_{iz} S_{jy} \cos(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &+ S_{jy} \sin(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{iz} S_{jx} \sin(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &- S_{jz} \end{aligned} \right] \\ &- \Im \sin(\omega_S t_1 + \gamma \Delta B_S(\mathbf{r}) t_1 + \gamma G \delta z) \\ &\times \left[\begin{aligned} &S_{jy} \cos(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{iz} S_{jx} \cos(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \sin(D_{ij} t_2) \\ &- S_{jx} \sin(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \cos(D_{ij} t_2) - 2I_{iz} S_{jy} \sin(\omega_S t_2 + \gamma \Delta B_S(\mathbf{r}) t_2) \sin(D_{ij} t_2) \end{aligned} \right] \end{aligned} \right\}. \quad (6)$$

We can use this expression to calculate the observable signal [84]. The final expression is given as

$$\begin{aligned}
 M_+^{total}(t_1 + t_2) &= Tr \left[\rho(t_1 + t_2) \gamma \hbar (I_i^+ + S_j^+) / V \right] \\
 &= \frac{i}{2} M_0^I e^{i(\omega_I + \gamma \Delta B_I(r))t_2} \left(\frac{\sqrt{2}}{2} + 1 \right) J_0(\xi_1) \times J_1\left(\frac{2}{3}\xi_2\right) e^{i(\omega_I + \gamma \Delta B_I(r) - \omega_S - \gamma \Delta B_S(r))t_1} \\
 &\quad + \frac{i}{2} M_0^I e^{i(\omega_I + \gamma \Delta B_I(r))t_2} \left(\frac{\sqrt{2}}{2} - 1 \right) J_0(\xi_1) \times J_1\left(\frac{2}{3}\xi_2\right) e^{i(\omega_S + \gamma \Delta B_S(r) - \omega_I - \gamma \Delta B_I(r))t_1} \\
 &\quad - i M_0^I e^{i\omega_I t_2} J_1(\xi_1) \times J_0\left(\frac{2}{3}\xi_2\right) \\
 &\quad + \frac{i}{2} M_0^S e^{i(\omega_S + \gamma \Delta B_S(r))t_2} \left(\frac{\sqrt{2}}{2} + 1 \right) J_0(\xi_2) \times J_1\left(\frac{2}{3}\xi_1\right) e^{i(\omega_S + \gamma \Delta B_S(r) - \omega_I - \gamma \Delta B_I(r))t_1} \\
 &\quad + \frac{i}{2} M_0^S e^{i(\omega_S + \gamma \Delta B_S(r))t_2} \left(\frac{\sqrt{2}}{2} - 1 \right) J_0(\xi_2) \times J_1\left(\frac{2}{3}\xi_1\right) e^{i(\omega_I + \gamma \Delta B_I(r) - \omega_S - \gamma \Delta B_S(r))t_1} \\
 &\quad - i M_0^S e^{i\omega_S t_2} J_1(\xi_2) \times J_0\left(\frac{2}{3}\xi_1\right),
 \end{aligned} \tag{7}$$

where M_0^I and M_0^S are the initial equilibrium magnetizations of spins I and S , respectively;

$J_0(\xi_1)$, $J_0(\xi_2)$, $J_1\left(\frac{2}{3}\xi_1\right)$, and $J_1\left(\frac{2}{3}\xi_2\right)$ are Bessel functions of order 0 and 1; $\xi_1 = -\gamma \mu_0 M_0^I t_2$

and $\xi_2 = -\gamma \mu_0 M_0^S t_2$ are arguments of Bessel functions. Eq. (7) shows the quantitative expression of 2D iZQC CRAZED signals of the solute spin S and solvent spin I . It can be seen that four signal peaks will be detected in the 2D spectrum, such as the cross-peak for S spin located at $(\omega_I + \gamma \Delta B_I(r) - \omega_S - \gamma \Delta B_S(r), \omega_S + \gamma \Delta B_S(r))$. These signals suffer from inhomogeneous broadening along the F_2 dimension, while they occur at the difference in Larmor frequency between I and S spins in the F_1 dimension. According to the iMQC treatment, cross-peaks are originated from distant dipolar interactions between spins in different molecules. The distant dipolar interactions survive from diffusional averaging within the dipolar correlation distance (d_c).

In the iZQC CRAZED sequence, the gradient pulse modulates the resonance frequency of spins as a linear function of the spatial position along the gradient axis. Hence the transverse magnetization is wound up into a helix, which represents a highly ordered state. It can be unwound subsequently by another gradient pulse with opposite polarity. The product of the magnitude G and duration δ of the gradient pulse during the evolution period τ is kept constant to maintain a fixed half-wavelength of the magnetization helix, or dipolar correlation distance, $d_c = \pi \gamma G \delta$. Typically, d_c ranges from 10 μm to a few mm and can be adjusted by changing the gradient pulse area [91, 96]. The low limit is approximate and essentially restricted by the diffusion properties of the sample in relation to the echo time of the pulse sequence. If the helix pitch is too short, diffusion will blur the helix so that the DDF cannot refocus the signals properly, and signal intensity will be lost [97]. This distance is much smaller than the typical sample size in an NMR experiment, and can be smaller than the anatomical detail that causes line broadening *in vivo*. In principle, only the inhomogeneous field inside the dipolar correlation distance affects the line-widths of iMQC spectra. Although the magnetic field in a tissue may be inhomogeneous due to susceptibility variations, the magnetic field within d_c can be relatively homogeneous, and $\Delta B_I(\mathbf{r})$ can be taken as equal to $\Delta B_S(\mathbf{r})$ (i.e. $\Delta B_I(\mathbf{r}) = \Delta B_S(\mathbf{r})$). Therefore, the above iZQC cross-peak referred to above for the S spin will be located at a frequency $\omega_I - \omega_S$ in the F_1 dimension, and will be free from the influence of field inhomogeneities beyond the dipolar correlation distance. Intuitively, a high-resolution one-dimensional (1D) spectrum free from the effects of field variations can be obtained by simply projecting cross-peaks onto the F_1 dimension. Similar analyses can be done for iDQCs and iSQC. To date, existing iMQC high-

resolution NMR methods have been designed based on three different low-order iMQCs for different applications. In the following, the iZQC, iDQC, and iSQC high-resolution approaches are described in detail, respectively.

2.2. iZQC high-resolution spectroscopic techniques

2.2.1. HOMOGENIZED sequence

Warren and coworkers first applied iZQCs to high-resolution NMR spectroscopy, and proposed an initial pulse sequence dubbed HOMOGENIZED (HOMOGeity Enhancement by Intermolecular Zero-quantum Detection) [41] to recover the spectral information of chemical shifts and J couplings from inhomogeneous fields. The HOMOGENIZED sequence is shown in Fig. 1. Two non-selective pulses $\pi/2$ and $\pi/4$ are used for signal excitation and coherence transfer, respectively, and the gradient pulse G is applied along the z direction to select the iZQC signal during indirect evolution period t_1 . An additional non-selective π pulse is inserted in the middle of the delay interval 2Δ to avoid signal dephasing before the intermolecular dipolar interaction has time to take effect. After the signal acquisition and data processing, a 2D spectrum results and spectral information free from the influence of field inhomogeneities is obtained along the F_1 dimension, since it is iZQC signals that evolve during the t_1 period.

Fig. 1

Fig. 1. HOMOGENIZED sequence for high-resolution iZQC spectra in inhomogeneous fields (Source: Adapted from Ref. [41])

The HOMOGENIZED method uses gradient echoes to detect iZQCs in a sample. The

availability of at least one concentrated component constitutes an indispensable condition for iZQC detection. Taking longitudinal relaxation into account, the signal expression for an $I+S$ spin system using the HOMOGENIZED sequence can be derived as

$$S_{HOMOGENIZED} = \frac{1}{4} \mathcal{M}_0 t_2 \left\{ \begin{aligned} & \sum_{\substack{m,n=I,S \\ m \neq n}} \frac{2\sqrt{2}}{3} M_0^m M_0^n e^{-t_2/T_1^m} e^{-i(\omega_m - \omega_n)t_1} e^{-i[\omega_m + \Delta B(r)]t_2} \\ & + \sum_{m=I,S} i(M_0^m)^2 e^{-t_2/T_1^m} e^{-i[\omega_m + \Delta B(r)]t_2} \end{aligned} \right\} + \frac{\sqrt{2}i}{2} \sum_{m=I,S} M_0^m (1 - e^{-t_1/T_1^m}) e^{-i[\omega_m + \Delta B(r)]t_2}, \quad (8)$$

where T_1^m is the longitudinal relaxation time of spin m . Eq. (8) shows that three types of signals are included in the 2D HOMOGENIZED spectrum: iZQC cross-peaks from different spins (the first term), iZQC cross-peaks from the identical spins (the second term), and SQC signals from the residual longitudinal magnetization (the third term).

Fig. 2

Fig. 2. iZQC spectra from HOMOGENIZED sequence for a mixture of butanone and acetone in a deshimmmed inhomogeneous field. All cross-peaks in the 2D spectrum are extended along the F_2 dimension by the field inhomogeneity, while the expanded view of the acetone-butanone triplet shows the enhancement of spectral resolution along the F_1 dimension. (Source: Adapted from Ref. [41])

Fig. 2 shows the HOMOGENIZED experimental result for a solution of 5% butanone in acetone under an intentionally deshimmmed inhomogeneous field with 85 Hz line-width. The spectral line is broadened along the F_2 dimension and no useful information can be obtained

from the projection along the F_2 dimension. However the iZQC cross-peaks between butanone and acetone in the 2D spectrum are free from inhomogeneous line broadening along the F_1 dimension, and the information on chemical shifts and J couplings can be obtained via the 1D projection onto the F_1 dimension. The expanded acetone-butanone cross-peak shows the elimination of inhomogeneity influence along the F_1 dimension and a triplet with 7 Hz J coupling splitting is recovered. Hence, HOMOGENIZED provides a potential tool for extracting high-resolution NMR information from 2D iZQC spectra in inhomogeneous fields.

2.2.2. Improved HOMOGENIZED versions

The HOMOGENIZED sequence provides a route to high-resolution 1D information and has shown its ability in several practical experiments. However, some flaws, such as strong t_1 -noise ridges from solvent-solvent iZQCs, solute-solute iZQCs, and residual longitudinal magnetization, remain technical obstacles for practical applications. Furthermore, conventional SQC signals may degrade desired signal separations and affect the regeneration of a high-quality 1D NMR spectrum. Based on the distinct understanding of iZQC mechanisms and experimental schemes, several improved versions have been proposed. Firstly, a HOMOGENIZED experiment with a solvent-selective $\pi/2$ pulse, named S90 [54], was suggested (Fig. 3a). In the S90 sequence, the second non-selective $\pi/4$ pulse in the HOMOGENIZED sequence is changed into a solvent-selective pulse. Since only solvent spins are involved in the coherence transfer step in the S90 sequence, unwanted solute-solvent iZQCs with negative frequencies in the F_1 dimension and solute-solute iZQCs can be removed and the resulting 2D spectra are more useful for the extraction of spectral information. When the S90 sequence is employed, the theoretical signal expression is given as following

$$S_{S90} = \frac{1}{3} \mu_0 t_2 M_0^I M_0^S e^{-i(\omega_S - \omega_I)t_1} e^{-i[\omega_S + \Delta B(r)]t_2} - \frac{i}{4} \mu_0 t_2 (M_0^I)^2 e^{-i[\omega_I + \Delta B(r)]t_2}. \quad (9)$$

Comparing with Eq. (8), it can be noticed that only solvent-solute iZQC (the first term) and solvent-solvent iZQC (the second term) are retained in Eq. (9). Even so, strong t_1 -noise ridge still exist in the S90 results. The reason is that the non-selective π pulse before acquisition refocuses not only signals from the desired solvent-solute iZQCs but also strong signals from residual solvent SQCs and solvent-solvent iZQCs. To further suppress strong t_1 -noise further, a SElective HOMoGENIZED (SEL-HOMOGENIZED) sequence [45] shown in Fig. 3b was designed, in which the non-selective π pulse is replaced by a WATERGATE [98] pulse sequence element, $(\pi)_{-x}^I - (\pi)_{-x}$ sandwiched between a crusher gradient pair. Since the solvent-selective π pulse in the WATERGATE element only selectively excites solvent spins, the net flip angle of the WATERGATE composite pulse is almost zero for solvent spins. Theoretically, strong solvent signals are dephased by the inhomogeneous field and the crusher gradient pair. In the meantime, spins outside the suppression band only experience one refocusing pulse, so their signals are refocused. Along the same lines, an $S90(WS)^2$ sequence with two additional WATERGATE modules was proposed to further suppress the solvent signals, as shown in Fig. 3c. The signal expression for SEL-HOMOGENIZED or $S90(WS)^2$ can be given as

$$S_{S90} = \frac{1}{3} \mu_0 t_2 M_0^I M_0^S e^{-i(\omega_S - \omega_I)t_1} e^{-i[\omega_S + \Delta B(r)]t_2}. \quad (10)$$

In Eq. (10), only the desired solvent-solute iZQC signals remain. Due to non-ideal experimental conditions, such as imperfect RF pulses and coherence transfer gradients, there is generally some residual solvent SQC signal, which affects the quality of iZQC spectra. It has been shown that a modified CRAZED sequence with three RF pulses can perform well in the suppression of interfering signals and yield “pure” iZQC signals without conventional SQC residuals [99, 100]. A three-pulse HOMOGENIZED sequence with an intermolecular double-quantum filter (DQF) [46], dubbed iDQF-HOMOGENIZED and abbreviated as iDH (Fig. 3d), was then designed to suppress strong solvent signals in 2D iZQC high-resolution spectra. Compared to the mechanism of solvent suppression in SEL-HOMOGENIZED and S90(WS)², the built-in DQF module in the iDQF-HOMOGENIZED serves as an effective way to eliminate the intense conventional solvent SQC signal. Furthermore, different coherence transfer pathways can be selected by suitable choice of RF pulse flip angles. Hence, theoretically not only conventional SQCs but also all orders of iMQC signals of solvent can be filtered out. In the following, the raising and lowering operator formalism is used to understand the mechanism of iDQF-HOMOGENIZED:

$$\begin{cases} I_{zi}I_{zj} \\ I_zS_z \end{cases} \xrightarrow{\frac{\pi}{2}I_x, \frac{\pi}{2}S_x} \begin{cases} \frac{1}{4}(I_i^+I_j^- + I_i^-I_j^+) \\ \frac{1}{4}(I^+S^- + I^-S^+) \end{cases} \xrightarrow{\pi I_x} \begin{cases} \frac{1}{8}\sin^2 \pi I_i^+I_j^+ \\ \frac{1}{4}\sin^2 \frac{\pi}{2} I^+S^+ \end{cases} \xrightarrow{\frac{\pi}{2}I_x} \frac{i}{4}I_zS^+ \xrightarrow{\pi I_x, \pi S_x, D_{IS}I_zS_z t_2} -\frac{i}{4}S^-. \quad (11)$$

The iZQC terms, $I_i^+I_j^-$ and $I_i^-I_j^+$ for solvent, I^+S^- and I^-S^+ for solute, are excited and evolve during the t_1 period. Only I^-S^+ can pass the DQF and be converted into an anti-phase single-quantum term I_zS^+ . In the spin echo period 2Δ and acquisition period t_2 , the Hamiltonian

due to long-range intermolecular dipolar interactions converts $I_z S^-$ into the observable in-phase term S^- . The final signal expression for iDQF-HOMOGENIZED is

$$S_{S90} = \frac{1}{4} \mu_0 t_2 M_0^I M_0^S e^{-i(\omega_S - \omega_I) t_1} e^{-i[\omega_S + \Delta B(r)] t_2}. \quad (12)$$

Fig. 3

Fig. 3. Improved HOMOGENIZED pulse sequences: (a) S90; (b) SEL-HOMOGENIZED; (c) S90(WS)²; (d) iDQF-HOMOGENIZED. Black bars indicate hard RF pulses, and Gaussian-shaped RF pulses are selective for the I spin (i.e. solvent spin).

An experimental comparison was performed on a sample of 10% butanone in cyclohexane to show the advantage of the SEL-HOMOGENIZED sequence over the HOMOGENIZED sequence. In experiments, the magnetic field was intentionally deshimmmed to produce a line-width of 50 Hz. Fig. 4 shows experimental results from the HOMOGENIZED and SEL-HOMOGENIZED sequences. All iZQC cross-peaks are extended along the F_2 dimension, but high-resolution information can be recovered from the F_1 dimension. In the 2D HOMOGENIZED spectrum (Fig. 4a), strong solvent signals and a complex spectral pattern are seen. In the 2D SEL-HOMOGENIZED spectrum (Fig. 4b), the solvent signal is significantly suppressed and the desired solvent-solute iZQC cross-peaks are distributed along the diagonal of the spectrum. After a 45° clockwise rotation, a projection along the F_1 dimension gives a 1D high-resolution spectrum free from the inhomogeneous line broadening (Fig. 4c). Expanded

views of peaks of the $-\text{CH}_3$ and $-\text{CH}_2\text{CH}_3$ groups of butanone are presented in Fig. 4d, where exact J couplings in the $-\text{CH}_2\text{CH}_3$ group can be determined.

Fig. 4

Fig. 4. 2D ^1H NMR spectra of 10% butanone in cyclohexane in an inhomogeneous field. (a) Spectrum acquired with HOMOGENIZED sequence, (b) spectrum acquired with SEL-HOMOGENIZED sequence, (c) the result obtained by 45° clockwise rotation of the spectrum of (b) and its 1D projection along the F_1 dimension, and (d) expanded views of the A~C peaks in (c) and their projections along the F_1 dimension. (Source: Adapted from Ref. [45])

A sample of a piece of grape sarcocarp fitted in a 5mm NMR tube was used to test the feasibility of the iDQF-HOMOGENIZED sequence for biological samples with intrinsic macroscopic susceptibility gradients and intense water signal. Experimental results are shown in Fig. 5. The line-width of the water resonance is 100 Hz in the conventional 1D NMR spectrum (Fig. 5a), and hardly any spectral information on solutes can be resolved. The 1D projection of the iDQF-HOMOGENIZED spectrum is shown in Fig. 5b. It can be seen that line-widths are greatly reduced after the spectrum is projected onto the horizontal axis. Furthermore, the water signal is effectively suppressed so that the two peaks of β and α glucose, which are circa 0.2 and 0.4 ppm away from the water resonance and completely concealed in the conventional 1D NMR spectrum, can be resolved. A weak triplet at 1.1 ppm from a methyl group with low concentration can also be observed, and the doublet of β glucose is well resolved (Fig. 5d). For comparison, a 1D spectrum of extracted grape juice acquired in a homogeneous field is presented in Fig. 5c. It can be seen that the projection of the iDQF-HOMOGENIZED spectrum of grape

sarcocarp is quite similar to the 1D high-resolution spectrum of grape juice and has reasonable signal-to-noise ratio (SNR).

Fig. 5

Fig. 5. ^1H NMR spectra of a piece of grape sarcocarp in an inhomogeneous field with (except for (c)) a line-width of 100 Hz. (a) Conventional single-pulse spectrum, (b) projection of the sheared 2D iDQF-HOMOGENIZED spectrum along the F_1 dimension, (c) conventional 1D spectrum of grape juice in a homogeneous field, and (d) part of the 2D iDQF-HOMOGENIZED spectrum with integral projections along both dimensions. Sine bell weighting functions were used for the iDQF-HOMOGENIZED spectrum, which is displayed in absolute value mode. Insets are expanded views. (Source: Adapted from Ref. [46])

Experimental comparisons between S90, SEL-HOMOGENIZED, S90(WS)², and iDQF-HOMOGENIZED sequences were performed on a piece of grape sarcocarp. The results are shown in Fig. 6a-d, together with their integral projections along the F_1 dimension. The corresponding spectra when the gradients are along the magic angle direction are given in Fig. 6e-h. To apply a gradient along the magic angle direction, gradients along the X and Z directions with a strength ratio of $1:\sqrt{2}$ are used, so that the resultant gradient is oriented along the direction $\phi=54.7^\circ$ with respect to the magnetic field B_0 . This suppresses iMQC signals, so magic angle gradients can be used to verify the purity of iMQC signals. All the 2D spectra are plotted with the same intensity scale and contour levels. The projection level is normalized to give a constant height of solvent peak. It can be seen that the strong water signal in S90 (Fig. 6a) is suppressed to a certain level by one or two WATERGATE blocks, and the peak of α -glucose located at 5.15

ppm becomes resolvable in the 2D SEL-HOMOGENIZED and S90 (WS)² spectra (Fig. 6b and c). Fig. 6e indicates that the solvent signal in S90 is mainly conventional SQC caused by T_1 relaxation in the t_1 period. Nevertheless, the WATERGATE block(s) at the end of sequences cannot effectively suppress the conventional solvent signal (Fig. 6f and g). The t_1 noise decreases dramatically in the iDQF-HOMOGENIZED spectrum (Fig. 6d), since almost all the solvent signal is filtered out by the DQF (Fig. 6h).

Fig. 6

Fig. 6. Comparison of performances of four improved HOMOGENIZED sequences on a piece of grape sarcocarp. The top row shows the 2D spectra and their integral projections for (a) S90, (b) SEL-HOMOGENIZED, (c) S90 (WS)², and (d) iDQF-HOMOGENIZED; the bottom row shows the magic-angle-gradient spectra corresponding to the top row. The projection levels of the top row are normalized for a same height of solvent resonance, and those of the bottom row are the same as the corresponding ones in the top row. (Source: Adapted from Ref. [46])

2.2.3. *iZQC spectroscopy with fast acquisition schemes*

In general, HOMOGENIZED and its improved versions all require 2D sampling of the full range of chemical shifts of solutes along both F_1 and F_2 dimensions. Moreover, spectral lines free of inhomogeneous broadening are achieved along the F_1 dimension. Therefore sufficient t_1 increments are required to obtain satisfactory spectral resolution, and longer acquisition times than for 1D sampling cannot be avoided. In practical applications, especially *in vivo* applications, long acquisition times will degrade the spectral resolution when magnetic fields are unstable or the organism being studied moves. Physiological motions such as heartbeat and breathing

irregularly distort every transient and contribute to t_1 -noise that ultimately broadens spectral line-widths in the indirect dimension. Hence, the improvement of 2D acquisition efficiency is important for practical applications of iZQC spectroscopy.

Fig. 7

Fig. 7. Pulse sequence for iZQC acquisition with four-fold speedup, seen as a superposition of four t_1 steps in a HOMOGENIZED sequence. (Source: Adapted from Ref. [42])

Warren and coworkers proposed a novel approach that enables four to thirty-two times improvement in the acquisition efficiency of 2D iZQC spectra [42]. Additionally, motion artifacts can be averaged for better spectral quality. The sequence can be seen as a superposition of several HOMOGENIZED experiments with different t_1 values (Fig. 7). The $\pi/2$ pulse produces iZQCs. During the iZQC evolution period a CSG is applied along the z direction. A small flip angle pulse transfers only a small fraction of the iZQCs into detectable signals. This leaves most coherences evolving in t_1 still available for next mixing pulse. In this way, the magnetization is partitioned into small amounts and each of them is characterized by a different t_1 evolution time. Spoiling gradients (S_a , S_b , S_c and S_d in Fig. 7) act as CSGs to distinguish the different concurrent evolutions. A small gradient following a mixing pulse winds the different magnetizations into a helix, and during the acquisition period only one of these helices is rephased at a given time. Fig. 8 demonstrates the result of this sequence on an aqueous solution of 5% butanone with eight t_1 steps per scan. This approach can be extended to other 2D pulse sequences. However, since the magnetization is partitioned into small amounts with different t_1 evolution time in this acquisition scheme, the desired iZQC signals are weaker than the residual

solvent signal and the intense solvent signal remains a great obstacle to its application.

Fig. 8

Fig. 8. 2D ^1H NMR spectrum of a sample of 5% butanone in water acquired with the fast iZQC pulse sequence with eight indirect acquisition points per scan. (Source: Adapted from Ref. [42])

Although the fast iZQC technique can speed up acquisition four to thirty-two fold, it is not suitable for *in vivo* applications due to the low sensitivity brought about by small-flip-angle excitation. Inheriting the merits of iDQF-HOMOGENIZED, three iDH2 sequences [59] (iDH2-2J, iDH2-1J, and iDH2-0J, see Fig. 9) were proposed by our group to shorten 2D acquisition time while preserving reasonable iZQC signal intensity. With the help of a substantial reduction in the spectral width of the F_1 dimension and of a fold-over correction (FOC) technique [101], the acquisition time for a 2D iZQC spectrum can be shortened by up to 50 times. Three types of iZQC spectra, with homodecoupling, original J -coupling splitting, and doubled J -coupling splitting respectively, can be obtained by appropriate manipulation of the t_1 period. Solvent suppression is also improved by the combined use of DQF and excitation sculpting (WS module). As previously, the raising and lowering operator formalism was used to elaborate the mechanism of iDH2 sequences.

$$I_z S_z \xrightarrow{\frac{\pi}{2} I_x, \frac{\pi}{2} S_x} \frac{1}{4} I^+ S^- \xrightarrow{\pi S_x} \frac{1}{4} I^+ S^+ \xrightarrow{\frac{\pi}{2} I_x} \frac{1}{4} I_z S^+ \xrightarrow{\pi I_x, \pi S_x, D_{FS} I_z S_x t_2} \frac{1}{4} S^-, \quad (13)$$

Neither longitudinal magnetization I_z nor two-spin solvent terms such as $I^+ I$, $I_z I_z$ in the t_1 period can be converted into +2-quantum terms by the solute-selective π pulse, so neither will pass

through the DQF period between the $(\pi)^S$ and $(\pi/2)^I$ pulses. The "Localization" module is used for spatial localization.

Fig. 9

Fig. 9. Schematic diagram for iDH2 sequences. (Source: Adapted from Ref. [59])

For the iDH2-2J sequence shown in Fig. 9a, the evolution period t_1 is divided into two equal parts by a solute-selective π pulse, and the resulting solvent-solute iZQC signals are located at

$$\begin{aligned} \text{iDH2-2J : } (\Omega_1, \Omega_2) &= [(\Omega_I - \Omega_S \pm \pi J_{kl}) / 2 + (\Omega_I + \Omega_S \pm \pi J_{kl}) / 2, \Omega_S \mp \pi J_{kl}] \\ &= (\Omega_I \pm \pi J_{kl}, \Omega_S \mp \pi J_{kl}) = (\omega_I + \Delta B \pm \pi J_{kl}, \omega_S + \Delta B \mp \pi J_{kl}), \end{aligned} \quad (14)$$

in which the chemical shift evolution of solute signals during the t_1 period is suppressed. When the transmitter frequency coincides with ω_I , that is $\omega_I = 0$, the spectral width of the F_1 dimension is determined only by the field inhomogeneity ΔB and the acquisition efficiency is improved. To obtain a 1D high-resolution integral projection, a shearing process is performed along the F_2 dimension by a counter-clockwise rotation of $\phi = -\arctan(\Delta B / \Delta B) = -\pi/4$. This operation transforms the frequency coordinates (Ω_1, Ω_2) into

$$(\Omega_1, \Omega'_2) = (\Omega_1, \Omega_2 - \Omega_1) = (\omega_I + \Delta B \pm \pi J_{kl}, \omega_S - \omega_I \mp 2\pi J_{kl}). \quad (15)$$

Then a high-resolution spectrum with doubled J -coupling splittings can be obtained by 1D projection onto the F_2 dimension.

For the iDH2-1 J sequence shown in Fig. 9b, normal evolution applies in the evolution period t_1 , and the resulting solvent-solute iZQC signals are located at

$$\text{iDH2-1}J: (\Omega_1, \Omega_2) = (\Omega_I - \Omega_S \pm \pi J_{kl}, \Omega_S \mp \pi J_{kl}) = (\omega_I - \omega_S \pm \pi J_{kl}, \omega_S + \Delta B \mp \pi J_{kl}) \quad (16)$$

For the iDH2-0 J sequence shown in Fig. 9c, the evolution period t_1 is replaced by a constant-time scheme with a hard π pulse placed between the first and second pulses, and the resulting solvent-solute iZQC signals are located at

$$\begin{aligned} \text{iDH2-0}J: (\Omega_1, \Omega_2) &= [-(-\Omega_I + \Omega_S \pm \pi J_{kl}) / 2 + (\Omega_I - \Omega_S \pm \pi J_{kl}) / 2, \Omega_S \mp \pi J_{kl}] \\ &= (\Omega_I - \Omega_S, \Omega_S \mp \pi J_{kl}) = (\omega_I - \omega_S, \omega_S + \Delta B \mp \pi J_{kl}). \end{aligned} \quad (17)$$

In both Eqs. (16) and (17), the chemical shifts of solutes are included in the F_1 dimension and it seems impossible to decrease the number of sampling points required in the evolution period t_1 and improve the acquisition efficiency by reducing the spectral width of the F_1 dimension. However, since all peaks occur in a band near the anti-diagonal (see Eqs. (16) and (17)), fast acquisition can be achieved without information loss by reducing the sampling rate in the t_1 period with and then applying FOC. The procedure for 1D high-resolution iDH2-0 J and iDH2-1 J spectra is as follows: undersampling, FOC, shearing, and projection. After FOC and shearing, Eqs. (16) and (17) become

$$\begin{aligned} \text{iDH2-1J: } (\Omega_1^{\text{FOC}}, \Omega_2') &= (\Omega_1^{\text{FOC}}, \Omega_2 - \Omega_1^{\text{FOC}}) = (\omega_l + \Delta B, \omega_s - \omega_l \mp \pi J_{kl}), \\ \text{iDH2-0J: } (\Omega_1^{\text{FOC}}, \Omega_2') &= (\omega_l + \Delta B \mp \pi J_{kl}, \omega_s - \omega_l). \end{aligned} \quad (18)$$

Clearly, high-resolution 1D spectra with normal J coupling splittings from iDH2-1J, and no J coupling splittings from iDH2-0J, can be achieved by projection of the respective 2D spectra onto the F_2 dimension.

Fig. 10 shows localized iDH2 results for an intact mudskipper (*Periophthalmus modestus*) post mortem. The sample was fitted into a 5 mm NMR tube. Fig. 10a shows the sagittal and axial spin-echo images with TE = 20 ms. The MRS studies were performed with three-dimensional (3D) localization. The voxel, which is marked on the images with dotted lines, was positioned near the spinal cord with a size of $3 \times 3 \times 10 \text{ mm}^3$. Fig. 10b, c and d are iDH2-1J, -2J, and -0J spectra, respectively. A PRESS spectrum was also acquired for comparison. It can be seen that the line-width is greatly reduced by the iDH2 sequences, and the J -scaling information is well retained for the doublet at 1.3 ppm with the iDH2-1J and iDH2-2J sequences.

Fig. 10

Fig. 10. Post-mortem studies of an intact mudskipper. (a) Sagittal and axial spin-echo images, (b) ~ (d) localized ^1H NMR spectra for iDH2-1J, -2J, and -0J respectively, (e) PRESS spectrum, and (f) High-resolution MAS spectrum of muscle tissues of the same mudskipper sample. Sine bell weighting functions were used for the iDH2 spectrum, which is shown in absolute value mode. (Source: Adapted from Ref. [59])

2.2.4. iZQC spectroscopy in unstable fields

As well as spatial variations, magnetic fields may also suffer from temporal variations. This often occurs in resistive or hybrid magnets. Although the magnetic fields they provide are stronger than those available from conventional superconducting magnets, their poor spatial homogeneity and temporal stability are both challenging for high-resolution spectroscopy. The HOMOGENIZED sequence proves useful in inhomogeneous magnetic fields, but its performance degenerates in unstable fields. The Carr-Purcell-Meiboom-Gill (CPMG) module with composite pulses can be effective in refocusing both spatial and temporal variations of magnetic fields. Warren and coworkers designed an acquisition scheme, named as CPMG-HOMOGENIZED, for measuring high-resolution iZQC spectra in unstable fields. In the CPMG-HOMOGENIZED sequence (Fig. 11) [43], the original single spin echo module with a non-selective π pulse in the middle of the delay interval 2Δ in the HOMOGENIZED sequence is replaced by the CPMG module.

Fig. 11

Fig. 11. (a) Experimental and (b) simulated 2D iZQC spectra for the composite CPMG-HOMOGENIZED sequence, plotted with a lowest contour level of 3 times the noise standard deviation. The CPMG-HOMOGENIZED sequence is shown inset. (Source: Adapted from Ref. [43])

Fig. 11 shows 2D experimental and simulated iZQC spectra for the CPMG-HOMOGENIZED sequence. The experimental sample was a mixture of methyl ethyl ketone and water with a volume ratio of 1:1. The experiment was performed in a 25T electromagnet with 1 kHz/s drift and 3 kHz line-width over 1 cm³ volume. The results show that the CPMG-

HOMOGENIZED detection approach can remove 99% of the effects of field inhomogeneity and generate a high-resolution liquid-state NMR spectrum at more than 1 GHz field strength. The effects of spatial inhomogeneity and temporal instability of the magnetic field on spectra are nearly eliminated, resulting in a resolution enhancement of circa 100 times. The J couplings during the t_1 period are preserved. Complementing advances in magnet technology with such methodological innovation in spin engineering provides unique and critical advantages. The shortcomings of the HOMOGENIZED sequence, however, remain in this sequence.

The CPMG-HOMOGENIZED sequence eliminates the perturbation of unstable field during the delay 2Δ . However, the effect of frequency variation during t_2 ($\int_0^{t_2} \Delta\omega(\mathbf{r}, t) dt$) still leads to random phase perturbation for each t_1 increment and thus produces t_1 -noise. Since iZQC spectroscopy provides high-resolution spectral information in the F_1 dimension, the chemical shift information in the F_2 dimension can be discarded. Based on this consideration, the iDQF-HOMOGENIZED sequence was modified to incorporate short time acquisition (STA) [66] and phase spectrum schemes to improve the SNR in unstable fields (Fig. 12) [50].

Fig. 12

Fig. 12. Modified iDQF-HOMOGENIZED sequence with short time acquisition scheme. (Source: Adapted from Ref. [50])

Spectra of an agar gel phantom containing 90 mM creatine hydrate (Cr) and 30 mM DL-lactic acid (Lac) acquired using the modified iDQF-HOMOGENIZED sequence in the presence of B_0 field fluctuations with a variation range of 60 Hz and a period of 4s are shown in Fig. 13. For the 2D spectrum obtained with the STA scheme, peaks are broadened along the F_2

dimension since the acquisition time is very short. The t_1 -noise is greatly reduced in the 2D spectrum and in its projection onto the indirect dimension. On the other hand, the projection cross-section theorem [102] states that the projection of a 2D phase-sensitive spectrum onto the indirect dimension forms an FT pair with the time-domain signals at $t_2 = 0$. Theoretically, an F_1 projection of the modified iDQF-HOMOGENIZED spectrum in phase-sensitive mode cannot only provide a phase-sensitive spectrum with improved line-widths, but also eliminate t_1 -noise due to unstable fields if t_2 acquisition begins at the echo maximum. The experimental result shown in Fig. 13c verifies this deduction. Although severe t_1 -noise remains in the 2D spectrum, it cancels out to result in a t_1 -noise free projection.

Fig. 13

Fig. 13. 2D NMR spectra from a modified iDQF-HOMOGENIZED sequence in an unstable field: absolute-value spectra (a) without and (b) with STA, and phased-mode spectra (c) without and (d) with STA. (Source: Adapted from Ref. [50])

2.2.5. 2D high-resolution iZQC spectroscopy

All the aforementioned iZQC methods are designed to provide 1D high-resolution information based on 2D acquisition. It is well known that 2D NMR spectroscopy can provide more information about structures and compositions. Moreover, the 1D NMR spectra of complex samples are generally congested, leading to severe signal overlap and complicating unambiguous spectral assignments. By analogy with the measurement of 1D high-resolution spectra using 2D acquisition in iZQC spectroscopy, it is possible to obtain high-resolution 2D spectra *via* 3D acquisition of iZQC signals. A pulse sequence, iZQC-COSY, was proposed by our group for

obtaining high-resolution 2D COSY spectra in inhomogeneous fields by 3D acquisition [49]. Since iZQC evolution periods in the sequence are insensitive to field inhomogeneities, high-resolution COSY spectra can be recovered from inhomogeneous fields by projecting the 3D data onto the indirectly acquired 2D plane.

Fig. 14

Fig. 14. The iZQC-COSY sequence for high-resolution 2D COSY spectrum in inhomogeneous fields *via* 3D acquisition. The coherence transfer pathway is presented to illustrate the sequence mechanism. Source: Adapted from Ref. [49])

The iZQC-COSY pulse sequence is shown schematically in Fig. 14. Four linear CSGs are applied along the z direction to select the coherence transfer pathway $0 \rightarrow 0 \rightarrow +2 \rightarrow +1 \rightarrow -1$. In this pathway, iZQC evolutions are included in two indirect detection periods t_1 and t_2 , and are both insensitive to field inhomogeneities. The amplitudes of the first and second CSGs, G_1 and G_2 , are sufficiently large to dephase all coherence orders except for ZQCs. The third and fourth CSGs, with an area ratio of $1:-2$, are applied to eliminate the residual conventional SQC signals. The second non-selective $\pi/2$ and fourth solvent-selective $\pi/2$ RF pulses are mixing pulses between two evolution periods. The fifth solvent-selective π RF pulse achieves coherence transfer for solvent spins. The sixth solvent-selective $\pi/2$ RF pulse rotates the solvent spins to the z direction to generate a DDF. A non-selective π pulse is inserted in the middle of two delay intervals (2Δ) to prevent the signals from being dephased by field inhomogeneities before the DDF takes effect. The excitation sculpting scheme [103] is used as a solvent-suppression module prior to acquisition. To achieve a high-resolution 2D COSY spectrum, we discard the direct

acquisition dimension (F_3 , which suffers from the field inhomogeneity) while retaining two indirect dimensions (F_1 and F_2 dimensions), and then perform an integral projection of the 3D spectrum onto the F_1 - F_2 plane. Eq. (19) in following provides a quantitative expression for the diagonal and cross-peaks in the projected high-resolution 2D iZQC-COSY spectrum for the S_k spin of an $I + S$ spin system.

$$M_+^{S_k}(t_1, t_2) = p \frac{M_0^S}{8} J_1 \left(\frac{2}{3} \mathcal{M}_0 M_0^I (2\Delta + t_3) \right) \times \left\{ \begin{aligned} & e^{i[(\omega_{S_k} + \pi J_{kl} - \omega_l)t_1 - (\omega_{S_k} - \pi J_{kl} - \omega_l)t_2]} e^{iJ_{kl}\Delta} + e^{i[(\omega_{S_k} - \pi J_{kl} - \omega_l)t_1 - (\omega_{S_k} + \pi J_{kl} - \omega_l)t_2]} e^{iJ_{kl}\Delta} \\ & + e^{i[(\omega_{S_k} + \pi J_{kl} - \omega_l)t_1 - (\omega_{S_k} + \pi J_{kl} - \omega_l)t_2]} e^{-iJ_{kl}\Delta} + e^{i[(\omega_{S_k} - \pi J_{kl} - \omega_l)t_1 - (\omega_{S_k} + \pi J_{kl} - \omega_l)t_2]} e^{-iJ_{kl}\Delta} \\ & - e^{i[(\omega_{S_k} + \pi J_{kl} - \omega_l)t_1 - (\omega_{S_l} - \pi J_{kl} - \omega_l)t_2]} e^{iJ_{kl}\Delta} + e^{i[(\omega_{S_k} - \pi J_{kl} - \omega_l)t_1 - (\omega_{S_l} - \pi J_{kl} - \omega_l)t_2]} e^{iJ_{kl}\Delta} \\ & + e^{i[(\omega_{S_k} + \pi J_{kl} - \omega_l)t_1 - (\omega_{S_l} + \pi J_{kl} - \omega_l)t_2]} e^{-iJ_{kl}\Delta} - e^{i[(\omega_{S_k} - \pi J_{kl} - \omega_l)t_1 - (\omega_{S_l} + \pi J_{kl} - \omega_l)t_2]} e^{-iJ_{kl}\Delta} \end{aligned} \right\}, \quad (19)$$

where p is a factor related to the integral projection.

Fig. 15

Fig. 15. 2D COSY spectra of butanone in cyclohexane. (a) Conventional spectrum in an inhomogeneous field with a line-width of 50 Hz; (b) projected iZQC-COSY spectrum in the same inhomogeneous field; (c) conventional spectrum in a well-shimmed field; and (d) simulated iZQC-COSY spectrum. The regions of the diagonal-peak (I) and cross-peak (II) in (b) to (d) are expanded for comparison. Gaussian weighting functions were used for the iZQC-COSY spectrum. (Source: Adapted from Ref. [49])

Experimental and simulated 2D COSY spectra of a sample of butanone in cyclohexane are presented in Fig. 15. In the conventional COSY spectrum acquired in an inhomogeneous field (Fig. 15a), all signals are extended along the diagonal direction. The J coupling multiplets are hardly resolved in either the 1D projection spectra or the 2D spectrum. Since the iZQC evolution (t_1 and t_2 periods) is intrinsically insensitive to magnetic field inhomogeneity, high resolution is achieved in the projected iZQC-COSY spectrum (Fig. 15b). Both chemical shifts and J splittings are preserved. Additionally, the strong solvent peak is suppressed. The diagonal peak centered at (0.95 ppm, 0.95 ppm) (labeled as I) and the cross-peak centered at (2.41 ppm, 0.95 ppm) (labeled as II) are expanded to illustrate the efficacy of the sequence at removing inhomogeneous line broadening. The line-width of the projection spectrum is reduced from 50 to 4 Hz. The simulated result shown in Fig. 15d is consistent with the experimental one. The conventional COSY spectrum in a well-shimmed field is shown in Fig. 15c. The experiment time for iZQC-COSY is long, due to the 3D acquisition, therefore it is necessary to improve the acquisition efficiency for practical applications.

To achieve high-resolution 2D J -resolved spectra in inhomogeneous fields with high acquisition efficiency, we introduced a spatially-encoded scheme [104, 105] into the iZQC technique [60]. Traditionally, numerous t_1 increments are required to obtain a 2D spectrum with a good resolution, resulting in long acquisition times. For the spatially encoded technique, the usual t_1 encoding is replaced by spatial encoding, in which signals are decoded during the detection period by an echo planar spectroscopic imaging (EPSI) scheme. By using the spatially encoded technique, a 2D J -resolved spectrum can be obtained with only two scans and the acquisition time is reduced to a few seconds. In the 2D iZQC J -resolved pulse sequence with spatially encoded scheme (Fig. 16), only one spatially encoded block is needed, and a selective

$\pi/2$ RF pulse replaces the second $\pi/2$ hard RF pulse. A bipolar gradient pulse pair, with adiabatic RF pulses repeated n_E times, is inserted before a single adiabatic pulse, together with a gradient pulse to alleviate the impact of molecular diffusion. We can observe that the iZQC (S^+I^- coherence with order $p = 0$) between solute S and solvent I evolves during the spatially-encoded period. The coherence S^+I^- is transferred to S^+I_z by the selective $\pi/2$ RF pulse. At the end of the spatially encoded period, the accumulated phase is given by

$$\begin{aligned}\varphi_{E2}(\mathbf{r}) &= \frac{4N_E[\omega_I + \Delta\omega_0(\mathbf{r})]^2}{R} + \frac{[\omega_I + \Delta\omega_0(\mathbf{r}) + \gamma\mathbf{G}_E \cdot \mathbf{r}]^2 + O_i^2}{R} \\ &\quad - \frac{4N_E[\omega_S + \Delta\omega_0(\mathbf{r})]^2}{R} - \frac{[\omega_S + \Delta\omega_0(\mathbf{r}) + \gamma\mathbf{G}_E \cdot \mathbf{r}]^2 + O_i^2}{R} \\ &= \frac{2(\omega_I - \omega_S)[(2n_E + 1)\gamma\mathbf{G}_E \cdot \mathbf{r} + \Delta\omega_0(\mathbf{r})]}{R} + \frac{\omega_I^2 - \omega_S^2}{R},\end{aligned}\tag{20}$$

where G_E and G_D are the amplitudes of the encoding and decoding gradient pulses, respectively. The sweep rate $R = \Delta\omega_{ad} / \tau_{ad}$, is given by the ratio between the sweep width $\Delta\omega_{ad}$ and the duration τ_{ad} of the adiabatic pulse. $\Delta\omega_0(\mathbf{r})$ is the spatially-dependent frequency induced by the inhomogeneous field at position $\mathbf{r}$, O is the offset of the adiabatic pulse, O_i is the initial frequency of the adiabatic pulse. The frequency $2(\omega_S - \omega_I)$ shown in the first term is the difference in chemical shift between the solute spin S and solvent spin I .

Fig. 16

Fig. 16. The 2D iZQCJ-resolved sequence with spatially encoded scheme. (Source: Adapted from Ref. [60])

Experiments were performed on a dilute solution of 97 mM ethyl 3-bromopropionate ($\text{BrCH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$) in acetone to illustrate the practical applicability of the iZQC J -resolved sequence in inhomogeneous fields. The results are shown in Fig. 17. It can be seen that all multiplet patterns (three triplets located at a, c and d and one quartet located at b) are clearly visible in the 2D spectra. The isolated spins (CH_3 of methanol located at e) are also detected. The whole spectral pattern is the same as the conventional J -resolved spectrum that would be obtained in a well-shimmed magnetic field. The solvent peak is located at 2.09 ppm, and mainly comes from the residual longitudinal magnetization that arises from to longitudinal relaxation during the evolution period. The average line-widths of the solute peaks, measured from the projection spectra along the chemical shift axis (F_1 dimension) are approximately 31.6 Hz (Fig. 17c) and 33.2 Hz (Fig.17f). The resolution in the F_2 dimension is satisfactory compared to the conventional 2D high-resolution J -resolved spectrum and should be sufficient for most J -resolved experiments (Fig. 17d vs. 17f). These results indicate that the spatially encoded pulse sequence is feasible in inhomogeneous fields with high acquisition efficiency.

Fig. 17

Fig. 17. Experimental results for a solution of 97 mM ethyl 3-bromopropionate in acetone. (a, c, e) Spectra acquired in a homogeneous field, and (b, d, f) spectra acquired in an inhomogeneous field with 110 Hz line-width. (a, b) Conventional 1D spectra, (c, d) conventional 2D water suppressed J -resolved spectra and their 1D projections, and (e, f) 2D iZQC J -resolved spectra and their 1D projections. Gaussian weighting functions were used throughout. (Source: Adapted from Ref. [60])

2.3. *iDQC high-resolution spectroscopic techniques*

The *iDQC*s share many of the same characteristics as *iZQC*s [106]. They also originate from the distant dipolar interactions between coupled spins within a distance d_c over which magnetic field variations are generally much smaller than the magnetic field variations across the whole sample. Hence, *iDQC*s similarly possess the potential for high-resolution spectroscopy in inhomogeneous fields. Moreover, additional phase cycling is not required to obtain “pure” *iDQC* signals and the signal intensities of *iDQC*s are approximately 30% stronger than those of *iZQC*s [87]. A series of *iDQC* sequences have been proposed to achieve high-resolution spectra in inhomogeneous fields.

2.3.1. *IDEAL sequence*

The first pulse sequence based on *iDQC*s for facilitating the recording of high-resolution information in inhomogeneous fields was established by our group with the name IDEAL (Intermolecular Dipolar-interaction Enhanced All Lines) [44]. The IDEAL sequence is shown in Fig. 18. The first non-selective $\pi/2$ pulse excites all orders of *iMQC*s, then a DQF module consisting of a solvent-selective $\pi/2$ pulse and a pair of gradient pulses is applied to select the desired solvent-solute *iDQC*s. A theoretical derivation similar to that in Section 2.1 can be carried out for the IDEAL sequence. The signals from the IDEAL sequence can be expressed as following

$$S_{\text{IDEAL}} = \frac{1}{3}(\gamma\mu_0 M_0^I t_2) M_0^S e^{-i[(\omega_S + \omega_I + 2\Delta B(r))t_1 - (\omega_S + \Delta B(r))t_2]} + \frac{i}{4}(\gamma\mu_0 M_0^I t_2) M_0^I e^{-i[(2\omega_I + 2\Delta B(r))t_1 - (\omega_I + \Delta B(r))t_2]}, \quad (21)$$

where the first term represents the desired solvent-solute iDQC signal and the second term the solvent-solvent iDQC signal. The solvent-solute iDQC cross-peak has a frequency range of $\omega_S + \omega_I + 2\Delta B(r)$ along the F_1 dimension. The corresponding frequency range in the F_2 dimension is $\omega_S + \Delta B(r)$. Hence, solvent-solute iDQC cross-peak will appear as separate streaks along a specific direction defined by $\phi = \arctan 2$, where ϕ is the angle with respect to the F_2 axis, due to the field inhomogeneity $\Delta B(r)$. Although the inhomogeneity broadens signals in both the F_1 and F_2 dimensions, projection of cross-peaks onto an axis perpendicular to the direction ϕ gives a 1D high-resolution spectrum free from inhomogeneous line broadening. Since the DDF of the solvent affects all solute spins equally, the relative areas and chemical shifts of peaks in the 1D projection from a 2D iDQC spectrum are the same as those obtained from a routine high-resolution 1D spectrum. The apparent scalar coupling constants in the projection spectrum are 1.5 times the conventional ones, which may help to obtain clearer multiplet patterns in weakly coupled systems, but may cause spectral peak overlap in strongly coupled systems.

Fig. 18

Fig. 18. IDEAL sequence for high-resolution NMR spectra in inhomogeneous fields. (Source: Adapted from Ref. [44])

To test the performance of the IDEAL sequence, we carried out an experiment on a mixture of 2-propanol and dimethylsulfoxide (DMSO). Fig. 19a shows the resulting spectrum. The intermolecular cross-peaks extend along the direction ϕ in the 2D spectrum. The trace indicates that J multiplets are narrow and well resolved. To obtain a 1D high-resolution spectrum, the data matrix needs to be tilted by $-\phi$ along with respect to the F_1 axis firstly, to bring signals into line along $F_1 = 0$, and then tilted by $-\phi$ along the F_2 axis to place the inhomogeneous broadening orthogonal to F_2 . After this shearing procedure, a projection along the F_2 axis yields a spectrum with inhomogeneous line broadening suppressed (Fig. 2(c)). The line-width is reduced from 105 to 2 Hz, remarkably similar to that in the conventional 1D high-resolution ^1H spectrum.

Fig. 19

Fig. 19. (a) IDEAL spectrum of a mixture of 2-propanol and DMSO in an inhomogeneous field with a line-width of 105 Hz. The -OH, -CH, and -CH₃ regions are shown expanded in both the F_1 and F_2 directions, with 1D cross-sections along the dashed lines; (b) conventional 1D ^1H spectrum in the inhomogeneous field; (c) integral projection of the sheared spectrum shown in (a); and (d) single-pulse spectrum in a well-shimmed field. (Source: Adapted from Ref. [44])

2.3.2. IDEAL-II sequence

As with the HOMOGENIZED sequence, the IDEAL sequence requires a long acquisition time for a 2D data set since the whole chemical shift range of solutes needs to be covered in the indirect dimension. This restricts its practical applications. To enhance the acquisition efficiency, we proposed a modified IDEAL sequence, named IDEAL-II [57]. As in IDEAL experiments, the 1D spectrum extracted from the 2D data set acquired with the IDEAL-II sequence retains

information on chemical shifts, J coupling constants, multiplet patterns, and relative peak areas, independent of magnetic field homogeneities in the limit of negligible diffusion effects. Although a 2D acquisition is still required, the acquisition time for this new sequence can be shortened substantially in comparison to the IDEAL and HOMOGENIZED sequences. The time saving is proportional to the ratio between the chemical shift range and the field inhomogeneity induced frequency range; this can be an order of magnitude or more, in a moderately inhomogeneous field.

Fig. 20

Fig. 20. IDEAL-II sequence for high resolution spectra in inhomogeneous fields. (Source: Adapted from Ref. [57])

The IDEAL-II sequence is shown schematically in Fig. 20. As in conventional 2D J -resolved spectroscopy, the evolution time t_1 in IDEAL-II is divided into two equal parts by the second and third RF pulses, which are applied one immediately after the other. The first and third RF pulses are non-selective, while the second RF pulse is selective for the solvent spins. A pair of CSGs with an area ratio of 1:2 is applied in the two $t_1/2$ evolution periods to select the desired coherence transfer pathway.

$$\begin{aligned} \begin{Bmatrix} I_{zi} I_{zj} \\ I_z S_z \end{Bmatrix} &\xrightarrow{\frac{\pi}{2} I_x, \frac{\pi}{2} S_x} \begin{Bmatrix} -\frac{1}{4} I_i^+ I_j^+ \\ -\frac{1}{4} I_i^+ S^+ \end{Bmatrix} \xrightarrow{\frac{\pi}{2} I_x, \pi I_x, \pi S_x} \begin{Bmatrix} \frac{i}{8} (I_i^- I_{zj} + I_{zi} I_j^-) \\ \frac{i}{4} I_z S^- \end{Bmatrix} \xrightarrow{\pi I_x, \pi S_x, \begin{Bmatrix} D_{IS} I_z S_z t_2 \\ D_{IS} I_{zi} I_{zj} t_2 \end{Bmatrix}} \begin{Bmatrix} \frac{i}{8} (I_i^- + I_j^-) \\ \frac{i}{4} S^- \end{Bmatrix}. \quad (22) \end{aligned}$$

The final expression for the solute spin S_k in the AX spin system can be given as

$$M_{+}^{S_k}(t_1 + t_2) = \frac{i(0.5t_1 + t_2)\mu_0\gamma M_0^I}{4} \left(\frac{kT}{\hbar\omega_I} \right)^2 M_0^S \{ e^{-i[\omega_I/2 - \pi J_{kl} + \gamma\Delta B(r)]/2} t_1 e^{i[\omega_{S_k} + \pi J_{kl} + \gamma\Delta B(r)]t_2} + e^{-i[\omega_I/2 + \pi J_{kl} + \gamma\Delta B(r)]/2} t_1 e^{i[\omega_{S_k} - \pi J_{kl} + \gamma\Delta B(r)]t_2} \}. \quad (23)$$

Eq. (23) shows that the iDQC cross-peak is a doublet located at $(\omega_I/2 - \pi J_{kl} + \gamma\Delta B(r))/2, \omega_{S_k} + \pi J_{kl} + \gamma\Delta B(r))$ and $(\omega_I/2 + \pi J_{kl} + \gamma\Delta B(r))/2, \omega_{S_k} - \pi J_{kl} + \gamma\Delta B(r))$. If the spectrometer reference frequency coincides with the resonant frequency of the I spin, i.e., $\omega_I = 0$, the iDQC cross-peak doublet between the I and S spins will have frequency coordinates $(-\pi J_{kl}, \omega_{S_k} + \pi J_{kl})$ and $(\pi J_{kl}, \omega_{S_k} - \pi J_{kl})$, and extend as separate streaks along a direction defined by $\phi = \arctg(2) = 63.4^\circ$, where ϕ is the angle of the spectral streaks with respect to the F_1 axis. Since the sizes of J coupling constants are much smaller than that of the field inhomogeneity, the spectral width needed for the F_1 dimension is approximate half the range of field inhomogeneity. This implies that all streaks in the 2D spectrum occupy only a narrow frequency range in the F_1 dimension, enabling a substantial decrease of the F_1 spectral width that needs to be sampled, and greatly reducing experiment time and data size. Although the ranges of F_1 streaks depends on the field inhomogeneity in both F_1 and F_2 dimensions, a projection of cross-peaks onto the F_2 dimension after a counter-clockwise rotation of 63.4° offers a 1D high-resolution spectrum without inhomogeneous line broadening. The projected spectrum in the F_2 dimension has the same multiplet patterns as a conventional 1D spectrum with a scale factor of $1 + \cot \phi = 3$ for the J coupling constants.

Fig. 21 shows the spectra of a piece of grape sarcocarp obtained using the IDEAL-II sequence. The line-width of the water resonance (near 4.8 ppm) is 100 Hz in the conventional 1D NMR spectrum (Fig. 21a). After projection of the 2D IDEAL-II spectrum onto the horizontal axis, several separated resonances of glucose are observable (Fig. 21b and c), similar to the spectrum of grape juice [107]. The line-width is reduced from 100 Hz to 4 Hz. In addition, a weak triplet peak from a methyl group with low concentration is observed at 1.1 ppm. Its J -coupling constant can be determined since the triplet is well resolved. Two peaks of β and α glucose, which are 0.2 and 0.4 ppm away from the water resonance respectively [107] are also observed. The experimental results suggest that line broadening due to intrinsic macroscopic susceptibility gradients can be almost completely removed by the IDEAL-II method.

Fig. 21

Fig. 21. 2D IDEAL-II measurement on a piece of grape sarcocarp: (a) Conventional 1D NMR spectrum with a line-width of 100 Hz for water, (b) 2D iDQC spectrum using the IDEAL-II sequence after a counter-clockwise rotation of 63.4° and projection; and (c) four expanded regions of (b). Insets are magnified 50 times for (a) and 10 times for (b) in the vertical direction. (Source: Adapted from Ref. [57])

2.3.3. Hadamard-IDEAL-II sequence

Besides applications in 1D spectroscopy, iDQCs can also be applied to obtain high-resolution 2D spectra, such as COSY. An iZQC method mentioned in Section 2.2 has been designed for high-resolution 2D COSY in inhomogeneous fields *via* 3D acquisition. The acquisition time is generally more than several hours, which is a great obstacle in time-critical

studies, such as studies of dynamic process. An alternative method for fast acquisition of 2D COSY spectra is based on the Hadamard encoded and decoded technique [58]. The basic strategy is to excite only the signal-bearing areas in the indirect detection dimension with soft polychromatic excitation pulses encoded by reference to a Hadamard matrix with order N . Then the number of t_1 increments needed for a conventional high-resolution COSY spectrum is replaced by a relatively small number N , and the evolution period t_1 in every scan is replaced by a constant duration, beneficial for shortening the acquisition time. Based on the Hadamard encoded technique and the IDEAL-II sequence, a new pulse sequence, named Hadamard-IDEAL-II, was designed for measuring high-resolution COSY spectra in inhomogeneous fields with time-efficient acquisition. The sequence is shown in Fig. 22. The first selective soft pulse is a polychromatic excitation pulse capable of exciting all interested signal-bearing areas with Hadamard encoding. The CSGG₁ is applied to select iZQCs during the constant period τ . The remaining parts of this sequence are the same as in IDEAL-II except for the solvent suppression module (WS).

Fig. 22

Fig. 22. Hadamard-IDEAL-II sequence for measuring high-resolution 2D COSY spectra in inhomogeneous fields with time-efficient acquisition. (Source: Adapted from Ref. [58])

To perform a Hadamard-IDEAL-II experiment, a 1D high-resolution spectrum is pre-acquired with the IDEAL-II sequence to determine the number of frequency sites (M) and their frequency values. According to the number M and the frequencies of the different sites, soft polychromatic pulses are produced by reference to a Hadamard matrix of order N ($N > M$). After

N scans and Hadamard decoding, a set of 2D spectra can be obtained. After shearing the decoded spectra and projecting them onto the F_2 dimension, we would have N 1D high-resolution spectra running in the F_2 dimension, which include all correlation information for reconstructing a high-resolution COSY spectrum.

A sample of 1-bromobutane and 4-bromoanisole dissolved in dichloromethane was utilized to testify the feasibility of the Hadamard-IDEAL-II sequence. Two COSY spectra were obtained using the Hadamard-IDEAL-II and conventional COSY sequences respectively in the same inhomogeneous field (106 Hz line-width), along with a conventional COSY spectrum acquired in a well-shimmed field. Experimental results are illustrated in Fig. 23. The 2D COSY spectra acquired with the conventional method in the well-shimmed and deshimmed fields are shown in Fig. 23a and b, respectively. The reconstructed COSY spectrum from the Hadamard-IDEAL-II sequence is shown in Fig. 23c. Four regions labeled as I, II, III and IV are expanded to show details. It can be seen that the COSY spectrum obtained with the Hadamard-IDEAL-II method (Fig. 23c) provides identical information to that in the conventional high-resolution COSY spectrum (Fig. 23a), except that apparent J coupling constants are magnified three-fold.

Fig. 23

Fig. 23. Experimental 2D COSY spectra of a sample of 1-bromobutane and 4-bromoanisole in dichloromethane. (a) Conventional spectrum obtained in a well-shimmed field, (b) conventional spectrum obtained in an inhomogeneous field, and (c) Hadamard-IDEAL-II spectrum obtained in the same inhomogeneous field as (b). Regions labeled as (I), (II), (III) and (IV) were expanded for comparison. Gaussian weighting functions were used for the Hadamard-IDEAL-II spectrum. (Source: Adapted from Ref. [58])

2.4. iSQC high-resolution spectroscopic techniques

2.4.1. IDEAL-III sequence

So far, high-resolution approaches based on iZQCs and iDQCs have been described. Similarly, iSQC is available for high-resolution applications in inhomogeneous fields. In a simple two-spin system ($I + S$), the transverse relaxation time of solute-solvent iZQCs or iDQCs can be expressed by $\frac{1}{T_2^{IS}} = \frac{1}{T_2^I} + \frac{1}{T_2^S}$ [88], in which T_2^I and T_2^S are the transverse relaxation time of SQCs of solvent and solute, respectively. Similarly, the apparent diffusion coefficients of solute-solvent iZQCs or iDQCs are expressed as $D^{IS} = D^I + D^S$ [88], in which D^I and D^S are the diffusion coefficients of solvent and solute. It is obvious that compared to conventional SQCs, iZQCs and iDQCs have larger apparent diffusion coefficients and smaller transverse relaxation times [88], whereas iSQC has similar diffusion and relaxation behaviors [87], hence iSQC offers different characteristics in the measurement of high-resolution spectra. Taking into account the properties of iSQC and the mechanisms of the iMQC high-resolution methods, a pulse sequence named IDEAL-III [48] was designed by our group which was the first application of iSQC to high-resolution spectroscopy in inhomogeneous fields. The IDEAL-III sequence is time-efficient, requiring only a narrow spectral width in the indirect dimension. As well as being applicable to chemical solutions under external non-ideal magnetic fields, IDEAL-III is also suitable for biological tissues and *in vivo* localized high-resolution applications.

Fig. 24

Fig. 24. Pulse sequences for high-resolution spectra in inhomogeneous fields *via* iSQC:(a)

IDEAL-III, and (b) IDEAL-III with PRESS localization. Filled vertical bars stand for non-selective RF pulses, sinc- and Gaussian-shaped pulses are selective RF pulses, dashed rectangles represent coherence selection gradients, shadowed rectangles represent slice-selective gradients, and vertical-line rectangles represent “W3” binomial π pulses. (Source: Adapted from Ref. [47])

The IDEAL-III sequence is shown schematically in Fig. 24a. In the sequence, the first and the third RF pulses are selective for solvent, while the second RF pulse is selective for solute. To select the coherence transfer pathway $0 \rightarrow +1 \rightarrow +2 \rightarrow +1 \rightarrow -1$, we use three linear CSGs with an area ratio of 1:0.7:−2.4. In this way, the desired solvent-solute iSQC are selected while both conventional SQCs and other iMQCs are filtered out. The excitation sculpting before acquisition and a specially designed phase cycling scheme further improve the solvent suppression efficiency. Additionally, a PRESS-like module [108] is integrated with the IDEAL-III sequence to obtain spatially localized iSQC spectra (Fig. 24b). Three slice-selective refocusing π RF pulses can both select a region of interest in a sample and refocus the iSQC signals, thus the non-selective π RF pulse in the non-localized IDEAL-III sequence can be omitted. As in the aforementioned sequences, the signal expression of the IDEAL-III for a solute spin S_k in an AX spin system can be derived as

$$M_{+}^{S_k}(z, t_1 + t_2) = \frac{-M_0^S \cos(2\pi J_{kl}\Delta)}{2} J_1 \left(\frac{2}{3} \mu_0 M_0^I (2\Delta + t_2) \right) \times \left\{ e^{i[-\omega_l t_1 + (\omega_{S_k} + \pi J_{kl})t_2 - \gamma \Delta B(r)t_1 + \gamma \Delta B(r)t_2]} + e^{i[-\omega_l t_1 + (\omega_{S_k} - \pi J_{kl})t_2 - \gamma \Delta B(r)t_1 + \gamma \Delta B(r)t_2]} \right\}. \quad (24)$$

From Eq. (24), it can be seen that only solvent-solute iSQC signal is selected and it splits

into two peaks at $(\omega_I + \gamma \Delta B(\mathbf{r}), \omega_{S_k} + \pi J_{kl} + \gamma \Delta B(\mathbf{r}))$ and $(\omega_I + \gamma \Delta B(\mathbf{z}), \omega_{S_k} - \pi J_{kl} + \gamma \Delta B(\mathbf{z}))$ due to the J coupling. If ΔB is the frequency range induced by the spatially dependent field inhomogeneity along the B_0 direction, the resonance frequency for the cross-peak would range over $\omega_{S_k} \pm \pi J \pm \gamma \Delta B/2$ in the F_2 dimension and $\omega_I \pm \gamma \Delta B/2$ in the F_1 dimension. If the spectrometer reference frequency coincides with the resonance frequency of I spin in the B_0 field, i.e. $\omega_I = 0$, the doublet intermolecular cross-peak between I and S spins will be at $(0, \omega_{S_k} + \pi J_{kl})$ and $(0, \omega_{S_k} - \pi J_{kl})$ and will extend as two separate streaks along a specific direction defined by $\phi = \arctg(1) = 45^\circ$ with respect to the F_1 axis. For all spins, all the streaks in the spectrum line up in the center of the F_1 dimension. Although the ranges of the streaks depend on the field inhomogeneity in both F_1 and F_2 dimensions, a projection of the cross-peaks onto the F_2 dimension produces a 1D high-resolution spectrum after the streaks are counter-clockwise rotated by $\pi/4$. As well as the influence of transverse relaxation and diffusion, the line-width of the projected spectrum mainly depends on the residual background inhomogeneous field within the dipolar correlation distance. The projected spectrum has the same multiplet patterns and splittings as a conventional 1D SQC spectrum.

Experiments on an *in vitro* sample of pig brain tissue fitted in a 5 mm NMR tube were performed on a Varian 500MHz NMR spectrometer to test the IDEAL-III sequence. The line-width of the water resonance (near 4.8 ppm) is 100 Hz in the conventional 1D ^1H NMR spectrum (Fig. 25a). Hardly any spectral information can be obtained from this spectrum due to the intense water signal and the line broadening caused by magnetic susceptibility gradients among the various structural components. The conventional 1D spectrum obtained utilizing the excitation sculpting scheme for solvent suppression is shown in Fig. 25b. The resolution remains very poor.

The 2D IDEAL-III spectrum after a counter-clockwise rotation of 45° and its integral projection along the F_2 axis are shown in Fig. 25c. It can be seen that the water signal is suppressed effectively and the weak metabolite signals are observable. The line-widths are greatly reduced and several multiplets of metabolites such as lactate (Lac, 1.31 ppm) and alanine (Ala, 1.47 ppm) can be resolved. However, the residual line-width in the projection spectrum is still circa 13 Hz (Fig. 25c), which may hinder the recognition of some peaks. For comparison, a 1D water-presaturated spin echo spectrum of pig brain tissue was also acquired using a Nano probe (Fig. 25d). The metabolites in the tissue were assigned according to the literature [109]. The Nano probe is specifically designed for narrow-line high-resolution MAS spectra. The line-width in Fig. 25d is 3.5 Hz in phased mode and most multiplets of the metabolites can be resolved. The main metabolites detected in the Nano spectrum can be observed in the projection of the IDEAL-III spectrum. Compared to the Nano MAS technique, the IDEAL-III sequence is more feasible and promising for *in vivo* and *in situ* NMR since most *in vivo* and *in situ* samples cannot be fitted into a Nano probe and spun.

Fig. 25

Fig. 25. ^1H NMR spectra of a sample of pig brain tissue: (a) Conventional 1D SQC spectrum; (b) 1D spectrum acquired with excitation sculpting; (c) 2D IDEAL-III spectrum after a counter-clockwise rotation of $\pi/4$ and integral projection along the F_2 axis; and (d) 1D water-presaturated MAS spin echo spectrum acquired with a Nano probe. Sine bell weighting functions were used for the IDEAL-III spectrum. (Source: Adapted from Ref. [47])

The IDEAL-III sequence with a localization module can be used for *in vivo* localized

studies. A comparison between localized IDEAL-III spectroscopy and conventional PRESS spectroscopy was performed on an *in vivo* Sprague Dawley rat. The experimental results are shown in Fig. 26. After first- and second-order auto-shimming, the line-width of solvent signal is still up to 250 Hz. No matter whether the voxel volume is $5 \times 5 \times 5 \text{ mm}^3$ or $10 \times 15 \times 20 \text{ mm}^3$ (to cover the whole cerebrum), no useful PRESS spectra can be obtained under such conditions. Therefore, all PRESS acquisitions were completed under voxel shimming condition. For the PRESS spectrum shown in Fig. 26c, the voxel volume is $5 \times 5 \times 5 \text{ mm}^3$. The voxel is displayed on the anatomic images in three orientations in Fig. 26a. There are only a few signals distinguishable, and most of the weak metabolite peaks are concealed due to the intrinsic field inhomogeneity in biological tissues *in vivo*. Although the magnetic field is voxel-shimmed, the resolution of the PRESS spectrum from the voxel of $10 \times 15 \times 20 \text{ mm}^3$ remains low (Fig. 26d), which might result from the magnetic susceptibility differences between various structural components or at the air/tissue, tissue/bone, and tissue/cerebrospinal fluid interfaces. The projection spectrum of 2D localized IDEAL-III projection acquired from the same voxel as for Fig. 26d is shown in Fig. 26e. Similar spectral resolution to Fig. 26c is obtained, and several metabolite peaks can be resolved. Obviously, 2D iSQC spectroscopy is less sensitive to intrinsic field inhomogeneity than the conventional PRESS technique.

Fig. 26

Fig. 26. ^1H MRS spectra of a rat brain. (a) and (b) Anatomic images of the two voxels investigated, with sizes of $5 \times 5 \times 5$ and $10 \times 15 \times 20 \text{ mm}^3$, in three orientations. (c) and (d) PRESS spectra acquired from the voxels indicated in (a) and (b), respectively. (e) 2D localized iSQC

projection spectrum acquired from the same voxel as (d). Sine bell weighting functions were used for the localized iSQC spectra. (Source: Adapted from Ref. [52])

2.4.2. iSQC-JRES sequence

Similarly to iZQCs and iDQCs, iSQCs can also be applied to obtain high-resolution 2D spectra. Based on iSQCs and a spin echo scheme, a pulse sequence, iSQC-JRES, was designed to recover high-resolution 2D J -resolved spectra from inhomogeneous fields *via* 3D acquisition. The iSQC-JRES sequence [48] is shown schematically in Fig. 27. The first and third RF pulses are selective for solvent, while the second RF pulse is selective for solute. Three linear CSGs with an area ratio of 1:0.7:−2.4 are applied to select the desired coherence transfer pathway. The excitation sculpting scheme before acquisition is used as a water suppression (WS) module to further suppress residual solvent signals. Compared to the direct detection period t_3 , both the indirect detection periods t_1 (iSQC evolution) and t_2 (spin echo evolution) are insensitive to field inhomogeneity. In addition, a constant-time scheme [110] is utilized to achieve J decoupling during the t_1 period, in which the minimum value for the echo time Δ is $-0.5t_1^{\max}$. To achieve a high-resolution 2D J -resolved spectrum, the direct acquisition dimension (F_3) is discarded while the two indirect dimensions (F_1 and F_2) are preserved by integral projection of the 3D spectrum onto the F_1 - F_2 plane. Since the DDF has an equal effect on all solute spins with the same spatial distribution and does not interfere with J couplings, the conventional 2D J -resolved spectral information and properties are preserved in the projection spectrum.

Fig. 27

Fig. 27. The iSQC-*J*RES sequence for measuring high-resolution 2D *J*-resolved spectra in inhomogeneous fields *via* 3D acquisition. (Source: Adapted from Ref. [48])

To test the feasibility of the iSQC-*J*RES sequence for spectral resolution enhancement, a sample of *in vitro* pig brain tissue in a 5 mm NMR tube was used. Experimental results are shown in Fig. 28. The line-width of the water resonance (near 4.8 ppm) is 60 Hz in the conventional 1D ^1H NMR spectrum (Fig. 28a). Little spectral information can be obtained from this spectrum. The 1D water-presaturated MAS spin echo spectrum acquired using a Nano probe is shown in Fig. 28b. The metabolites are assigned according to the literature [109]. The conventional 2D water-presaturated *J*-resolved spectrum after a clockwise rotation of 45° and its *J* decoupled spectrum along the F_2 dimension are shown in Fig. 28c. Since the spectral resolution is degraded by the intrinsic macroscopic susceptibility gradients, it is difficult to obtain spectral information from this spectrum. The 2D projection of the spectrum acquired using the iSQC-*J*RES sequence is presented in Fig. 28d. It can be seen that the main metabolites detected in the MAS spin echo spectrum (Fig. 28b) can be observed in the iSQC-*J*RES spectrum (Fig. 28d). Compared to Fig. 28c, the *J* decoupled projection spectrum of Fig. 28d along the F_1 dimension has much higher resolution. Most *J* coupling information for the metabolites such as lactate (Lac, 1.31 ppm), alanine (Ala, 1.47 ppm), and glutamate (Glu, 3.75 ppm) can be obtained. The spectral resolution enhancement and solvent suppression in this measurement suggest the potential of the proposed method for *in vivo* studies. However, the line-width in the projection spectrum remains 18 Hz, which may hinder the differentiation of some peaks. The acquisition time for the iSQC-*J*RES sequence remains long, and the spatially encoded scheme can also be applied to the iSQC method for high-resolution 2D *J*-resolved spectrum with high acquisition efficiency [111].

Besides the spatially encoded scheme, a fast acquisition module for 2D J -resolved spectroscopy has recently been proposed [112] and may be useful for iMQC high-resolution 2D J -resolved spectra.

Fig. 28

Fig. 28. NMR spectra of samples of pig brain tissue. (a) Conventional 1D SQC spectrum and the expanded region for the metabolites, (b) 1D water-presaturated MAS spin echo spectrum acquired with a Nano probe, (c) conventional 2D water-presaturated J -resolved spectrum after a clockwise rotation of 45° and its projection along the F_2 axis, and (d) iSQC- J RES 2D projection spectrum and its 1D projection along the F_1 axis. Gaussian weighting functions were used for the localized iSQC spectra. (Source: Adapted from Ref. [48])

3. High-resolution NMR spectroscopy based on nutation echoes

NMR experiments are typically performed with samples immersed in a magnet bore shimmed to produce a highly homogeneous static field. However, there are many cases where the sample cannot be placed inside the magnet bore, e.g. because the sample size is larger than the magnet bore, or the sample cannot be moved away from its natural environment. In such circumstances, it would be useful if a mobile magnet could be moved over an otherwise inaccessible object. Efforts in this direction have led to crucial developments of portable NMR systems [113, 114]. Such NMR systems have been utilized in well-logging, materials science, and biomedicine [115-117].

The advantage of such *ex situ* analyses is that the limitations of sample size and

transportability no longer prevail, but an accompanying handicap appears: the magnetic fields are unavoidably spatially inhomogeneous. Such magnetic field variations are usually orders of magnitude larger than those created by microscopic structures of the detected samples, so chemical shift information cannot be extracted from the conventional NMR spectra. This imposes restriction on the applications of single-sided NMR systems [117-120]. Although the Hahn spin-echo [121] and Carr-Purcell [122] sequences can remove inhomogeneous line broadening, they also completely refocus chemical shifts. This situation was changed in 2001 when a novel approach was introduced by Pines and coworkers [65]. This method utilizes matched gradients of the magnetic field and RF field to generate so-called nutation echoes and then recover chemical shifts, opening a way for high-resolution *ex situ* NMR.

3.1. Nutation echoes

If the gradient of the static magnetic field, G_0 , and the gradient of the effective RF field, G_1 , are parallel and their magnitudes are proportional, such that $G_1(\mathbf{r}) = k \cdot G_0(\mathbf{r})$ over a certain region of a sample, a nutation echo can be generated by a single RF pulse, as shown in Fig. 29a. The RF gradient pulse spreads the magnetization out in the yz plane, which can be expressed as

$$\mathbf{M}(\mathbf{r}, \tau) = M_0 \cos[\omega(\mathbf{r})\tau] \hat{z} - M_0 \sin[\omega(\mathbf{r})\tau] \hat{y}, \quad (25)$$

where M_0 is the initial z magnetization, $\omega(\mathbf{r})$ is the nutation frequency produced by the gradient $G_1(\mathbf{r})$, and $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors along the x , y and z direction respectively. During the following time interval t_1 , the transverse component of the magnetization evolves under the influence of the magnetic field gradient $\Delta\omega(\mathbf{r})$ and the chemical shift Ω .

$$\begin{aligned} \mathbf{M}_a(\mathbf{r}, t_1) = & M_0 \cos[\omega_1(\mathbf{r})\tau] \hat{\mathbf{z}} + M_0 \sin[\omega_1(\mathbf{r})\tau] \sin[\Delta\omega_0(\mathbf{r})t_1 + \Omega t_1] \hat{\mathbf{x}} \\ & - M_0 \sin[\omega_1(\mathbf{r})\tau] \cos[\Delta\omega_0(\mathbf{r})t_1 + \Omega t_1] \hat{\mathbf{y}}. \end{aligned} \quad (26)$$

If $\omega(\mathbf{r}) = \omega(0) + k \cdot \Delta\omega(\mathbf{r})$ for a spin packet at position $\mathbf{r}$, a nutation echo is formed after a time $t_1 = k\tau$

$$\begin{aligned} \mathbf{M}_a(\mathbf{r}, t_1) = & M_0 \cos[\Delta\omega(\mathbf{r})t_1 + \omega_1(0)\frac{t_1}{k}] \hat{\mathbf{z}} \\ & + 0.5M_0 \{ \cos[\Omega t_1 - \omega_1(0)\frac{t_1}{k}] - \cos[\Omega t_1 + \omega_1(0)\frac{t_1}{k} + 2\Delta\omega_0(\mathbf{r})t_1] \} \hat{\mathbf{x}} \\ & + 0.5M_0 \{ \sin[\Omega t_1 - \omega_1(0)\frac{t_1}{k}] - \sin[\Omega t_1 + \omega_1(0)\frac{t_1}{k} + 2\Delta\omega_0(\mathbf{r})t_1] \} \hat{\mathbf{y}}. \end{aligned} \quad (27)$$

From Eq. (27), it can be seen that half of the magnetization in the xy plane is independent of $\Delta\omega(\mathbf{r})$, and the other half is dependent on $\Delta\omega(\mathbf{r})$. This means that at the beginning of the refocusing period, the magnetization is dephased in the xy plane, and at the end of the refocusing period ($t_1 = k\tau$), half of the transverse magnetization is recovered while the other half is dephased. The nutation echo is modulated by the chemical shift Ω and the overall modulation frequency $\omega(0)/k$. The complete nutation echo can be recovered, if the inhomogeneous RF pulse is followed by an ideal $\pi/2$ pulse which is phased shifted by $\pi/2$ with respect to the RF gradient pulse. This pulse brings the defocused magnetization from the yz plane into the xy plane before the full magnetization is refocused. The magnetization can be expressed as

$$\mathbf{M}_b(\mathbf{r}, t_1) = M_0 \cos[\Omega t_1 - \omega_1(0)\frac{t_1}{k}] \hat{\mathbf{x}} + M_0 \sin[\Omega t_1 - \omega_1(0)\frac{t_1}{k}] \hat{\mathbf{y}}. \quad (28)$$

The SNR is enhanced by this sequence. However, it requires an ideal $\pi/2$ pulse for all spins over the detected sample, regardless of the RF field strength. Such a $\pi/2$ rotation can be realized with constant rotation composite pulses which are insensitive to B_1 [123].

Fig. 29

Fig. 29. Nutation echo based pulse sequences: the ideal $\pi/2$ pulse (approximated by a constant rotation composite pulse) is depicted in black, and inhomogeneous RF pulses are depicted in white. (a) Sequence without an ideal $\pi/2$ pulse; (b) sequence with an ideal $\pi/2$ pulse. (Source: Adapted from Ref. [66])

3.2. High-resolution spectroscopic techniques using nutation echoes

3.2.1. 1D high-resolution schemes in indirect domain

Both Eqs. (27) and (28) show that 2D NMR spectra with high resolution information along the indirect domain can be obtained by the simple pulse sequences shown in Fig. 29a and b. Fig. 30 displays two 2D spectra of *p*-anisaldehyde, obtained in a 4.2 T magnetic field with a constant gradient of 0.25 mT/cm perpendicular to the field direction. The spectrum in Fig. 30a was obtained using the sequence of Fig. 29a. The inhomogeneous RF pulse was followed immediately by acquisition; the t_1 delay was followed by the t_2 delay, thus before FT, signals corresponding to t_1 delay and t_2 should be separated. In the indirect dimension, four proton peaks in an intensity ratio of 3:2:2:1, corresponding to the methyl group, two types of aromatic proton, and the aldehyde proton, respectively, are clearly resolved. The direct dimension shows the inhomogeneous line broadening due to the B_0 field inhomogeneity; the signal intensity is proportional to the receptivity of the RF coil. The spectrum in Fig. 30b was obtained by the pulse

sequence shown in Fig. 29b under identical experimental conditions. In the indirect domain, four signals are well resolved. The direct domain displays a distortion which can be ascribed to the non-uniform excitation profile of the composite $\pi/2$ pulse performed in the inhomogeneous B_0 field..

Fig. 30

Fig. 30. 2D spectra of *p*-anisaldehyde in a magnetic field gradient of 0.25 mT/cm obtained using the pulse sequences shown in Fig. 29a and b. Both spectra were acquired immediately after the pulse(s) and processed using a shearing transformation corresponding to a ratio of $k = t_1/\tau=6.2$. (Source: Adapted from Ref. [66])

3.2.2. 1D high-resolution schemes in the direct domain

In combination with stroboscopic acquisition, the nutation echo pulse sequence can provide resolved NMR spectra in the direct detection dimension (Fig. 31a). In this pulse sequence, the magnetization is initially dephased by a β pulse with a duration of τ_β before the free evolution period τ_w . At the end of this period, a nutation echo forms immediately after this point. Another constant rotation composite $\pi/2$ pulse tips the magnetization back into the yz plane and the cycle starts again. If the acquisition is restricted to successive echoes, the resulting free induction decay (FID) leads to a high-resolution spectrum.

The pulse sequence is based on the repeated application of composite z -rotation pulses, $[\beta(\mathbf{r})]_{-z} = (\pi/2)_y [\beta(\mathbf{r})]_x (\pi/2)_{-y}$. The $\pi/2$ pulses which rotate spins from the yz plane to the xy plane to induce signals are accomplished by constant rotation composite pulses. A train of

composite z -rotation pulses is shown in Fig. 31b. During the free evolution period, spins are dephased due to the magnetic field gradient $\Delta\omega(\mathbf{r})$. The phase at time t_1 is given by

$$\varphi_1(\mathbf{r}) = \Omega \tau_{dw} / 2 + \Delta\omega_0(\mathbf{r}) \tau_{dw} / 2. \quad (29)$$

The first $\pi/2$ pulse tips spins into the yz plane and leads to a phase

$$\varphi_2(\mathbf{r}) = \pi/2 - \Omega \tau_{dw} / 2 - \Delta\omega_0(\mathbf{r}) \tau_{dw} / 2. \quad (30)$$

The local RF pulse $\omega(\mathbf{r})$ rotates spins in the yz plane during τ_β , and the phase $\varphi_3(\mathbf{r})$ is

$$\varphi_3(\mathbf{r}) = \pi/2 - \Omega \tau_{dw} / 2 - \Delta\omega_0(\mathbf{r}) \tau_{dw} / 2 + \omega_1(\mathbf{r}) \tau_\beta. \quad (31)$$

The second $\pi/2$ pulse tips the spins into the xy plane and leads to a phase

$$\varphi_4(\mathbf{r}) = -[\varphi_3(\mathbf{r}) - \pi/2] = \Omega \tau_{dw} / 2 + \Delta\omega_0(\mathbf{r}) \tau_{dw} / 2 - \omega_1(\mathbf{r}) \tau_\beta. \quad (32)$$

The resulting phase at time t_5 is given by

$$\begin{aligned} \varphi_5(\mathbf{r}) &= \varphi_4(\mathbf{r}) + \Omega \tau_{dw} / 2 + \Delta\omega_0(\mathbf{r}) \tau_{dw} / 2 = \Omega \tau_{dw} + \Delta\omega_0(\mathbf{r}) \tau_{dw} - \omega_1(\mathbf{r}) \tau_\beta \\ &= \Omega \tau_{dw} + \Delta\omega_0(\mathbf{r}) \tau_{dw} - [\omega_1(0) + k \cdot \Delta\omega_0(\mathbf{r})] \tau_\beta = \Omega \tau_{dw} - \omega_1(0) \tau_\beta + \Delta\omega_0(\mathbf{r}) (\tau_{dw} - k \tau_\beta). \end{aligned} \quad (33)$$

When the condition $\tau_w = k \cdot \beta$ is satisfied, the phase does not depend on the magnetic field gradient $\Delta\omega(\mathbf{r})$ and the nutation echo takes place at a time $\tau_w/2$ after the z -rotation pulse is applied. The constant phase shift $[\alpha(0) \cdot \beta]$ can be corrected by adjusting the phases of the RF pulses or by a shift of the reference frequency in the spectrum after data processing.

Fig. 31

Fig. 31. (a) Pulse sequence for direct detection of resolved NMR spectra in inhomogeneous fields, combined composite z -rotation pulses with stroboscopic acquisition; (b) diagram describing spin evolution under a train of z -rotation pulses. (Source: Adapted from Ref. [65])

3.2.3. 2D high-resolution methods

If the refocusing in the indirect dimension is combined with a coherence transfer superimposed on the nutation echo and stroboscopic acquisition, 2D high-resolution spectra can be obtained. A COSY pulse sequence with refocusing B_1 gradient pulses in two dimensions is shown in Fig. 32a. The pulse sequence can be extended to heteronuclear spectroscopy. A HETCOR sequence with heteronuclear decoupling in the indirect dimension is shown in Fig. 32b. To achieve heteronuclear decoupling in the indirect dimension, a nonselective 180° pulse is applied to carbon spins at the center of the free evolution period t_1 . Direct dimension acquisition of ^{13}C signals is combined with an INEPT transfer to enhance sensitivity. The HETCOR pulse sequence was tested with a sample of ethanol in a tube of 10 mm diameter. A static B_0 field gradient of 0.4 mT/cm was applied, resulting in a total 70 ppm line-width. Fig. 32c shows the refocused HETCOR spectrum of ethanol.

Fig. 32

Fig. 32. Nutation echo based pulse sequences: (a) COSY; (b) HETCOR. (c) HETCOR spectrum of ethanol- d_1 . (Source: Adapted from Ref. [66])

3.2.4. Improved methods

Single-sided magnets pose significant challenges for the nutation echo techniques depicted above. One of the main reasons is that the static magnetic field gradient imposes a field variation over the sample that is many times larger than the RF field gradient. Large chemical shift offsets imply that only a small fraction of the sample satisfies the equation $G_1(\mathbf{r}) = k \cdot G_0(\mathbf{r})$, and the echo, with greatly reduced amplitude, is difficult to detect in practice. This problem can be alleviated by using a double passage of adiabatic pulses to modulate the phase of the transverse magnetization [69].

As shown in Fig. 33a, two full-passage adiabatic pulses are applied immediately one after the other. Both pulses have the same duration, while their peak amplitudes are different in order to produce a spatial modulation. The first adiabatic pulse induces a modulation of the transverse magnetization that mainly depends on the local RF field. The beginning of the second adiabatic pulse reverses this modulation. The accumulated phase during the first adiabatic pulse is not completely canceled during the second one when the amplitudes of two adiabatic pulses are different. The difference is roughly proportional to the amplitude of the local RF field, resulting in a spatial modulation. Because the frequency sweep during the second passage is identical to the first one, the offset dependence is canceled out. Fig. 33a shows a FID recorded in the presence of a 20 kHz/cm field gradient. The combined presence of matched RF and static field gradients leads to a sharp nutation echo taking place at 1 ms. The echo position is controlled by

the relative amplitudes of the adiabatic full passages. The combination of adiabatic double passage with stroboscopic acquisition provides high-resolution spectra in the presence of field gradients, as shown in Fig. 33c and d. A free evolution delay is inserted between the excitation pulse and adiabatic double passage, and the stroboscopic acquisition is restricted to the peaks of echoes (open circles). Because the approach depends on spatial matching between the inhomogeneity of the main magnetic field and the RF field, the gradual decay of amplitude in subsequent echoes mainly derives from the repeated accumulation of errors in refocusing.

Fig. 33

Fig. 33. Experiments on a mixture of acetone and benzene: (a) Nutation echo induced by the pulse sequence in which an adiabatic double passage is applied immediately after the 90° excitation pulse. (b) Fourier transform of (a). (c) Nutation echoes induced by a pulse sequence that has a free evolution period immediately after the excitation pulse and subsequent free evolution periods after each double passage of the pulse train. (d) Fourier transform of the FID constructed from the peaks of the echoes [open circles in (c)]. (Source: Adapted from Ref. [69])

A “shim pulse”, which uses adiabatic pulses applied synchronously with field gradients, can be used to correct artifacts originating from magnet and gradient imperfections. To make spins evolve in an apparently homogeneous field $B_{0,app}(\mathbf{r})$, one should design a pulse with the desired $\varphi(\mathbf{r})$. During a free-precession delay with a length t_{dw} , the apparent field is given by [72]

$$B_{0,app}(\mathbf{r}) = B_0(\mathbf{r}) - \frac{\varphi(\mathbf{r})}{\gamma t_{dw}}, \quad (34)$$

where $B_0(\mathbf{r})$ is the magnetic field produced by the magnet. Eq. (34) indicates that $B_{0,app}(\mathbf{r})$ can be controlled by time modulation of the RF field and auxiliary coil currents.

It is well known that an adiabatic pulse applied to transverse magnetization induces a phase shift depending on the frequency offset $\Delta\omega(\mathbf{r})$,

$$\phi(\mathbf{r}) = \int_0^{t_p} \sqrt{[\gamma B_1(t)]^2 + \{\omega_1(t) - [\Omega + \Delta\omega_0(\mathbf{r})]\}^2} dt, \quad (35)$$

where $B_1(t)$ and $\omega_1(t)$ represent the amplitude and frequency modulations of an RF pulse, respectively. A second adiabatic pulse reverses this phase shift and restores the magnetization to its initial state. However, a residual phase shift survives after an adiabatic double passage if the field inhomogeneity changes between the first and second pulses. Eq. (35) shows that $\phi(\mathbf{r})$ depends on $\Delta\omega(\mathbf{r})$, which also can be controlled by the coils producing the B_0 inhomogeneity and/or gradients. Fig. 34 shows that the combination of adiabatic pulses with a gradient along the z axis can remove the influence of z and z^2 inhomogeneous fields. The total gradient waveform is a superposition of a constant baseline for linear correction and a smoothly modulated waveform numerically optimized for quadratic correction.

Fig. 34

Fig. 34. NMR spectrum correction using shim pulses on a sample of 5 mg/mL thiamine (vitamin B_1) hydrochloride in $^2\text{H}_2\text{O}$. (a) Pulse sequence for recording 1D NMR spectra with shim pulses and stroboscopic acquisition. (b) ^1H NMR spectra obtained by the pulse sequence shown in (a):

inhomogeneous magnetic field without applied gradients (i); inhomogeneous magnetic field with linear correction (ii); inhomogeneous magnetic field with linear and quadratic correction (iii); homogeneous magnetic field without applied gradients (iv); spectrum obtained with a simple $\pi/2$ pulse (v). (Source: Adapted from Ref. [72], *D. Topgaard, R.W. Martin, D. Sakellariou, C.A. Meriles, A. Pines, "Shim pulses" for NMR spectroscopy and imaging, Proc. Natl. Acad. Sci. U. S. A., 101 (2004) 17576-17581. "Copyright (2004) National Academy of Sciences, U.S.A."*).

It can be concluded that if the spatial distribution of magnetic field is known, the effects of B_0 inhomogeneity may be suppressed by nutation echoes using matched B_0 and B_1 gradients. Nutation echoes can also be used to map the spatial distribution of B_1 gradients of an arbitrary RF coil geometry with the aid of a small probe sample. Diffusion coefficients and spin-lattice relaxation times can be measured from the same experimental data set of nutation echoes.

4. High-resolution NMR spectroscopy based on spatially-encoded techniques

4.1. Spatially-encoded NMR spectroscopic techniques

The prototype spatially-encoded NMR spectroscopic technique was proposed by Frydman and coworkers [105], and enables the acquisition of a 2D NMR spectrum within a single scan. The technique was dubbed ultrafast NMR (UFNMR) [124]. In UFNMR, the usual t_1 encoding in canonical 2D experiments is replaced by spatial encoding, which is decoded during the detection period by an EPSI scheme.

The original pulse sequence for UFNMR spectroscopy is shown in Fig. 35. The spatially-encoded block consists of a train of bipolar gradients and selective excitation pulses. The N_1

discrete selective excitation pulses synchronized with positive gradients trigger the evolution of spins located at different position z , and negative gradients refocus the phase shifts caused by positive gradients. At the end of the spatially encoded block, the resulting phase is [104]

$$\varphi_E(z) = \frac{\omega_1 t_1^{\max} z}{L} - \frac{\omega_1 t_1^{\max}}{2}, \quad (36)$$

where $t_1^{\max}$ is the interval between the first and last excitation pulses, ω is the chemical shift of spins in the indirect dimension, and L is the sample length. The phase dependence on position z prevents the signal from being recorded by the canonical acquisition method. However, a train of bipolar gradient pulses ($\pm G_D$), as shown in Fig. 35a, will demodulate the linear term $\omega_1 t_1^{\max} z / L$ and lead to the formation of echoes (Fig. 35b) during the acquisition period. The time at which an echo appears is proportional to ω . As shown in Fig. 35c, the positive (A and C) and negative (B and D) gradient data sets are separated. The separate positive/negative gradient data sets are Fourier transformed along the t_2 dimension, so the chemical shift ω can be measured along the direct dimension of the detected spins (Fig. 35d). The constant term in Eq. (36) only affects the phases of the echoes.

Fig. 35

Fig. 35. (a) The original-spatially encoded pulse sequence affords a 2D NMR spectrum within a single scan. (b) The signals viewed in 2D $k - t_2$ space. (c) The positive (A and C) and negative (B and D) gradient data sets are separated. (d) Each data set shown in (c) is Fourier transformed along the t_2 dimension, with the dwell time being $2T_D$. (Source: Adapted from Ref. [105], L .

Frydman, T. Scherf, A. Lupulescu, The acquisition of multidimensional NMR spectra within a single scan, Proc. Natl. Acad. Sci. U. S. A., 99 (2002) 15858-15862. "Copyright (2002) National Academy of Sciences, U.S.A.").

For the original pulse sequence which uses selective pulses to excite different groups of spins, it is impossible to excite all spins fully unless one uses pulses with a perfect rectangular square excitation profile in the frequency domain. This results in significant signal loss, governed by the excitation pulse profiles. With the aid of linearly swept (chirped) RF pulses, one can easily modulate the signal phase of the whole sample from one position to another sequentially in time. Three continuous spatially-encoded schemes that have been proposed are real-time amplitude modulation ($\pi/2 - \pi/2$) [125], real-time phase modulation ($\pi/2 - \pi$) [126], and constant-time phase modulation ($\pi - \pi$) [104]. Because the $\pi/2$ pulse must be carefully calibrated to obtain an accurate $\pi/2$ excitation with a given adiabaticity parameter (e.g. 0.068), the pulse sequence with constant-time phase modulation is most convenient to perform [127, 128]. When the discrete spatially-encoded block is replaced by the constant-time phase modulation block, the resulting encoded phase becomes

$$\varphi_E = \frac{4\gamma\omega_1 G_E z}{R}. \quad (37)$$

The constant term $\omega_{l_1}^{\max} / 2$ in Eq. (36) is eliminated in Eq. (37).

4.2. Phase correction in spatially-encoded spectroscopic techniques

As shown in Eq. (36), the phase $\varphi_E(z)$ accumulated by spins experiencing the discrete spatial encoding process is position-dependent. When the inhomogeneous field and the relative phases of RF pulses are considered, the phase becomes

$$\varphi_E(z) = \frac{[\omega_1 + \Delta\omega_0(z)]t_1^{\max}}{L}z - \frac{\omega_1 t_1^{\max}}{2} + \phi_{RF}(z). \quad (38)$$

For simplicity, the magnetic field gradient $\Delta\omega(z)$ is assumed to be along the z axis, as used in the EPSI signal acquisition. If the phase of RF pulse becomes $\phi_{RF}(z) = -\Delta\omega_0(z)t_1^{\max}z/L$, the phase shown in Eq. (38) is free from the effect of the magnetic field gradient, and the echo along the indirect dimension reflects the chemical shift of spins. In Fig. 36b, results obtained in fields with two kinds of inhomogeneity along the z axis are shown. The spatially-resolved chemical shift information (Fig. 36c) demonstrates that both linear and nonlinear z inhomogeneities have been compensated by discrete excitation pulses. Even in inhomogeneous fields with 10-100 ppm line-width, spectral resolution on the order of 0.1 ppm can be reached. To compensate for field inhomogeneities along arbitrary axes, one needs to combine phase correction with multiple-axis spatial encoding.

Fig. 36

Fig. 36. A strategy capable of yielding high-resolution spectra by utilizing phase correction in spatially-encoded spectroscopic techniques. (a) Phase manipulation of RF pulses in the original spatially encoded pulse sequence. (b) Fields with z and z^2 inhomogeneities respectively measured

by 2D EPSI NMR, illustrated for n-butylchloride/ CDCl_3 . (c) High-resolution spectra obtained using the scheme shown in (a), from the inhomogeneous fields shown in (b). (Source: Adapted from Ref. [74]).

4.3. Tracking differences in chemical shifts by spatially-encoded spectroscopic techniques

Spin echo correlated spectroscopy (SECSY) has long been utilized to obtain 1D high-resolution chemical shift differences for coupled spins. Since the scalar coupled protons detected in this experiment belong to the same molecule, the field inhomogeneity they experience is identical and can thus be cancelled out, leaving just the chemical shift difference as the evolution frequency in F_1 . Conventional SECSY has two main drawbacks: first, the prolonged acquisition time needed due to the requirement for numerous t_1 increments; and second, the direct dimension remains sensitive to field inhomogeneity. As an alternative to circumvent these two drawbacks, Pelupessy *et al.* [129] proposed an ultrafast SECSY experiment with a J -resolved detection scheme. The acquisition time is shortened by orders of magnitude by the utilization of spatial encoding, and the direct dimension offers high-resolution scalar coupling information.

For tracking chemical shift differences between J -coupled spins, Pelupessy *et al.* put identical spatial encoded modules before and after the second $\pi/2$ pulse, as shown in Fig. 37a [129]. A bipolar gradient pair with linearly swept adiabatic pulses repeated n_E times is used to attenuate translational diffusion effects [130]. For an $I+S$ AX spin system, the coherence S_k^- evolves before the second $\pi/2$ pulse, and $S_{kz}S_l^+$ evolves after the second $\pi/2$ pulse. At the time labeled 1 in Fig. 37a, the phase is given by

$$\varphi_{E1}(\mathbf{r}) = -\frac{4n_E[\omega_{S_k} + \Delta\omega_0(\mathbf{r})]^2}{R} - \frac{[\omega_{S_k} + \Delta\omega_0(\mathbf{r}) + \gamma\mathbf{G}_E \cdot \mathbf{r}]^2 + O_i^2}{R}. \quad (39)$$

The coherence can then be transferred by a simple $\pi/2$ pulse from S_k^- to S_l^+ through the spin-spin coupling, relying on the scalar coupling constant J_{kl} . At the time labeled 2 in Fig. 37a, the total accumulated phase is given by

$$\begin{aligned} \varphi_{E2}(\mathbf{r}) &= \frac{4N_E[\omega_{S_k} + \Delta\omega_0(\mathbf{r})]^2}{R} + \frac{[\omega_{S_k} + \Delta\omega_0(\mathbf{r}) + \gamma\mathbf{G}_E \cdot \mathbf{r}]^2 + O_i^2}{R} \\ &\quad - \frac{4N_E[\omega_{S_l} + \Delta\omega_0(\mathbf{r})]^2}{R} - \frac{[\omega_{S_l} + \Delta\omega_0(\mathbf{r}) + \gamma\mathbf{G}_E \cdot \mathbf{r}]^2 + O_i^2}{R} \\ &= \frac{2(\omega_{S_l} - \omega_{S_k})[(2N_E + 1)\gamma\mathbf{G}_E \cdot \mathbf{r} + \Delta\omega_0(\mathbf{r})]}{R} + \frac{\omega_{S_l}^2 - \omega_{S_k}^2}{R}. \end{aligned} \quad (40)$$

The chemical shift difference $(\omega_{S_l} - \omega_{S_k})$ between spins S_k and S_l is encoded not only by the gradients $\mathbf{G}_E$ but also by the magnetic field gradient $\Delta\omega(\mathbf{r})$. After decoding by the gradient $\mathbf{G}_D$, with an amplitude $(2N_E + 1)$ times as large as that of the encoding gradient, the phase is

$$\varphi_{E2}(z) = \frac{2(\omega_{S_l} - \omega_{S_k})[(2N_E + 1)\gamma\mathbf{G}_E \cdot \mathbf{r} + \Delta\omega_0(\mathbf{r})]}{R} + \frac{\omega_{S_l}^2 - \omega_{S_k}^2}{R} + t_D[(2N_E + 1)\gamma\mathbf{G}_E \cdot \mathbf{r} + \Delta\omega_0(\mathbf{r}) + \omega_{S_l}]. \quad (41)$$

At $t_D = 2(\omega_{S_k} - \omega_{S_l}) / R$, an echo forms because the phase is independent of the encoding and decoding gradients and of the magnetic field gradient $\Delta\omega(\mathbf{r})$. A block consisting of a decoding gradient followed by a π pulse is repeated n_D times to collect a train of echoes. An FT

of this echo train provides a 2D spectrum. The J -modulation occurs during the detection period [131], adding a second dimension that reveals multiplets due to scalar couplings. The phase has the same form as that in Eq. (20). The frequency offset $2(\omega_{S_k} - \omega_{S_l})$ shown in the first term is the chemical shift difference between two J -coupled spins ω_{S_k} and ω_{S_l} , so the pulse sequence can only be used for measurements on J -coupled systems.

The robustness of the pulse sequence shown in Fig. 37a with respect to field inhomogeneities was verified by experiments on a sample of 5% propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$) in CDCl_3 . Fig. 37d and e were obtained in homogeneous and inhomogeneous fields, respectively, as illustrated by the conventional one-pulse spectra shown in Fig. 37b (homogeneous field with 2 Hz line-width) and Fig. 37c (inhomogeneous field with 2400 Hz line-width at 600 MHz). The resolution of the spectrum shown in Fig. 37d is almost the same as that in Fig. 37e. The resonances labeled S_k and S_l (e.g. a and b) in Fig. 37d come from the SQC evolution of S_k and S_l during the spatially encoded blocks before and after the second $\pi/2$ pulse respectively; peak splitting patterns along the direct dimension demonstrate that the two spins are scalar-coupled. Because each coherence transfer from spin S_k to S_l is accompanied by a transfer in the reverse direction, the spectrum is symmetric in the spatially encoded dimension with respect to $\varphi = 0$. The central peak at $\varphi = 0$ arises from magnetization that is refocused but not transferred from one spin to another. Comparing the cross section shown in Fig. 37e with the 1D spectrum shown in Fig. 37b, we can observe that the SNR of the spatially-encoded spectrum is reduced due to the intrinsically low sensitivity of ultrafast techniques.

Fig. 37

Fig. 37. Experiments on a sample of propanol: (a) Pulse sequence combining spatial encoding with coherence transfer; (b) 1D spectrum obtained by a single-pulse excitation in a homogeneous field; (c) as (b) but in an inhomogeneous field with circa 4 ppm line-width; (d) 2D spectrum obtained by the pulse sequence shown in (a) in the homogeneous field; (e) as (d) but in the same inhomogeneous field as that for (c). Both spectra were recorded with $N_E = 1$ and $N_D = 64$, gradients applied simultaneously along the z and y directions with strengths $G_E = 0.8$ G/cm and $G_D = 2.4$ G/cm. The adiabatic pulses had a wideband uniform rate and smooth truncation (WURST) profile and a sweep width of 14 kHz in 6 ms. (Source: Adapted from Ref. [129])

The extension of the above principle to multidimensional spectroscopy, such as SECSY and SETOCSY, is straightforward, and sheared spectra obtained by ultrafast SECSY and SETOCSY sequences can provide the same information as COSY and TOCSY. As shown in Fig. 37d and e, chemical shifts are symmetric in the spatially encoded dimension with respect to $\varphi = 0$, so Zhang *et al.* proposed ultrafast SECSY and SETOCSY methods using half of the spectral width, which improves the spectral resolution along the spatially encoded dimension [76]. Fig. 38 demonstrates the ultrafast SECSY pulse sequence and the data processing procedure used to reconstruct a COSY spectrum.

Fig. 38

Fig. 38. (a) Ultrafast 2D SECSY pulse sequence, (b) ultrafast 2D half-SECSY spectrum, (c) expanded zero-filled 2D SECSY spectrum from (b), (d) spectrum from (c) after a shearing along the F_1 axis by $\Omega'_1 = \Omega_1 + \Omega_2$, (e) spectrum from (d) after symmetrization about the diagonal, (f) reconstructed 2D COSY spectrum in which the signal intensity at every position is equal to the

higher of that in (d) and in (e), (g) conventional COSY spectrum. (Source: Adapted from Ref. [76])

According to a previous report [76], the echo time t_D and the spectral resolution in the F_1 dimension $\Delta\nu_1$ can be written as

$$t_D = \frac{4G_E(\omega_{S_l} - \omega_{S_k})}{R(G_D + \Delta B_0)} - \frac{G_P t_P + \Delta B_0 t_P}{G_D + \Delta B_0} + 2N\tau_D - 2N\tau_D \frac{\Delta B_0}{G_D + \Delta B_0}, \quad (42)$$

$$\Delta\nu_1 = \frac{0.605\pi\Delta\nu_{ad}}{\gamma G_E L \tau_{ad}},$$

where N is an integer less than N_D . The second term is dependent on the field inhomogeneity ΔB_0 and pre-acquisition purge gradient G_P , which can be adjusted to position echoes in the middle of each acquisition period τ_D . Eq. (42) shows that the spectral resolution $\Delta\nu_1$ in the F_1 dimension is independent of the inhomogeneous field, but that the echo time t_D suffers from the effect of field inhomogeneity (the fourth term), resulting in line broadening along the F_2 dimension. Fortunately, a shearing of the acquired data prior to FT can remove the effect of the fourth term if the inhomogeneous field ΔB_0 varies linearly along the z axis.

Fig.38b demonstrates that the spectral peaks obtained by the pulse sequence shown in Fig. 38a are located at $(\omega_{S_l} - \omega_{S_k}, \omega_{S_k} \pm \pi J_{kl})$. When $\omega_{S_l} = \omega_{S_k}$, the peak is located at $(0, \omega_{S_k})$. Fig. 38c is the zero-filled 2D SECSY spectrum from Fig. 38b. After data processing, peaks will appear at $(\omega_{S_l} \pm \pi J_{kl}, \omega_{S_k} \pm \pi J_{kl})$, $(\omega_{S_k} \pm \pi J_{kl}, \omega_{S_k} \pm \pi J_{kl})$, and $(\omega_{S_l} \pm \pi J_{kl}, \omega_{S_l} \pm \pi J_{kl})$, as shown in Fig. 38d. Cross-peaks indicate the existence of J -coupling between spins S_k and S_l .

Because a COSY spectrum is symmetric with respect to the diagonal, the spectrum shown in Fig. 38d is symmetrized with respect to the diagonal to obtain the spectrum shown in Fig. 38e; the stronger signal of Fig. 38d and e at each position is selected to build the COSY spectrum shown in Fig. 38f. Compared to the conventional COSY spectrum shown in Fig. 38g, the number of peaks in the reconstructed 2D COSY spectrum is reduced by half due to the decoupling along the F_1 dimension. Note that this ultrafast COSY method is limited to samples with linear field variation along the z axis.

5. Other high-resolution NMR spectroscopic techniques

Besides the high-resolution NMR methodologies mentioned above, there remain many other pulse sequences and data processing methods aimed at recovering spectral information from experiments in inhomogeneous fields. Here, we give a brief description.

A coherence transfer echo is a kind of echo occurring when coherence is transferred from one spin to another between defocusing and refocusing. The two transitions involved may belong to different nuclei and may have different coherence orders. As a special case, total spin coherence transfer echo spectroscopy (TSCTES) uses the property that the transition between two states that has the maximum total coherence order is insensitive to chemical shift differences and scalar couplings, and can be used to generate coherence transfer echoes that refocus the effects of field inhomogeneity [132, 133]. However, this technique is only suitable for J -coupled spin systems, and is useless for uncoupled spins. Peaks of isolated spins will be lost in the resulting spectra. For heteronuclear coherence transfer echoes, the evolution of one nuclear species under an inhomogeneous field can be refocused when the coherence is transferred to a second species [134]. The second nuclear species behaves as a probe for local environment. This

method has been applied successfully to amorphous solids, yielding high-resolution information even in the presence of isotropic chemical shift distribution [135].

It is well known that the evolution of intramolecular ZQCs is insensitive to field inhomogeneities [136]. Through J coupling effects, intramolecular ZQCs can be excited and selected using appropriate sequences of RF and gradient pulses. This allows information on chemical shift differences and J couplings to be retrieved from inhomogeneous fields. Several high-resolution spectroscopic methods based on intramolecular ZQCs have been proposed for obtaining the desired information. Rothman and coworkers have proposed a 2D indirect ZQC detection method to recover high-resolution NMR information from *in vivo* rat brains at 11.74T [137]. However, intramolecular ZQC methods cannot provide information on isolated spins. Moreover, unlike in normal spectral analyses, the chemical shifts of J -coupled spins are generally not directly available. Note that intramolecular ZQC techniques are different from iZQC techniques, although both techniques achieve improved spectral resolution along the indirect dimension. The detection of intramolecular ZQCs is strictly limited to J -coupled spin systems, whereas the detection of iZQCs relies on long range dipolar interactions between spins in different molecules, and is thus not limited to J -coupled spin systems.

In conventional NMR spectroscopy, decoherence of the SQC signal due to internal interactions (homogeneous transverse relaxation) and to unknown variations of the magnetic field (inhomogeneous broadening) limits the spectral resolution [138]. The faster transverse decay results in broader spectral lines. Long-lived coherences (LLC's) based on coherent superpositions between singlet and triplet states can lead to significant resolution enhancement in NMR spectroscopy [139-141]. Transitions between singlet and triplet states, which belong to different irreducible representations of the permutation group, are normally symmetry-forbidden

and therefore have vanishing probabilities. For a simple system with two J -coupled spins $I = 1/2$ and $S = 1/2$ (e.g., two nonequivalent protons), the LLC's can be defined as a linear combination $Q_{\text{LLC}} = I_x S_x$ and $2I_y S_z - 2I_z S_y$ in the product basis. It is obvious that LLC's belong to the class of ZQCs, and thus are insensitive to field inhomogeneities. More importantly, LLC's have very long lifetimes T_{LLC} , much longer than the transverse relaxation time T_2 of conventional SQCs, and hence can yield extremely narrow linewidths. Compared to the homogeneous linewidths from T_2 relaxation, the linewidths of LLC's can be further improved by a factor $k = T_{\text{LLC}}/T_2$ where $3 \leq k \leq 9$ [141], and ultra-high resolution within 1 Hz can be obtained. Hence, LLC's provides a means for homogeneous linewidths that are significantly narrower than in conventional spectroscopy. High-resolution NMR methods based on LLC's may be useful for structural and dynamic NMR studies of biological macromolecules with very short relaxation times.

A high-resolution NMR method exploiting the intermolecular NOE was established by Balbach and coworkers [142]. An approximation $\Delta B(\mathbf{r}) = \Delta B(\mathbf{r}')$ can be made for solvent spins at position $\mathbf{r}$ and solute spins at position $\mathbf{r}'$ that show short-range dipolar interactions. Therefore, the cross-peaks in NOESY spectra measured in inhomogeneous fields display streaks inclined at 45° and a 1D high-resolution spectrum can be obtained by taking cross-sections or projections after shearing. However, the intensities of cross-peaks in an NOESY spectrum are dependent on cross relaxation rates, which depend on spatial proximity and local dynamics. Intermolecular cross-relaxation rates between solute and solvent are low and variable, so the sensitivity and quantification offered by such methods are both poor.

Reference deconvolution is a post-processing method using the shape of a single resonance line to measure the actual frequency distribution. This frequency is produced by a small-scale

local B_0 inhomogeneity and then deconvolves its distribution from the whole spectrum [143, 144]. The drawbacks of reference deconvolution are that a significant improvement in spectral line-width imposes a severe penalty in SNR, and that the line-shape improvement obtainable is limited by the nature of the inhomogeneously broadened line-shape.

The chemical shift imaging (CSI) method is based on the simple fact that the variation of magnetic field within a voxel is much smaller than that across the whole sample volume. When voxel spectra are realigned, a spectrum with much narrower lines compared to the total volume 1D spectrum may be obtained [145]. The limitation of the CSI method is that long acquisition times are usually required since many phase-encoding steps are necessary to achieve satisfactory spatial resolution. In addition, the sensitivity loss, which depends on the spatial resolution, restricts applications of CSI.

Imaged deconvolution [146] offers a compromise between pure reference deconvolution and pure spectroscopic imaging. Reference deconvolution is not practical in the case of severe line broadening, whereas spectroscopic imaging methods like CSI place impractical demands on both hardware and experimental time. The imaged deconvolution method reaches the optimal compromise between the two methods to render the acquisition of high-resolution spectra in highly inhomogeneous fields possible.

Using Fourier decomposition it is possible to design sequences of RF pulses and time-varying linear gradients in three dimensions [147] that can pre-compensate field inhomogeneity-induced phase differences between spins at different spatial locations [129], and hence allow high resolution spectral information to be obtained. The resulting pulse sequences may be long in terms of time, data can only be acquired one point at a time, and some signals may be weak since relaxation effects are neglected in the treatment.

6. Summary

High-resolution NMR spectroscopy has been widely applied in many fields as a non-invasive tool. Generally, magnetic fields for high-resolution NMR should be homogeneous, with spatial variations below 10^{-8} . To circumvent this rigorous homogeneous field requirement, researchers have proposed a variety of approaches based on different mechanisms. This paper reviews various spectroscopic techniques for obtaining high-resolution NMR information. First, we briefly introduce the necessity for and significance of high-resolution NMR techniques; then, three main types of techniques based on iMQC and similar signals are systematically described to exemplify high-resolution NMR methodologies. Theoretical mechanisms and experimental verifications are presented to demonstrate the robustness with respect to the magnetic field inhomogeneity of the methodologies discussed. Finally, other experimental high-resolution NMR approaches and data processing methods are briefly described. Each method has characteristic advantages and disadvantages in practical applications, and no one method is applicable to all practical circumstances. The different methods provide complementary strategies for high-resolution NMR spectra in inhomogeneous fields.

Methods based on iMQCs can be applied to solutions in inhomogeneous fields and to biological samples with intrinsic macroscopic susceptibility effects and in *in vivo* localized MRS. Both 1D and 2D spectra can be obtained by iMQC methods. The main advantage of iMQC methods lies in their exploitation of identical interactions for all the different spin groups in a molecule, thus simultaneously retaining all the desired spectral information, including chemical shifts, J couplings, multiplet patterns. The prerequisite for implementation of iMQC based techniques is the availability of at least one concentrated chemical component to generate a DDF. Fortunately, solution-state chemical samples or biological samples usually contain a

concentrated solvent that can provide an intrinsic DDF. Some tentative applications of iMQC methods have been explored and presented. More developments focusing on improving sensitivity may open up perspectives for broader applications of iMQC methods.

Methods based on nutation echoes provide an efficient way to recover chemical shift information in the presence of inhomogeneous fields in *ex situ* situations. In contrast with iMQC methodology which is more suitable for slightly and moderately inhomogeneous fields, nutation echo methodology is applicable even in extremely inhomogeneous fields. This effectiveness facilitates high-resolution applications on single-sided portable magnetic resonance systems in high-resolution *ex situ* NMR experiments. However, this methodology can only provide information on chemical shifts; J coupling information cannot be obtained due to the lack of spectral resolution. Furthermore, much work remains to be done on magnet and the RF coil design to allow this methodology to be applied to actual *ex situ* situations. Large gradients or imperfect non-linear field correlations are issues that should be taken into account in the initial stages of designing high-resolution open NMR systems. Applications in fields such as biomedical diagnostics and food analysis would undoubtedly benefit from further developments in the technology.

Enhanced by the spatially encoded technique, a series of single- and multi-scan compensation approaches have been proposed for measuring high-resolution NMR spectra in inhomogeneous fields. Spatially-encoded techniques have obvious speed advantages, but both sensitivity and resolution are sacrificed. Therefore they are not suitable for low-sensitivity and low-field applications. Multi-scan alternatives combine spatially-encoded compensation with traditional 2D NMR approaches to enhance the poor sensitivity and scale the spectral information. It is particularly useful in scenarios characterized by relatively high heterogeneities,

limited shimming capabilities, and relatively low magnetic fields where chemical shift displacements are at a premium. The main challenge of spatially-encoded methods is the extension of field inhomogeneity compensation from one spatial dimension to three.

Overall, existing high-resolution NMR spectroscopic methodologies are systematically discussed and the underlying mechanisms of different approaches are presented with experimental verifications. Improvements for all these methodologies are being actively explored.

Acknowledgements

This work was supported by the NNSF of China under Grants 11174239, 11275161, 11375147, 11205129, 11204256, and U1232212, and the Prior Research Field Fund for the Doctoral Program of Higher Education of China under Grant 20120121130003.

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Fig.1

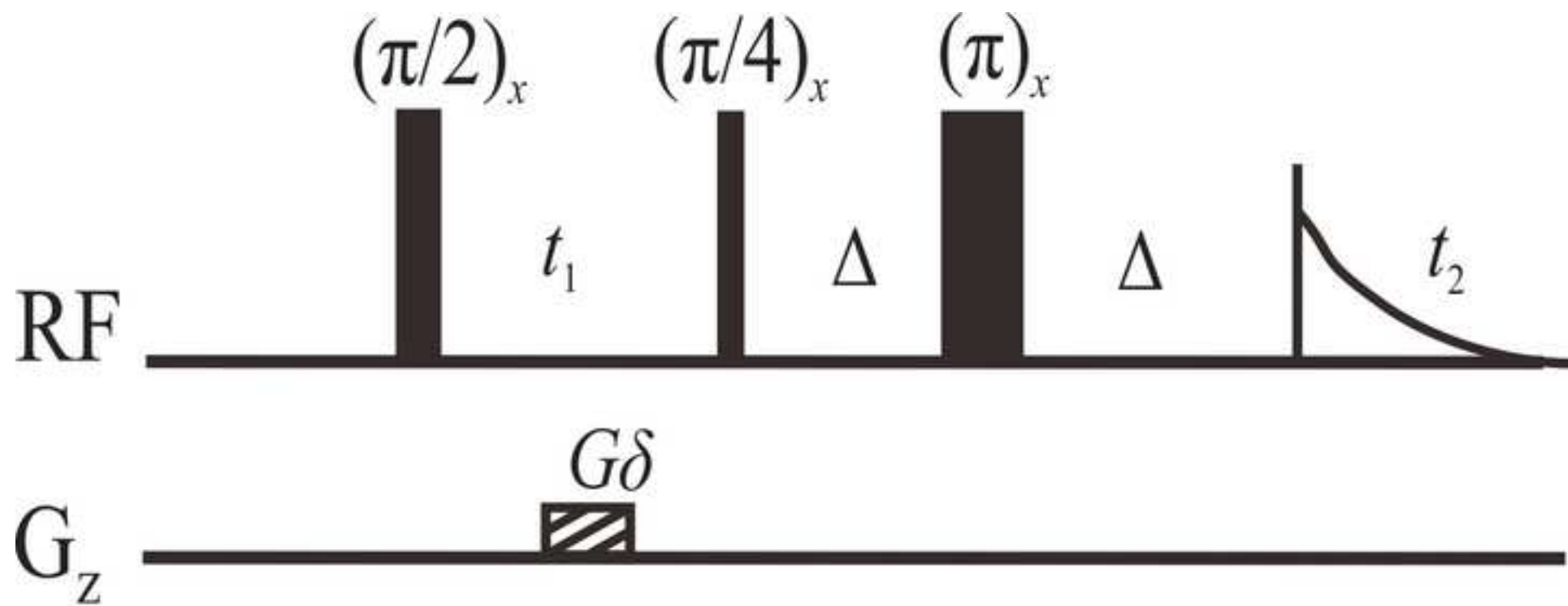


Fig.2

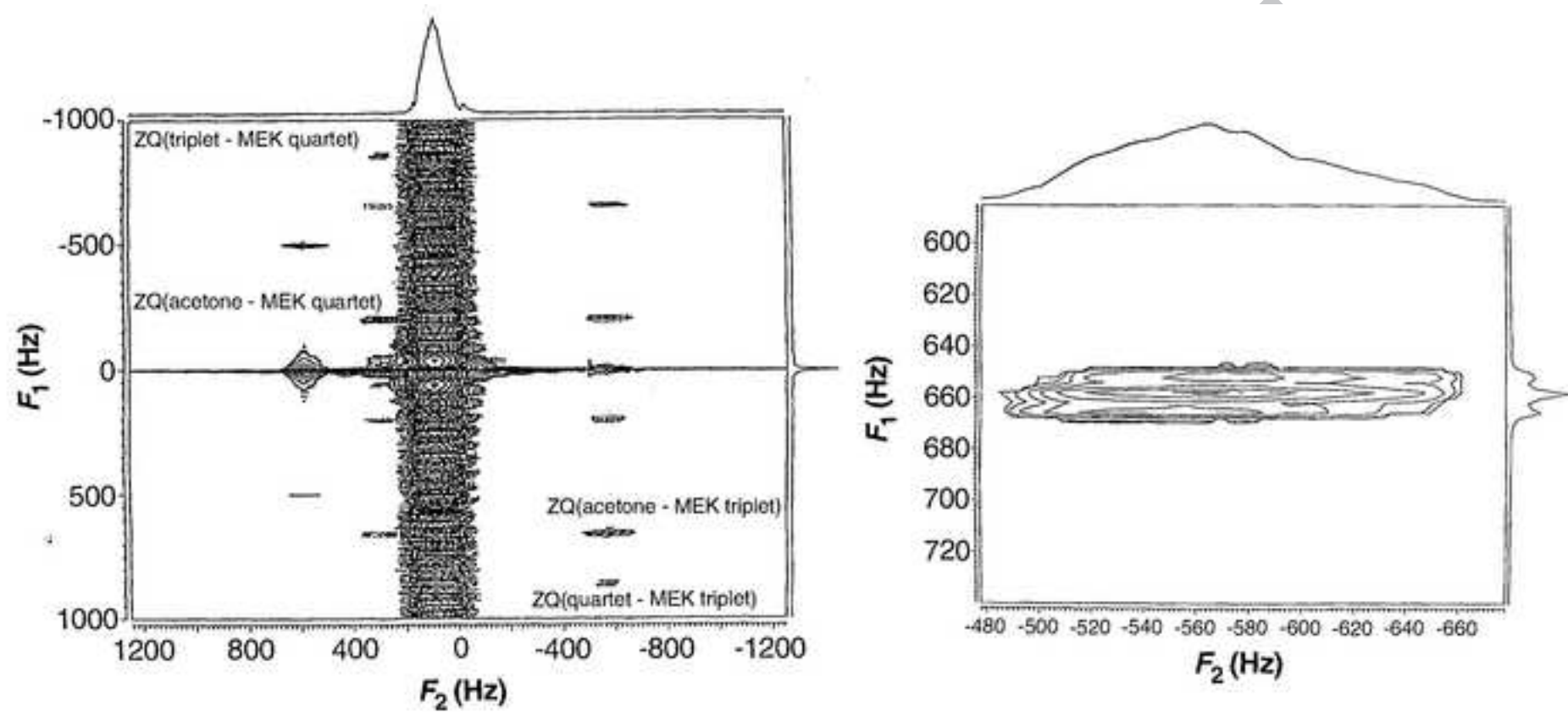
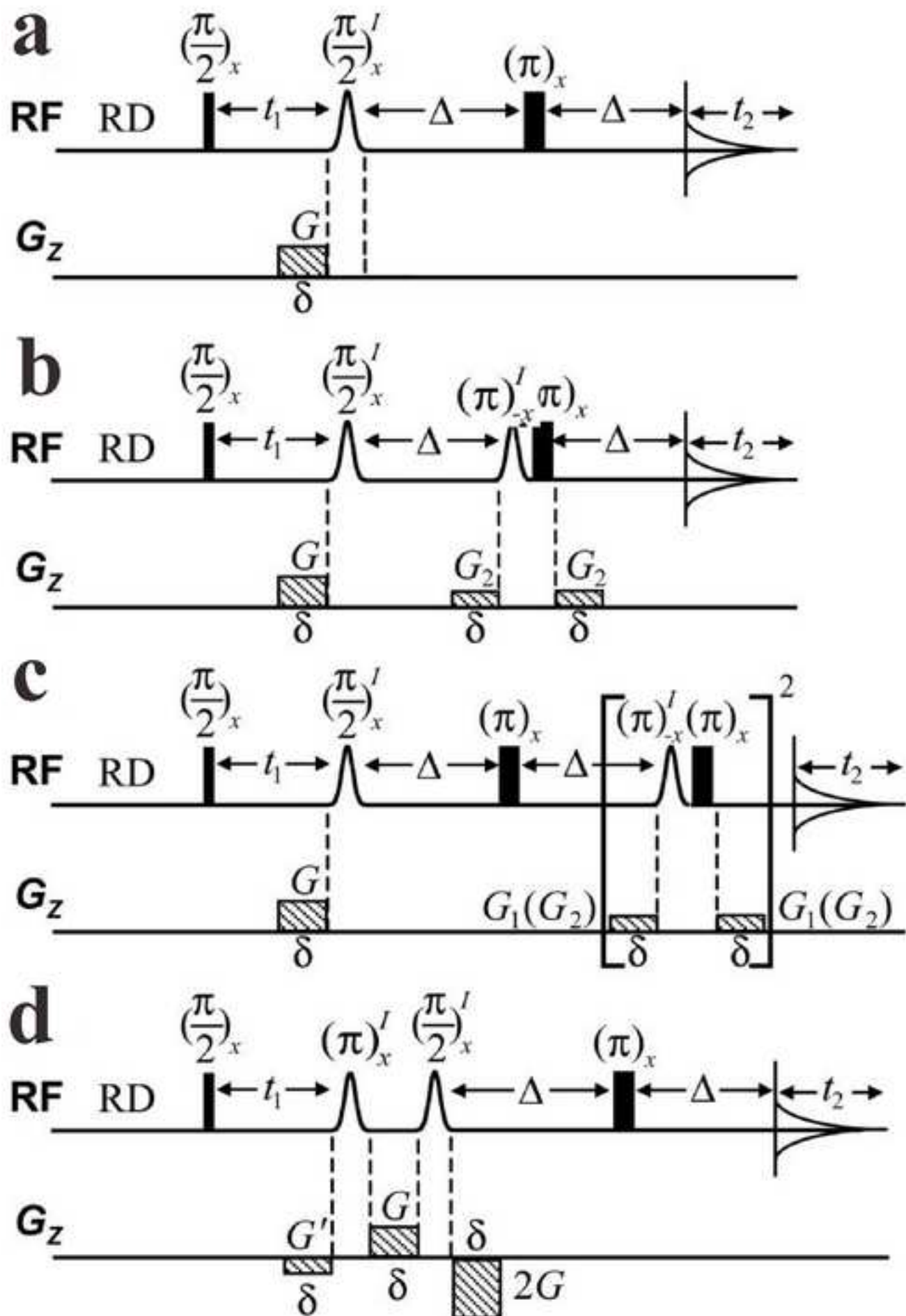


Fig.3



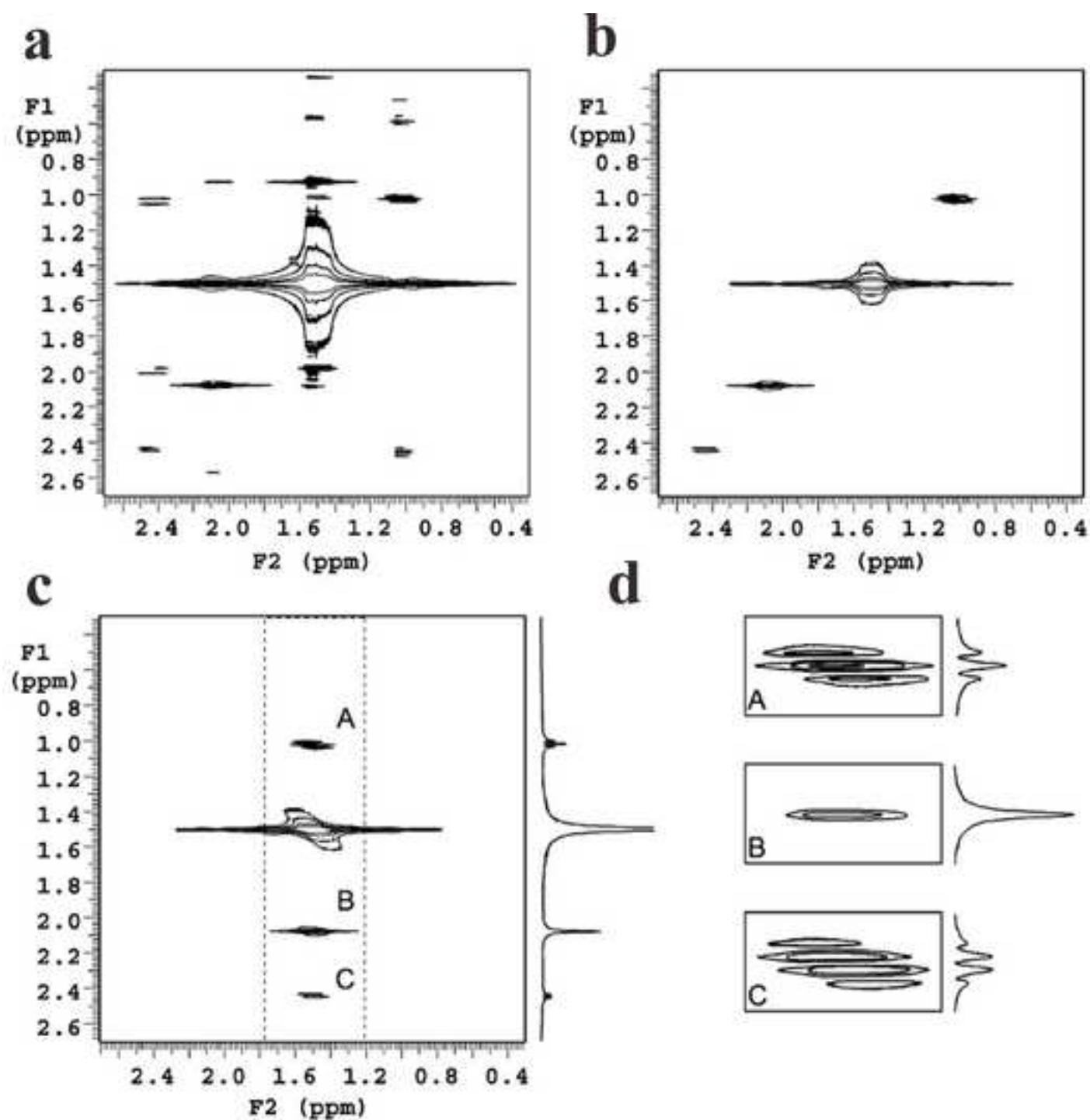


Fig.5

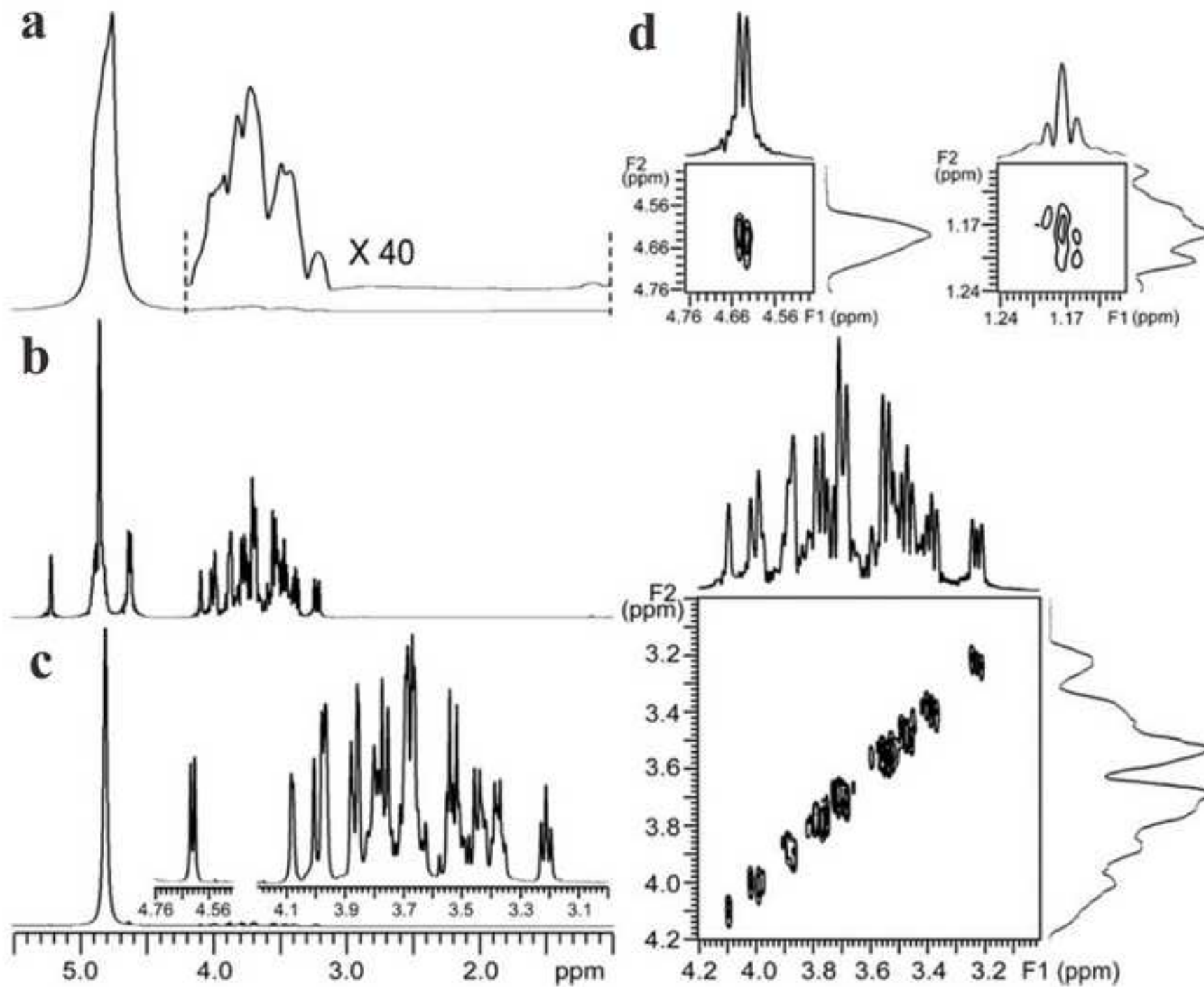


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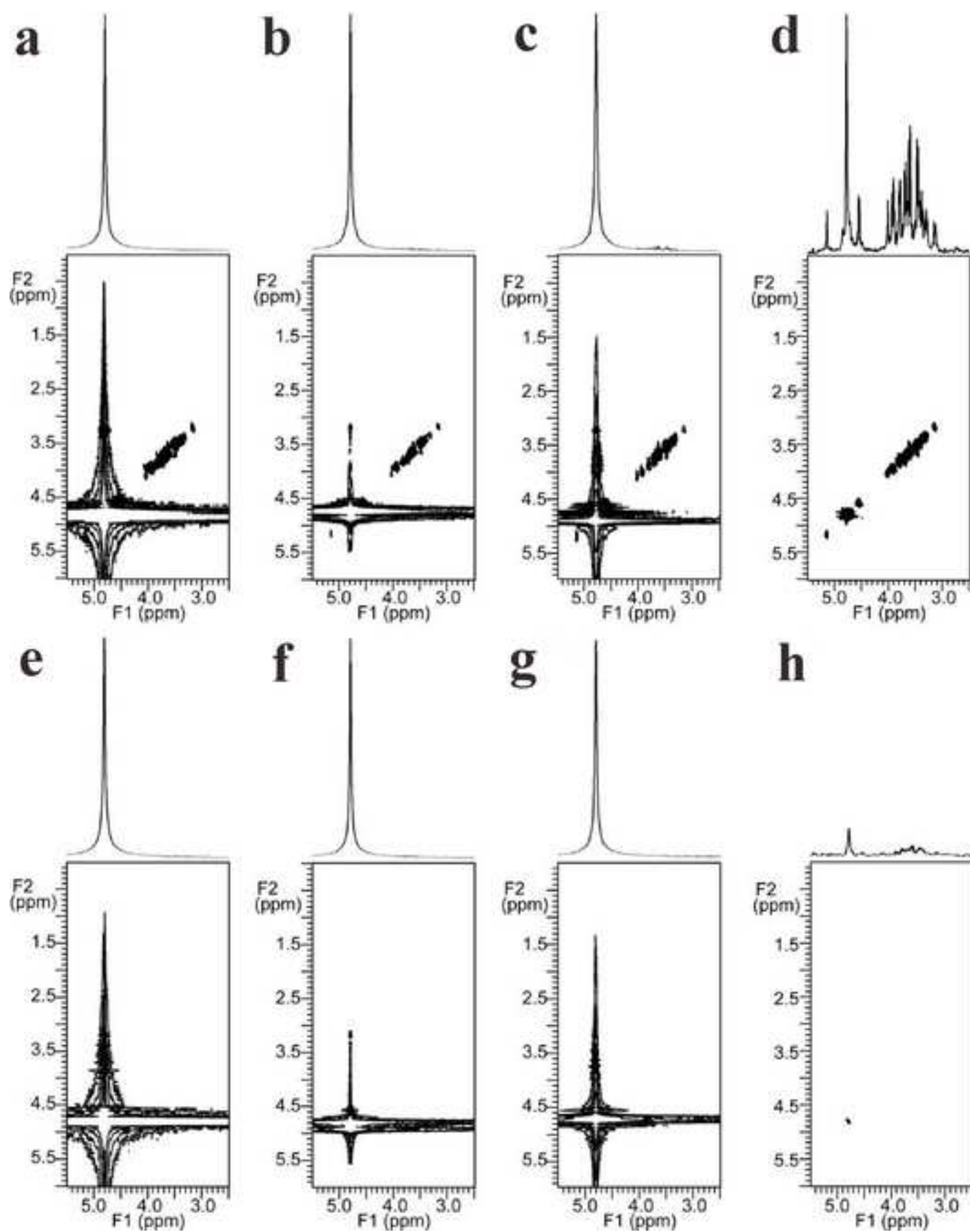


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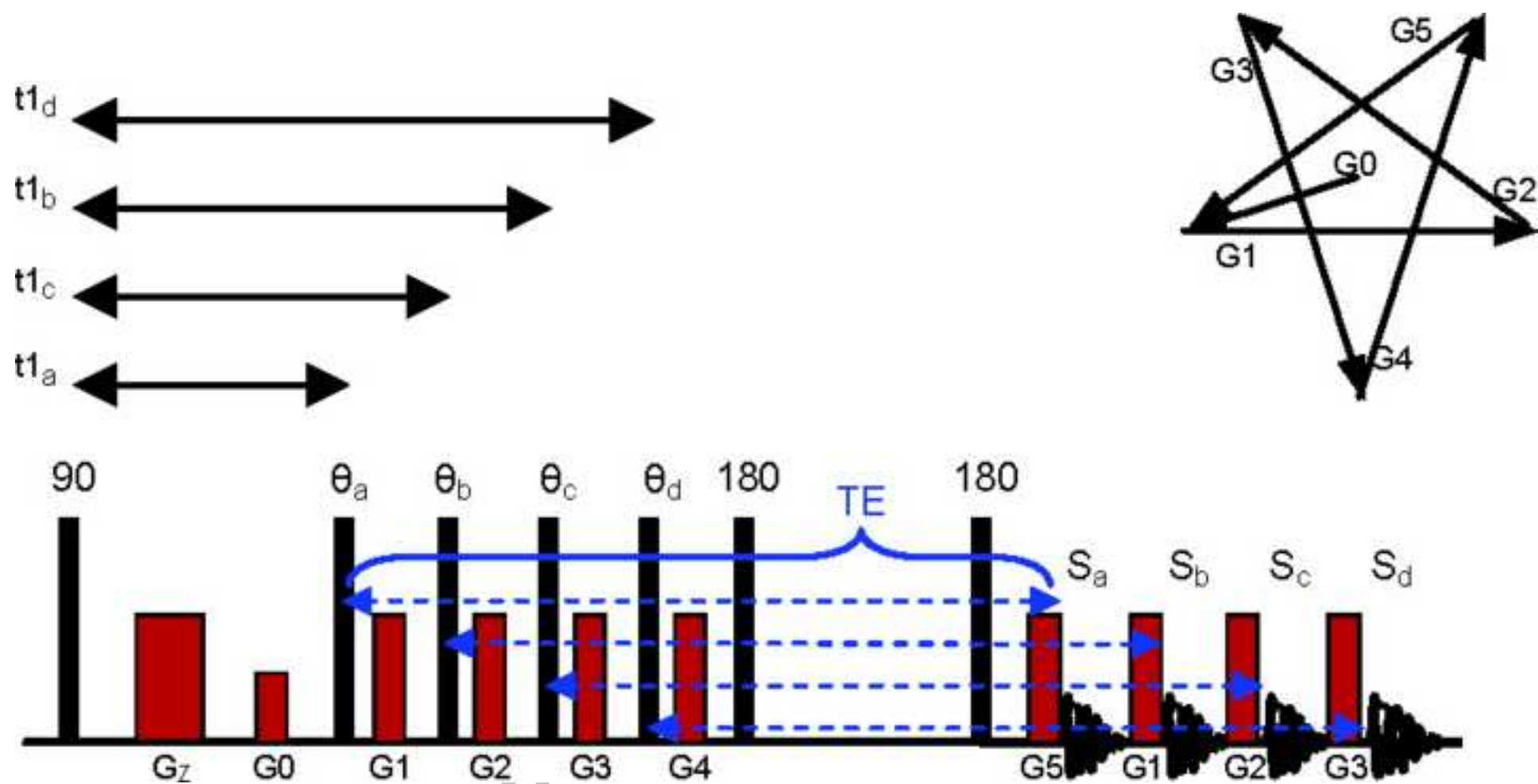


Fig.8

ACCEPTED MANUSCRIPT

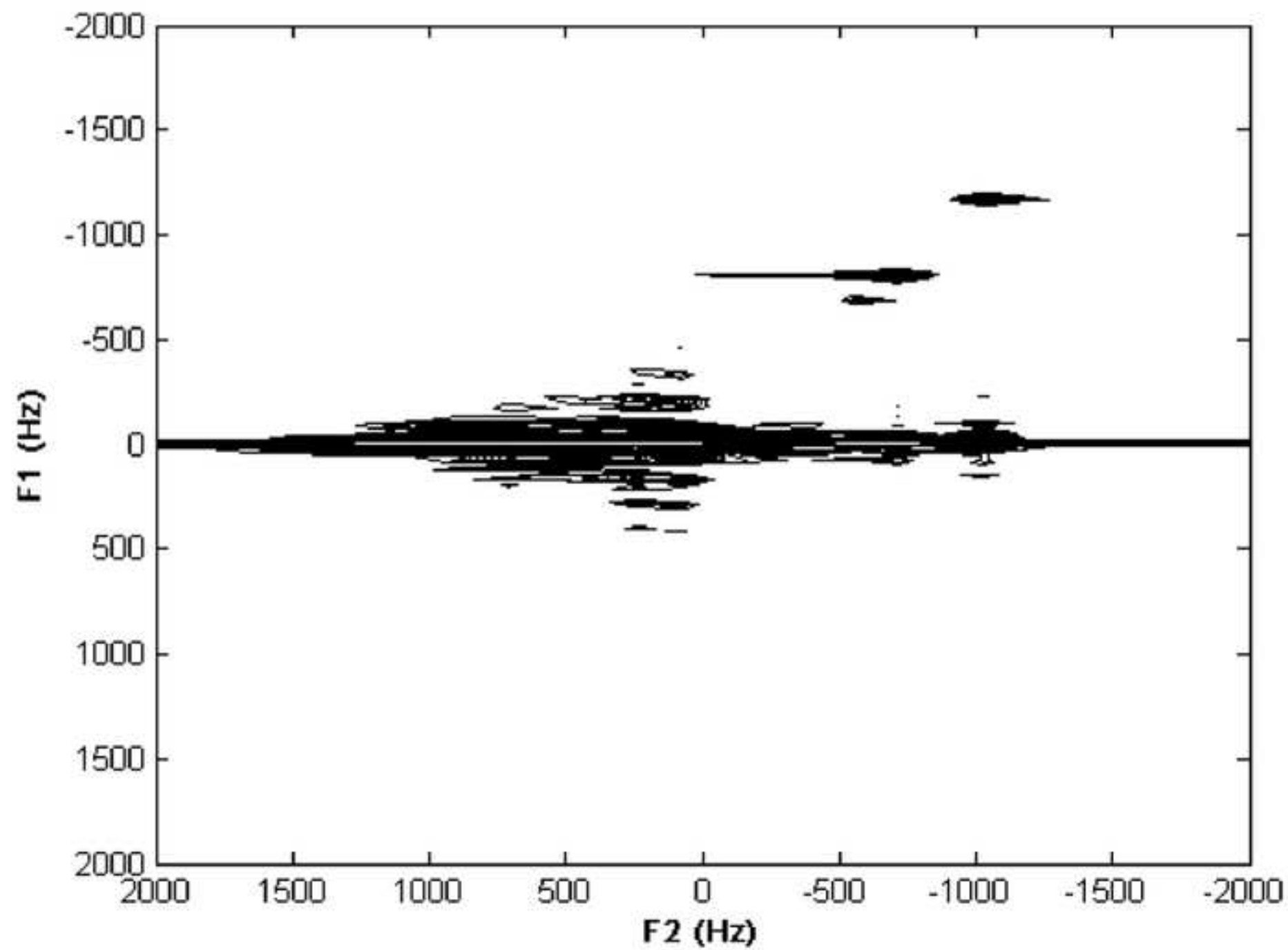


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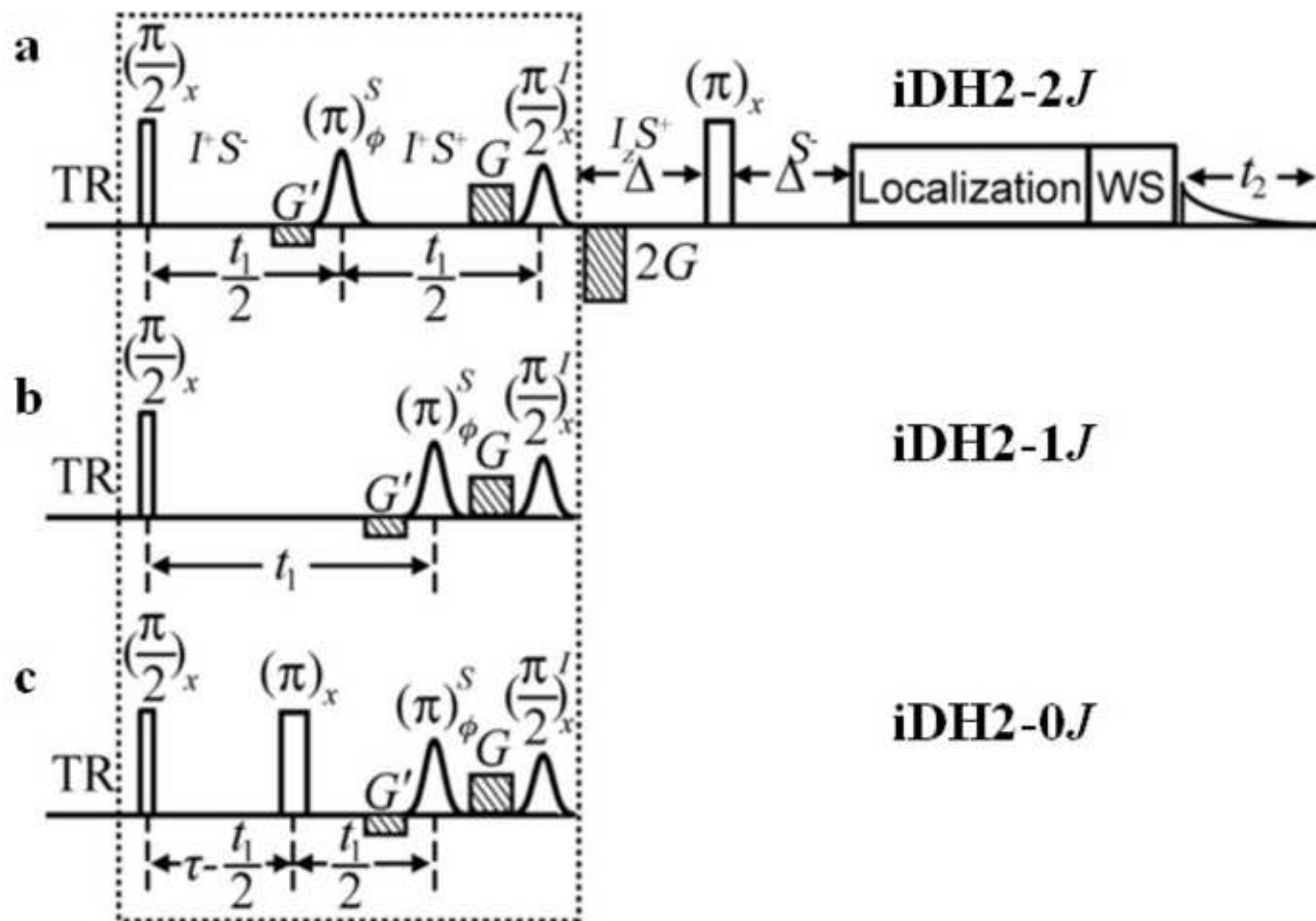


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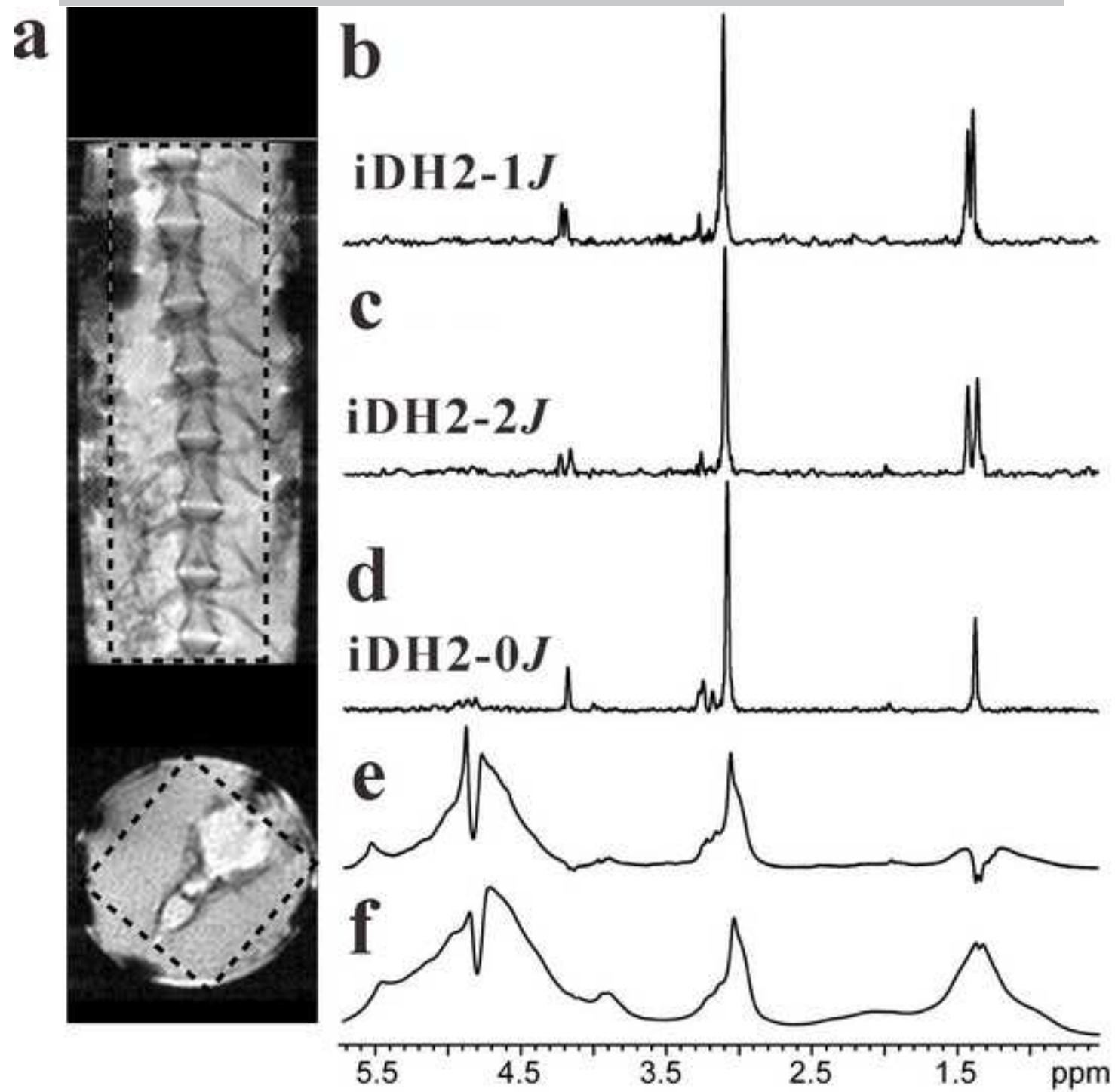


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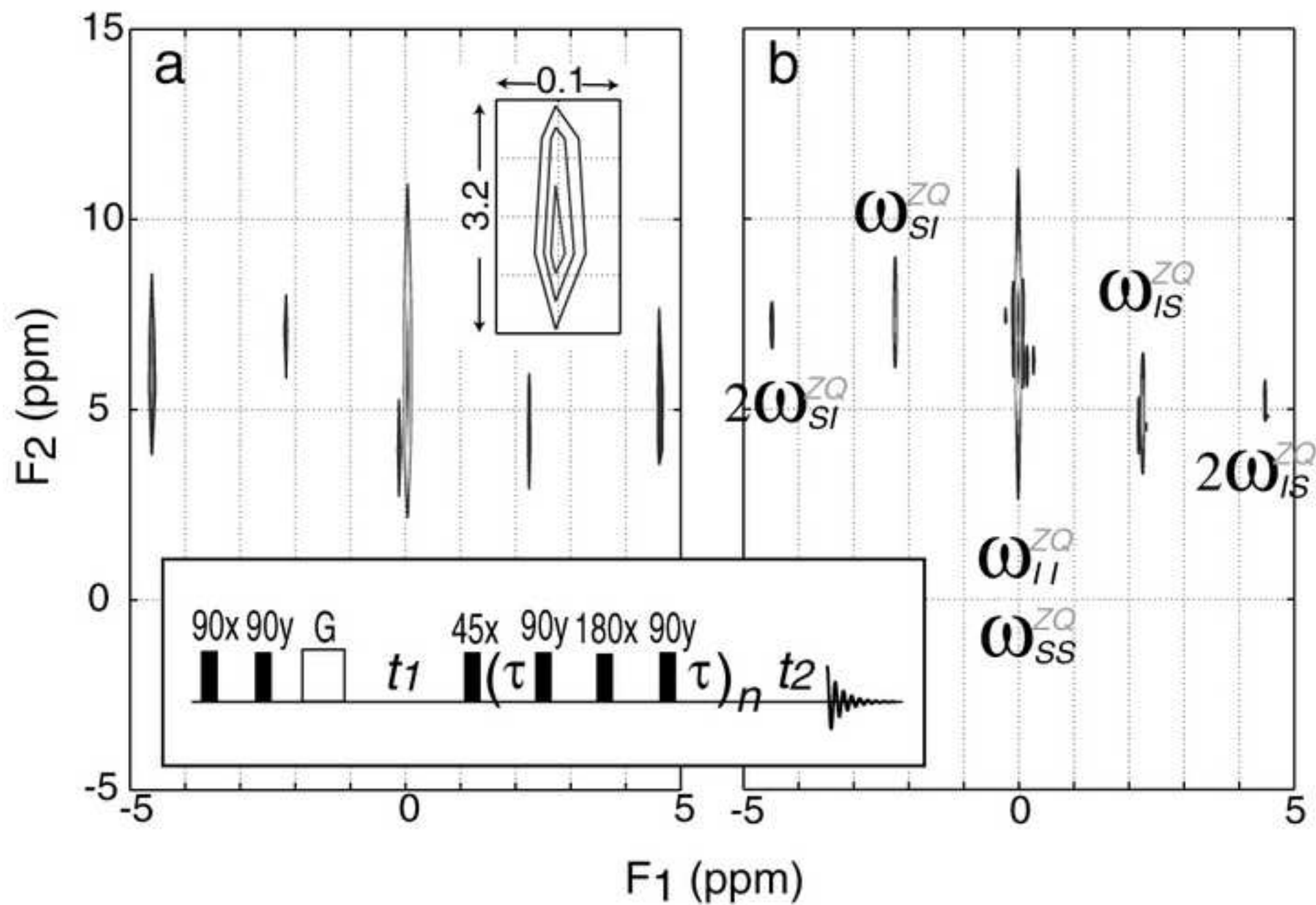
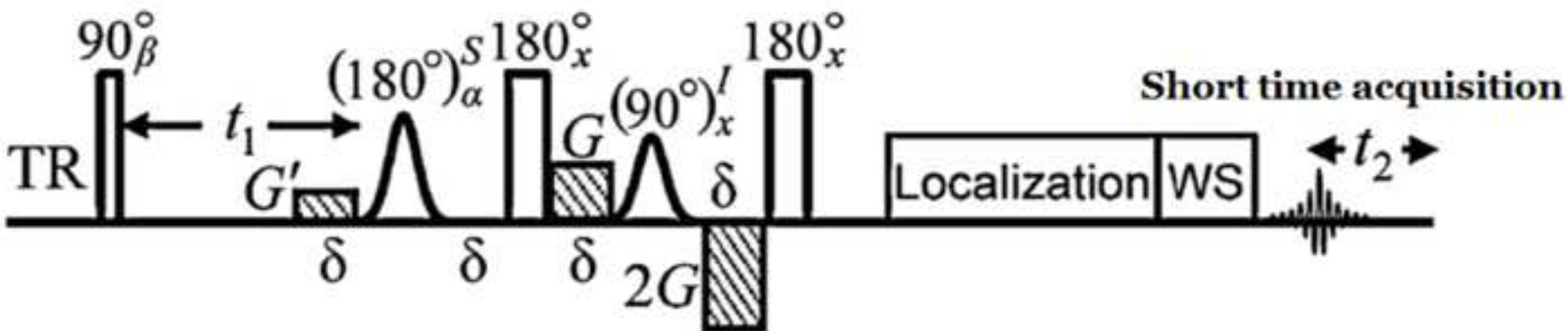


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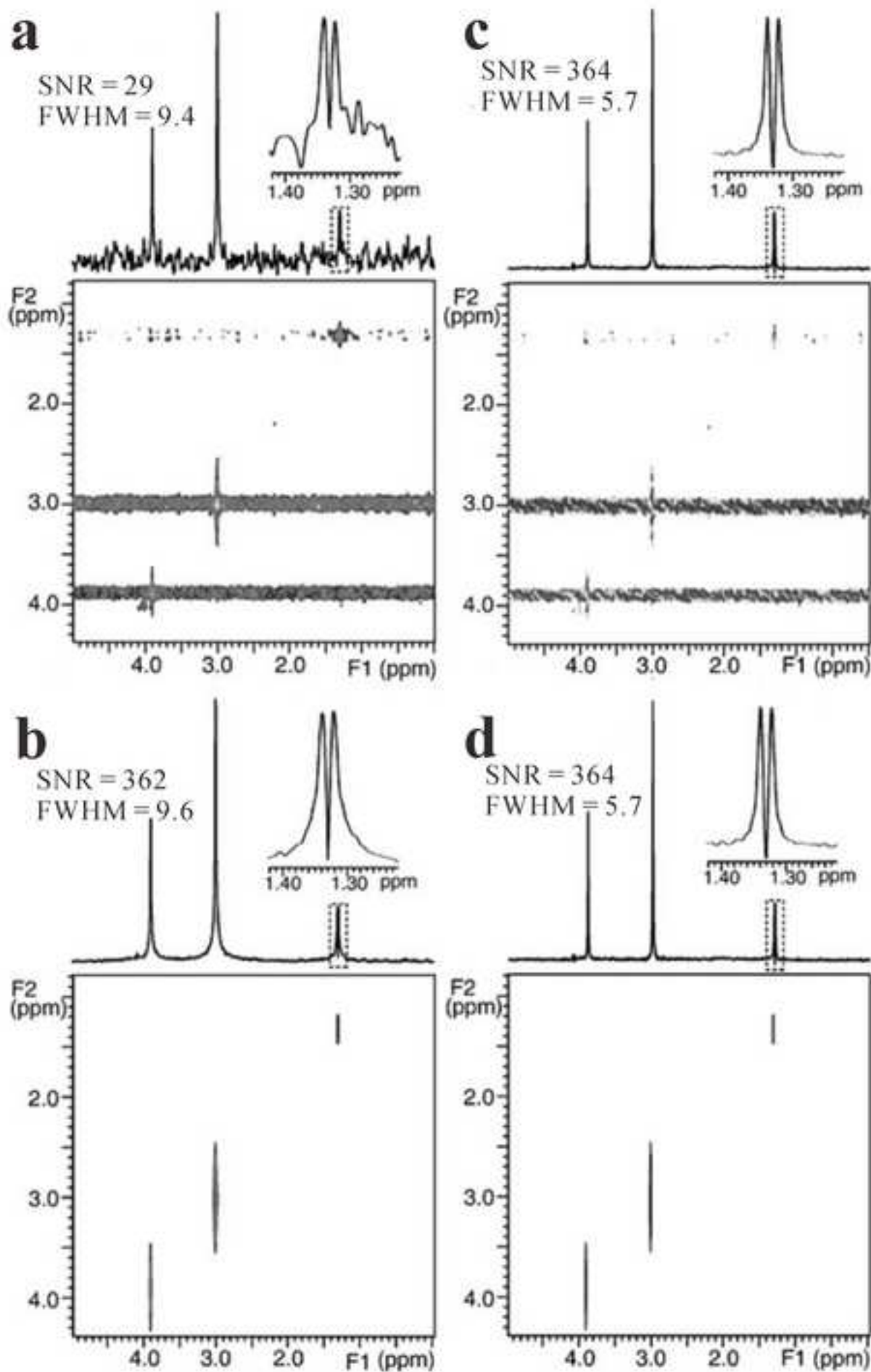


Fig.14

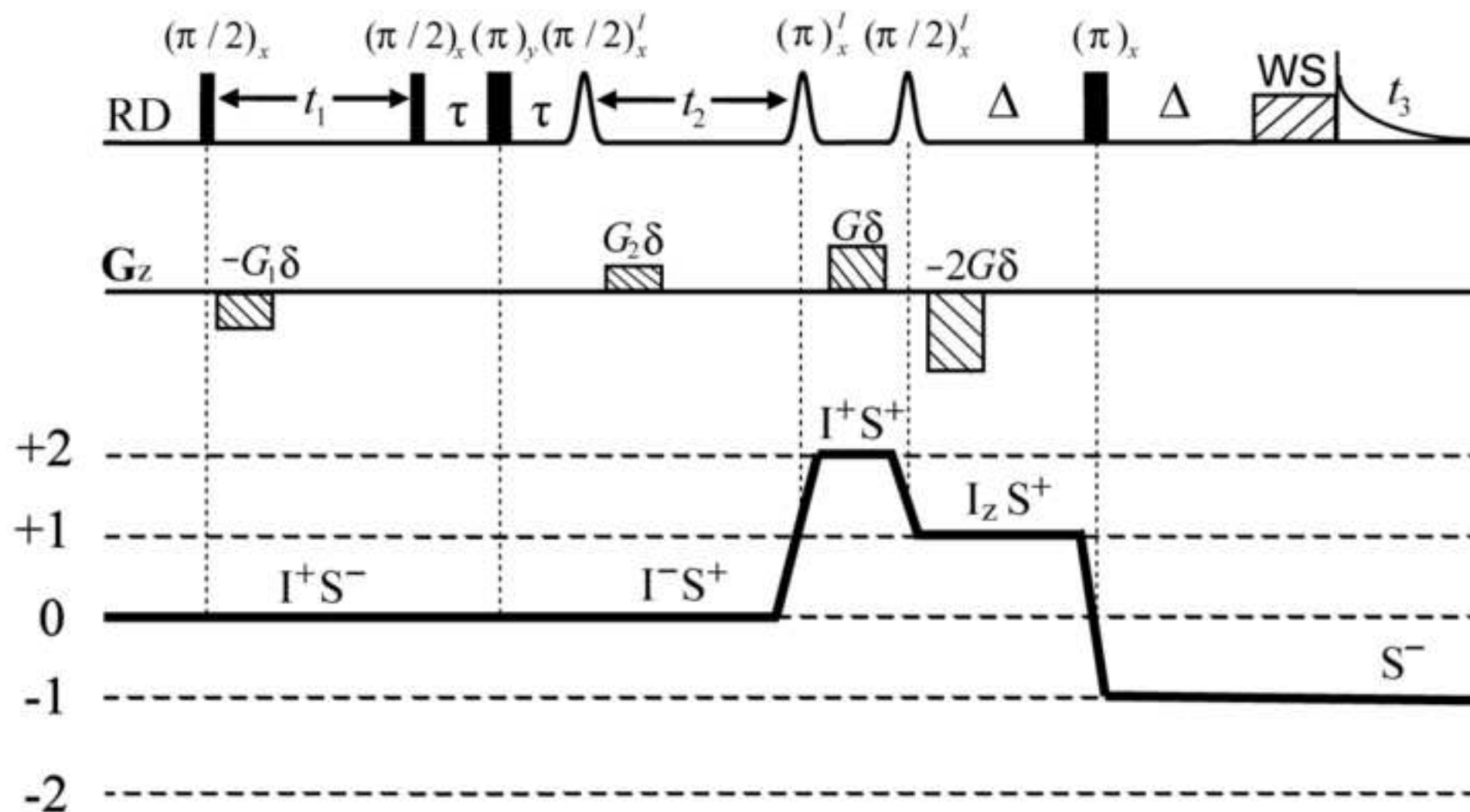


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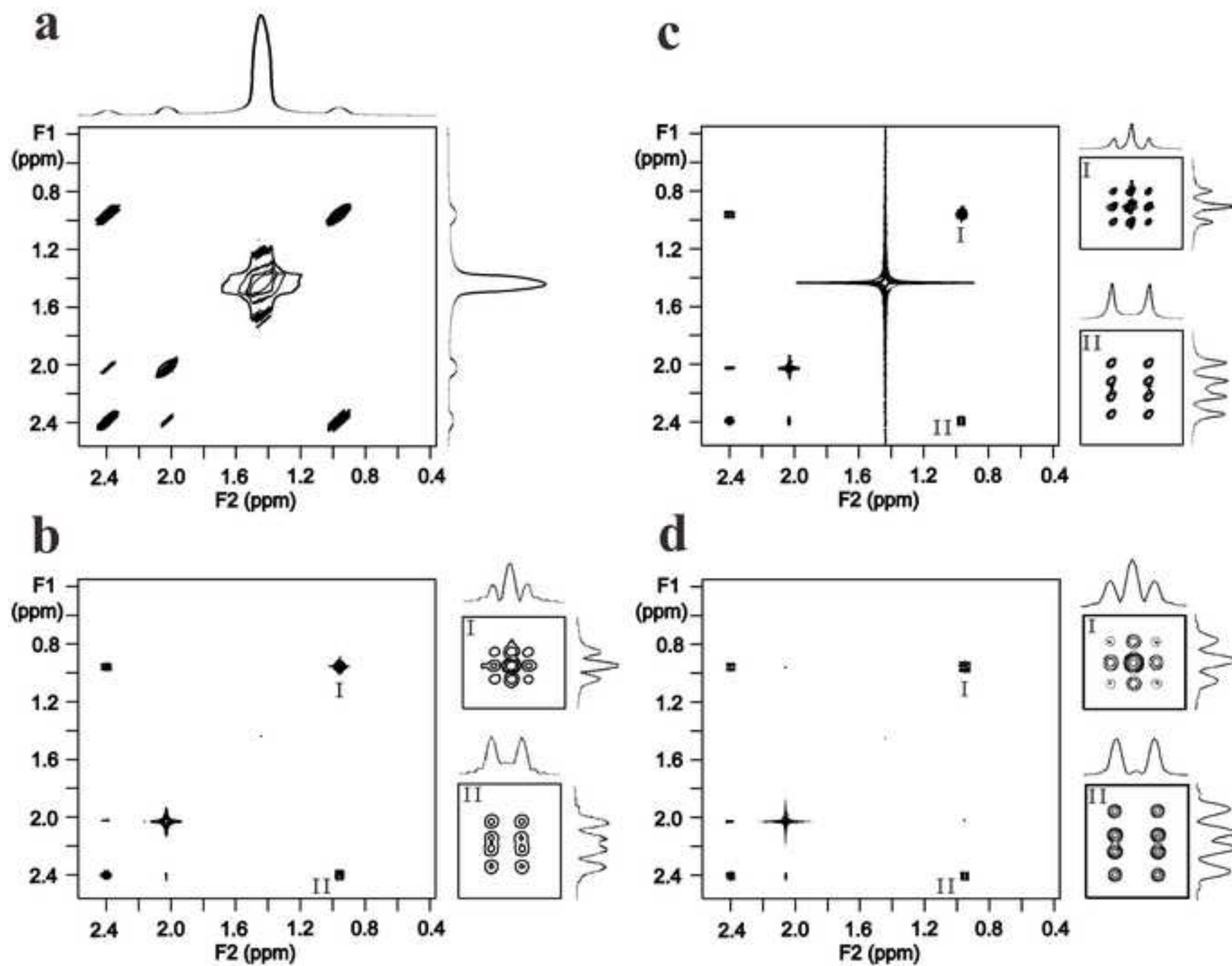
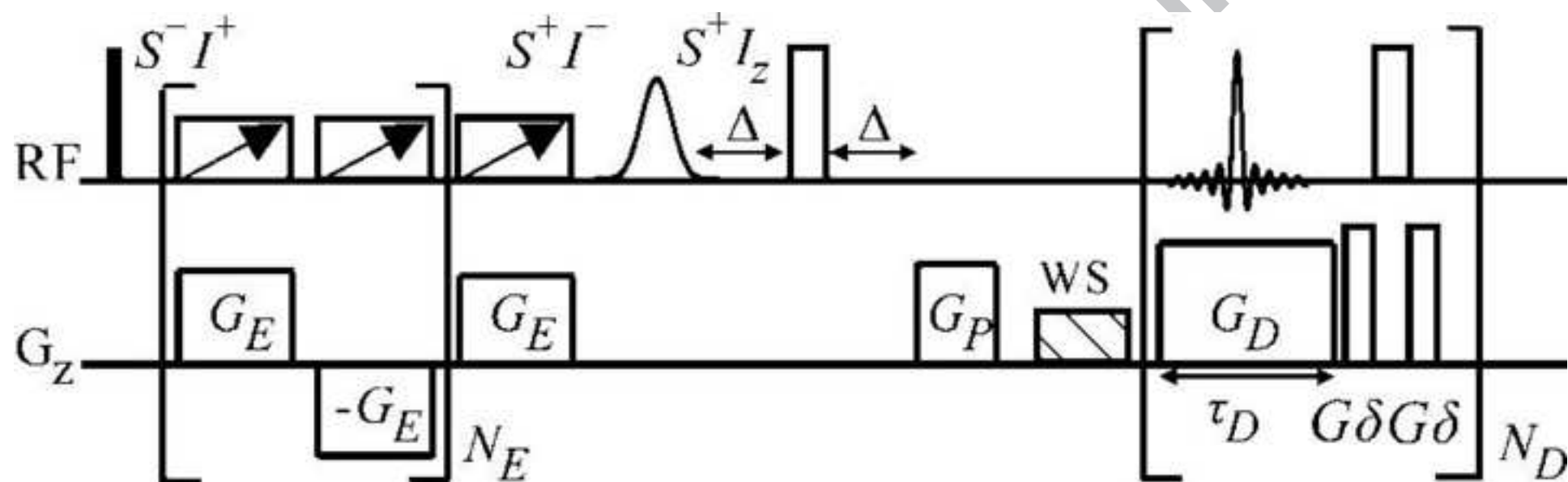


Fig.16



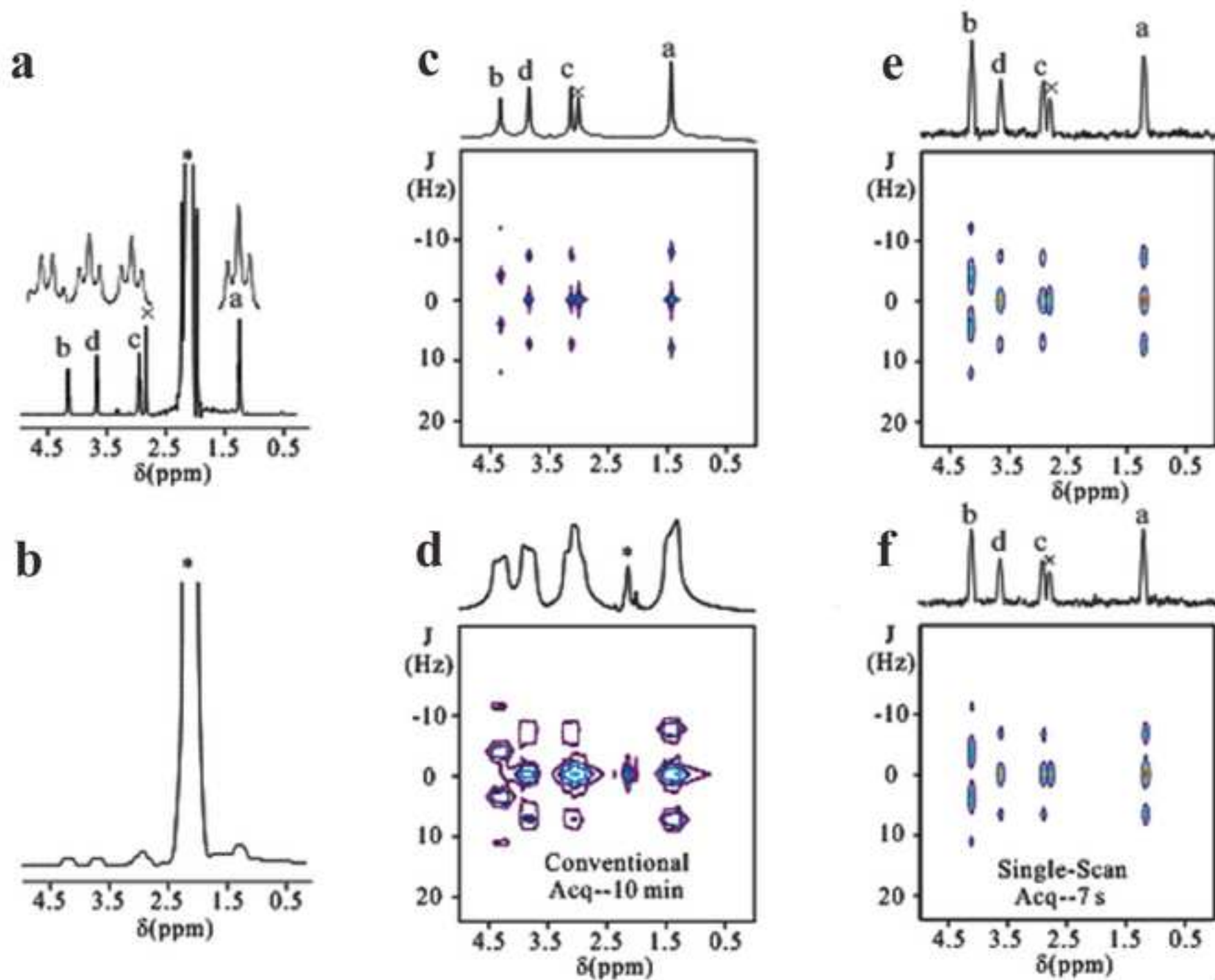
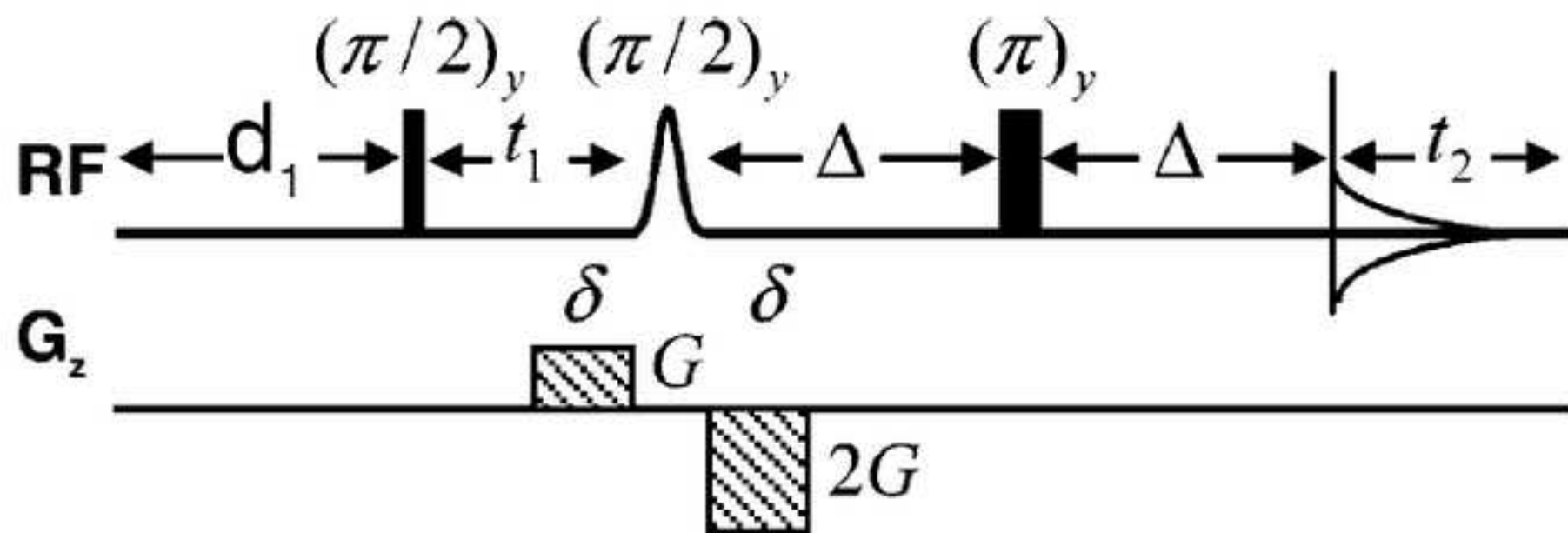


Fig.18



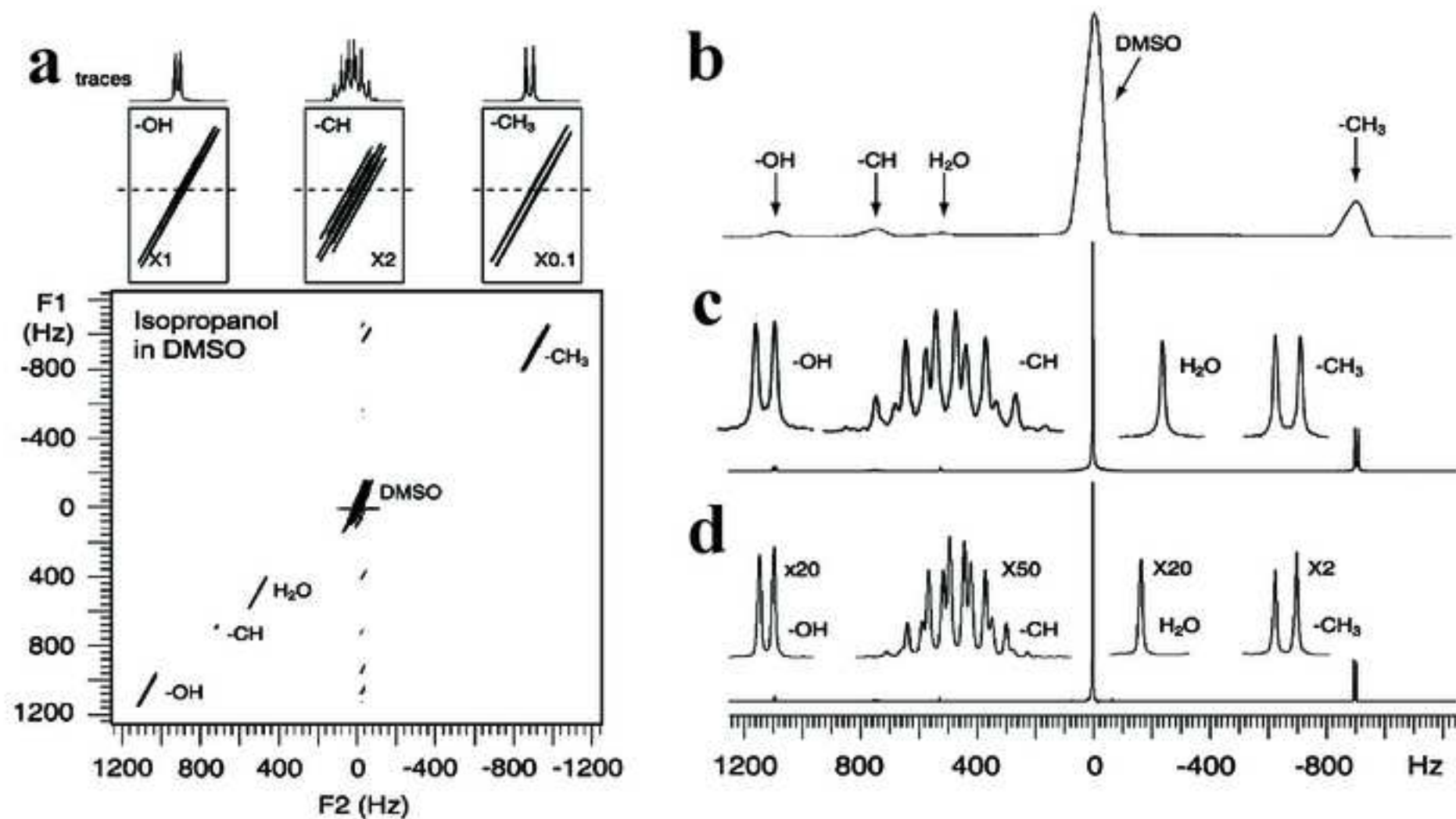
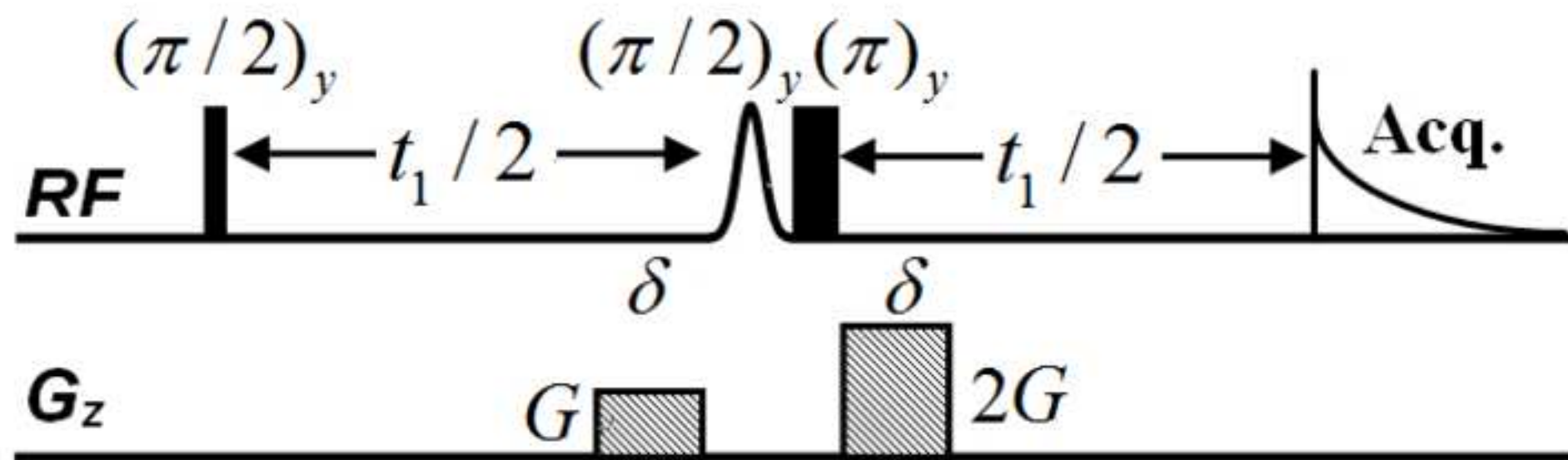
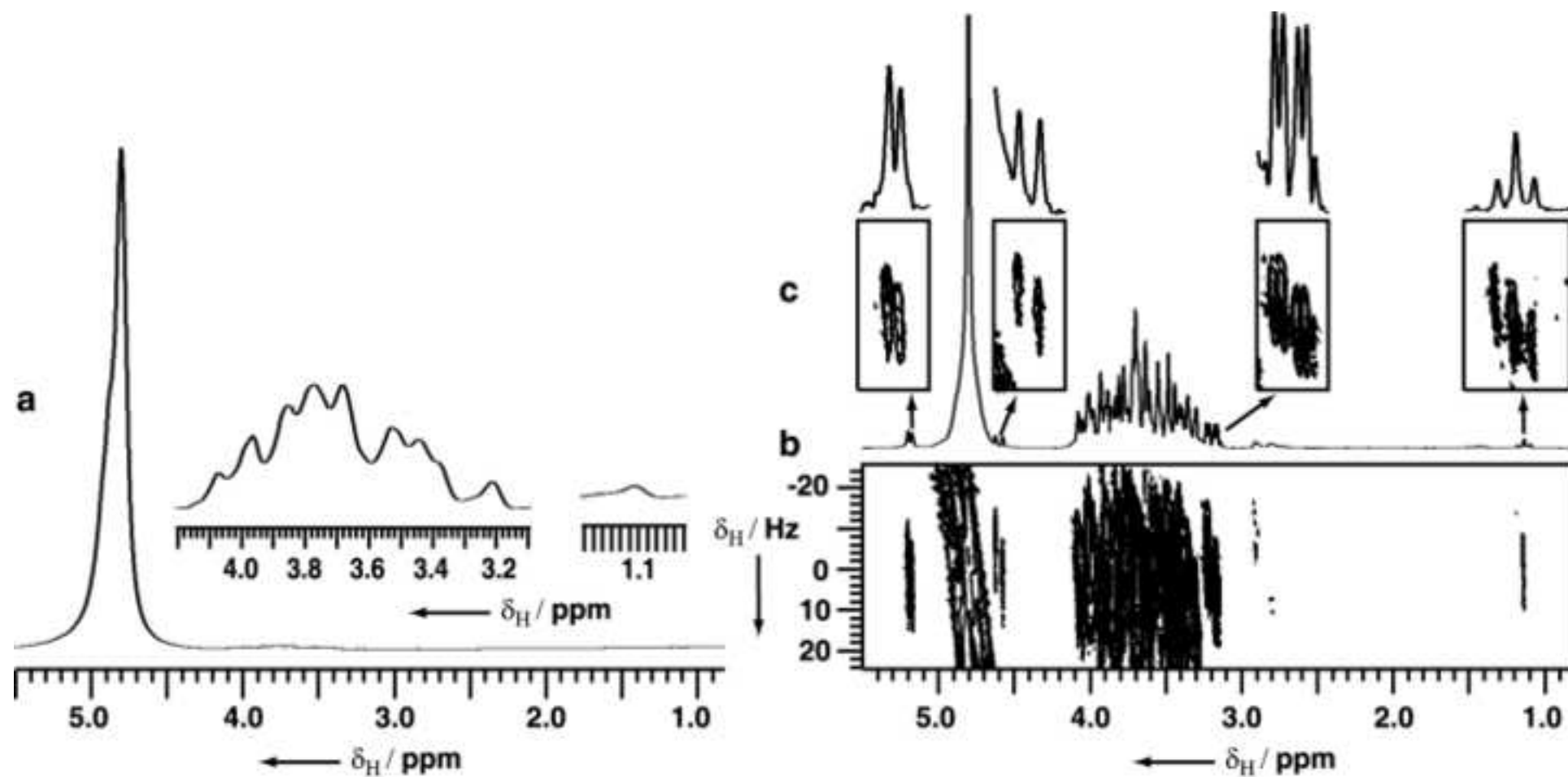
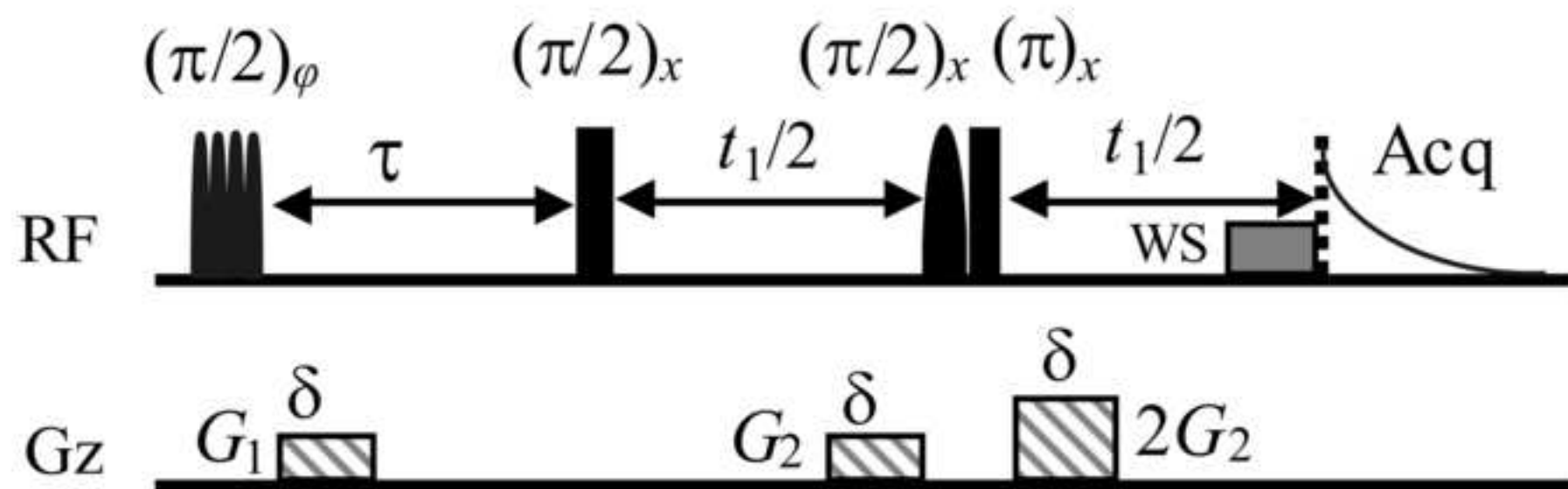
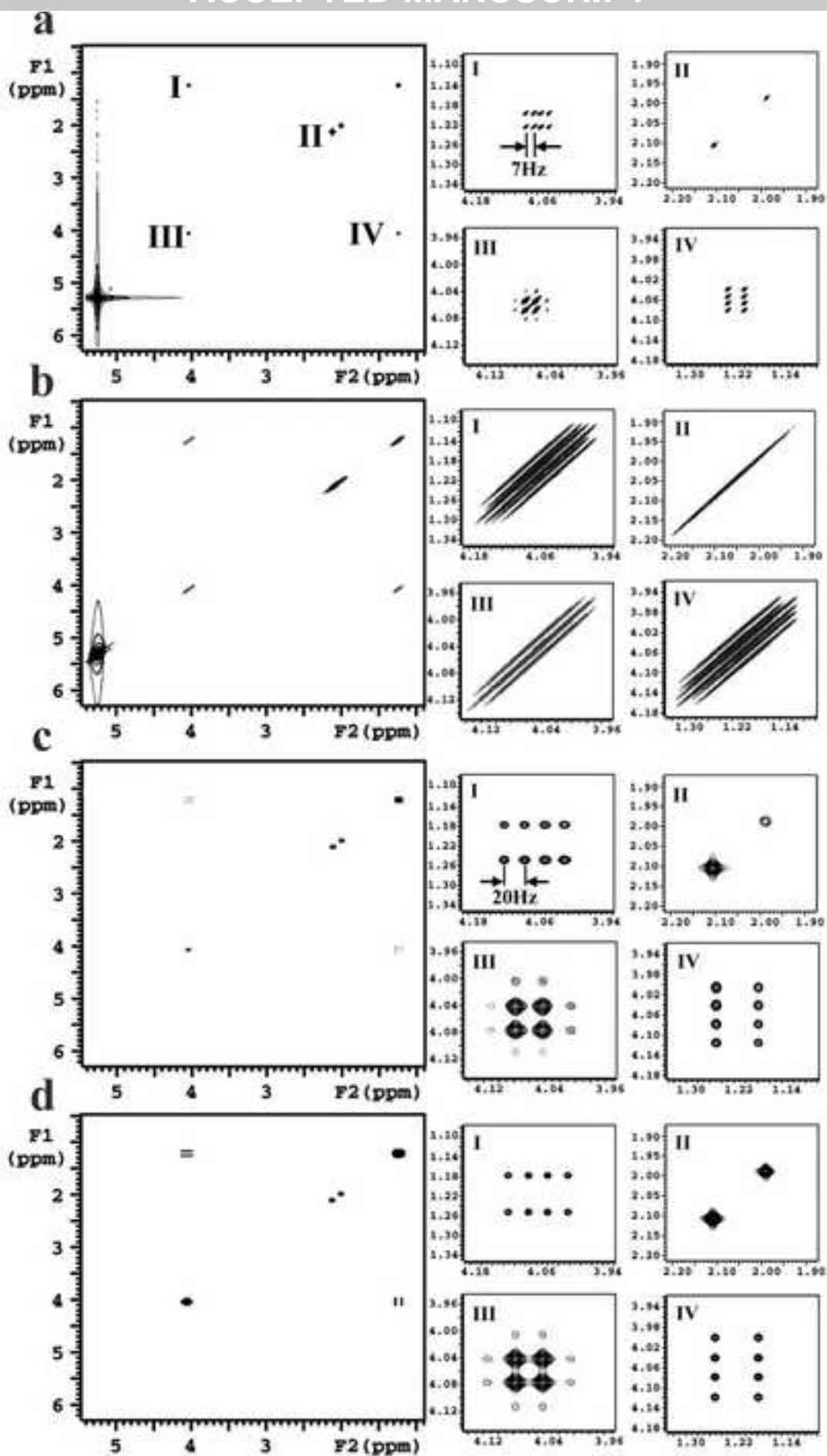


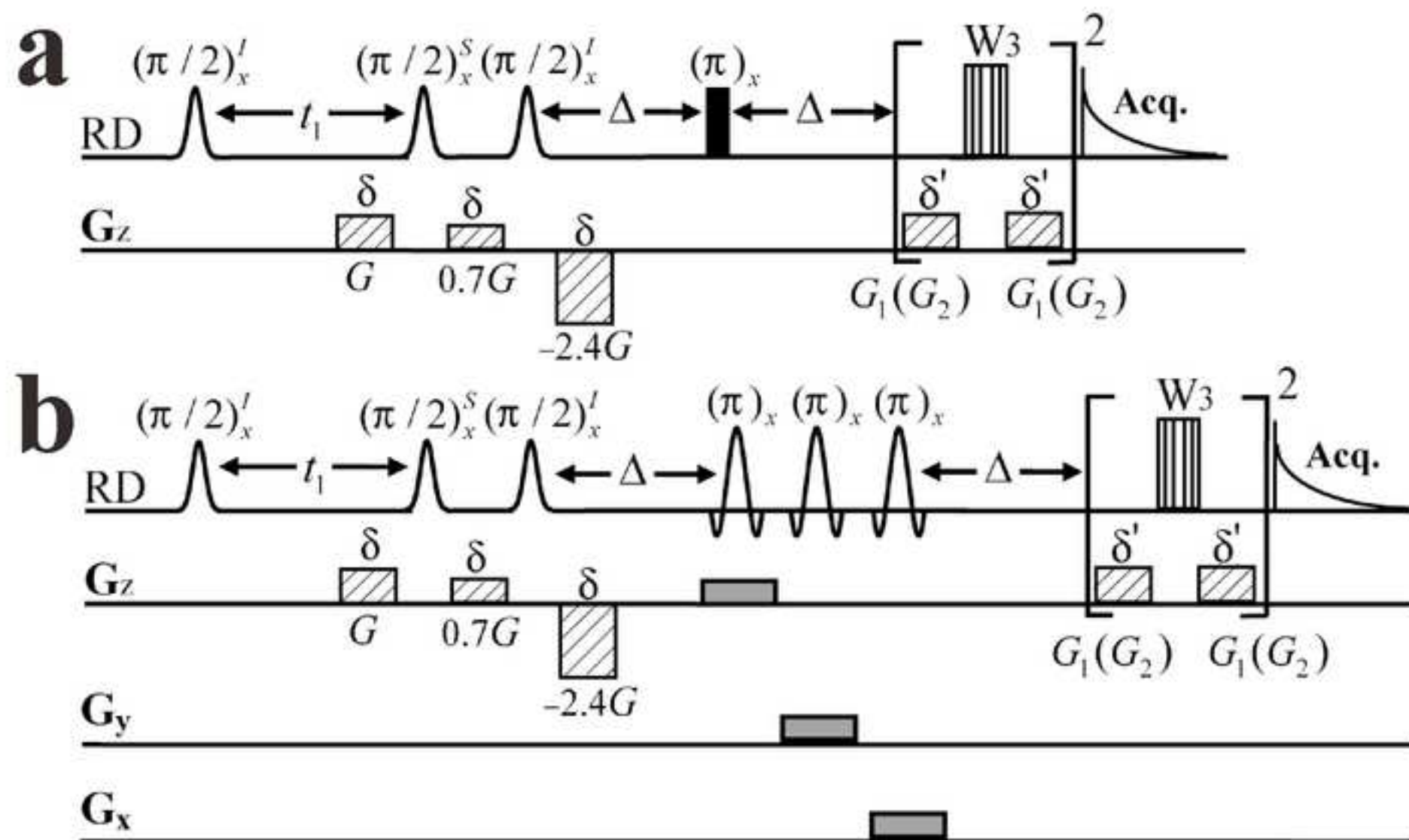
Fig.20

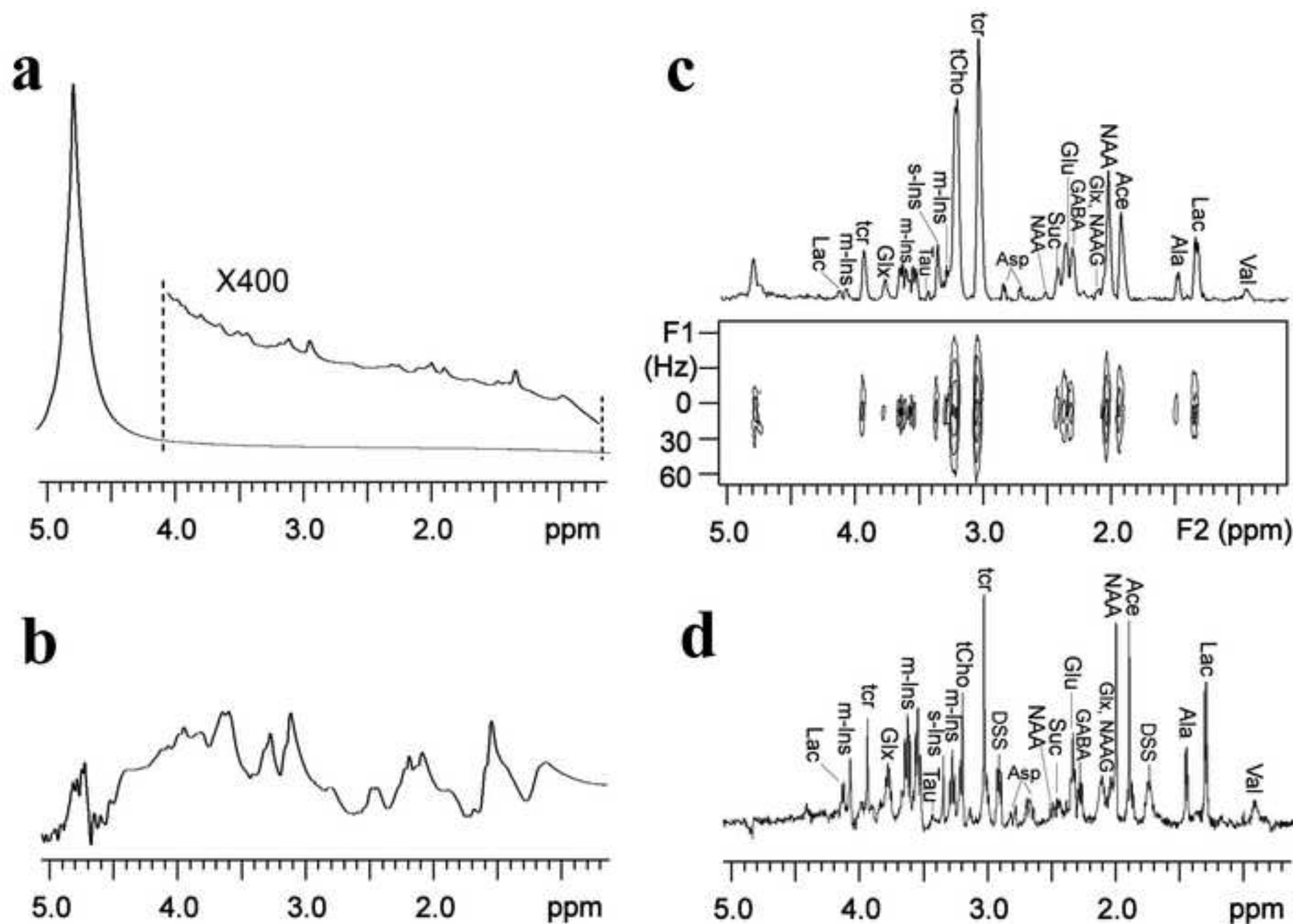


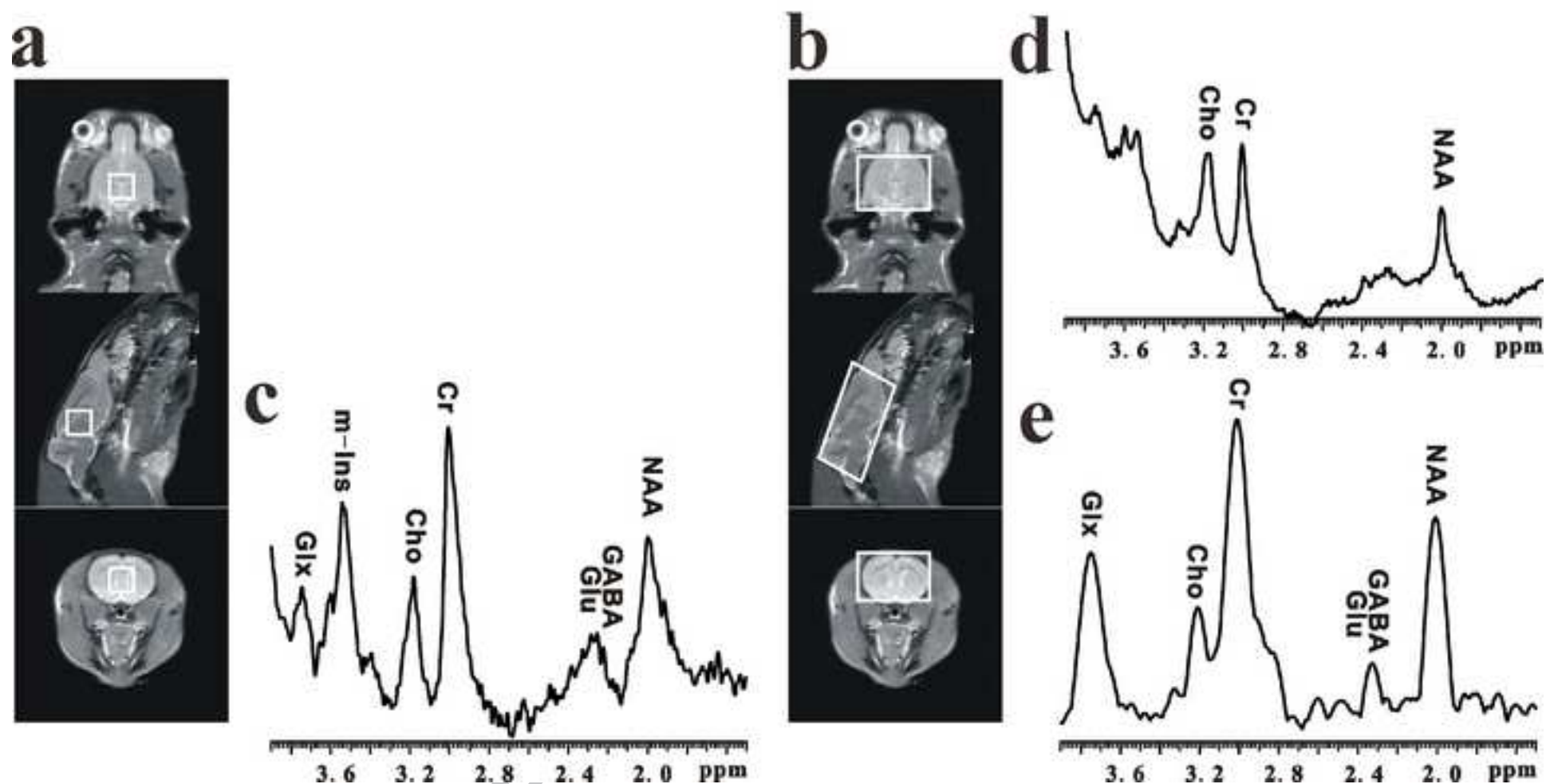


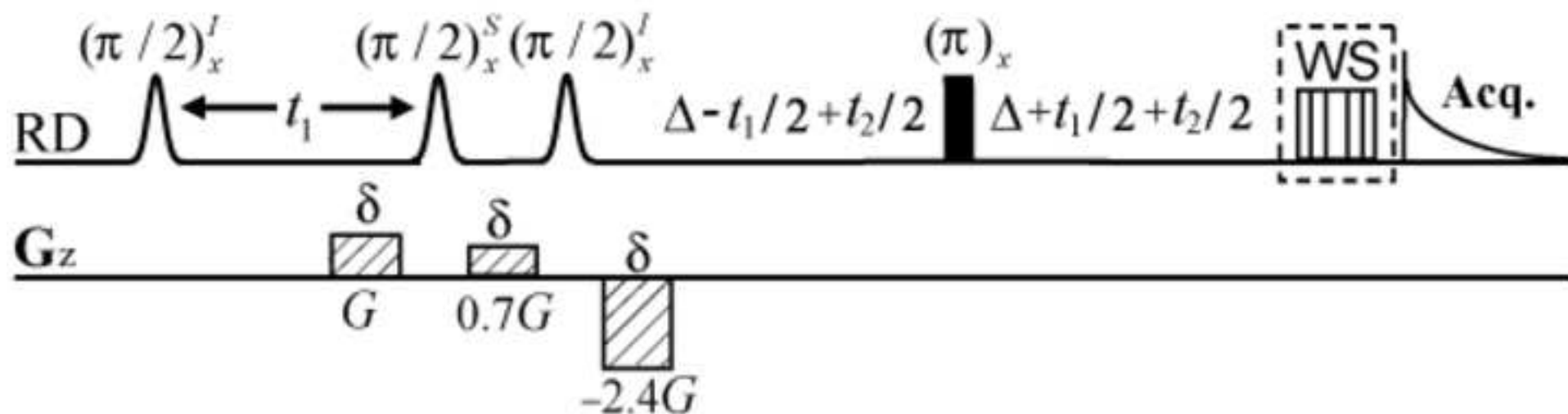












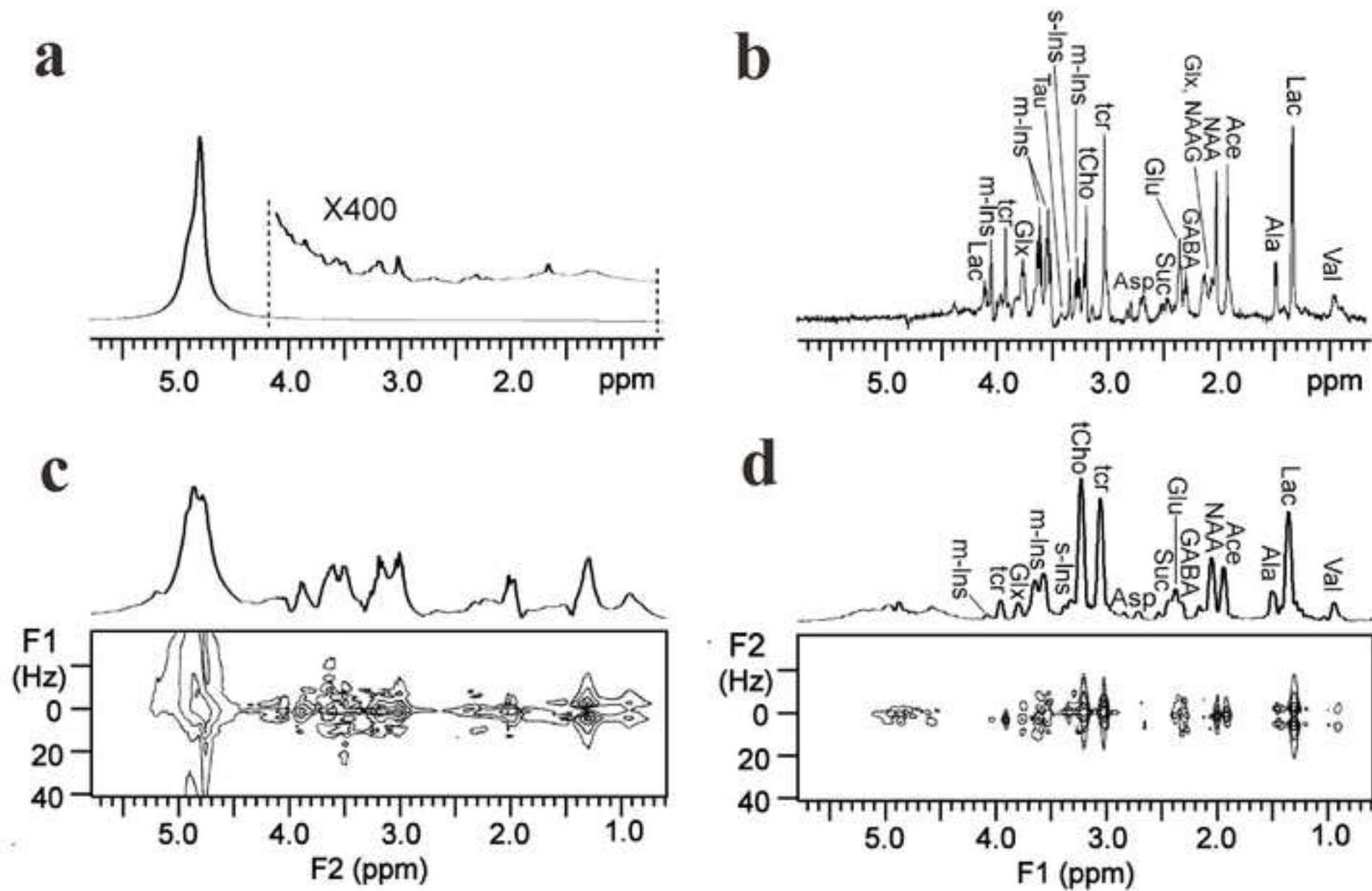
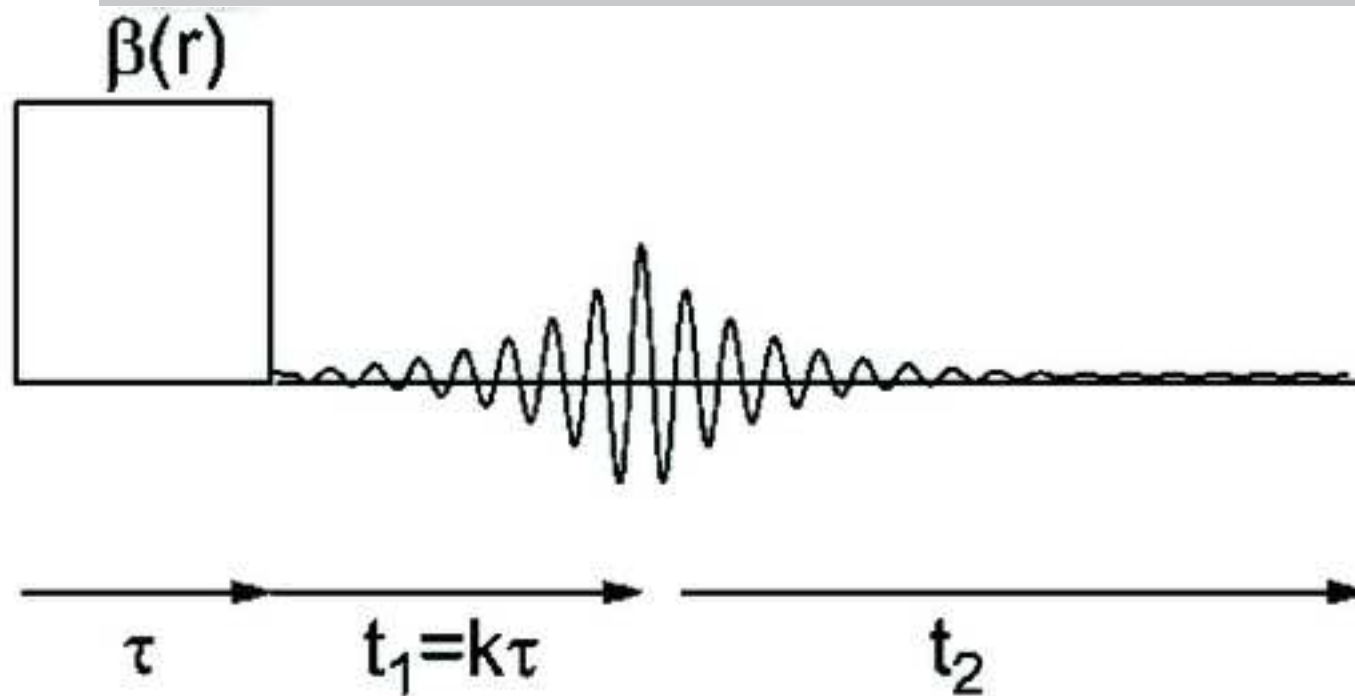
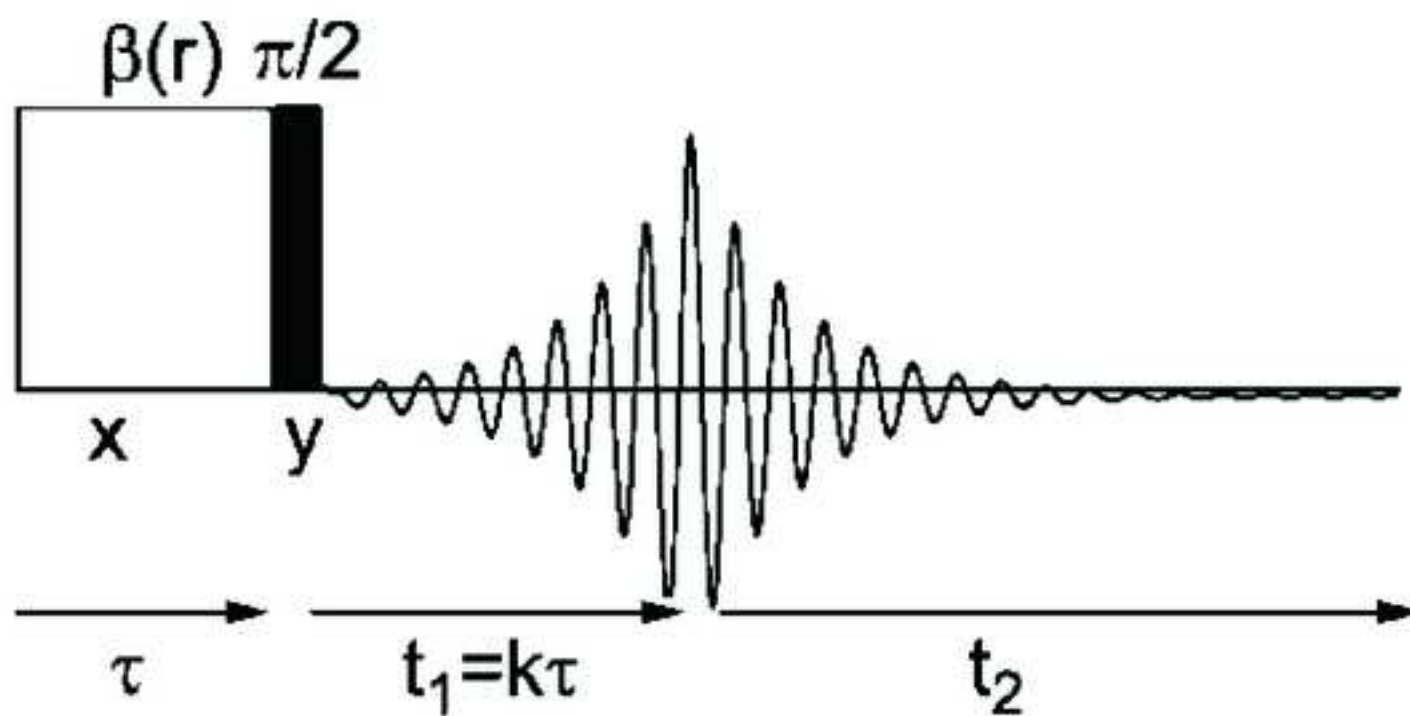


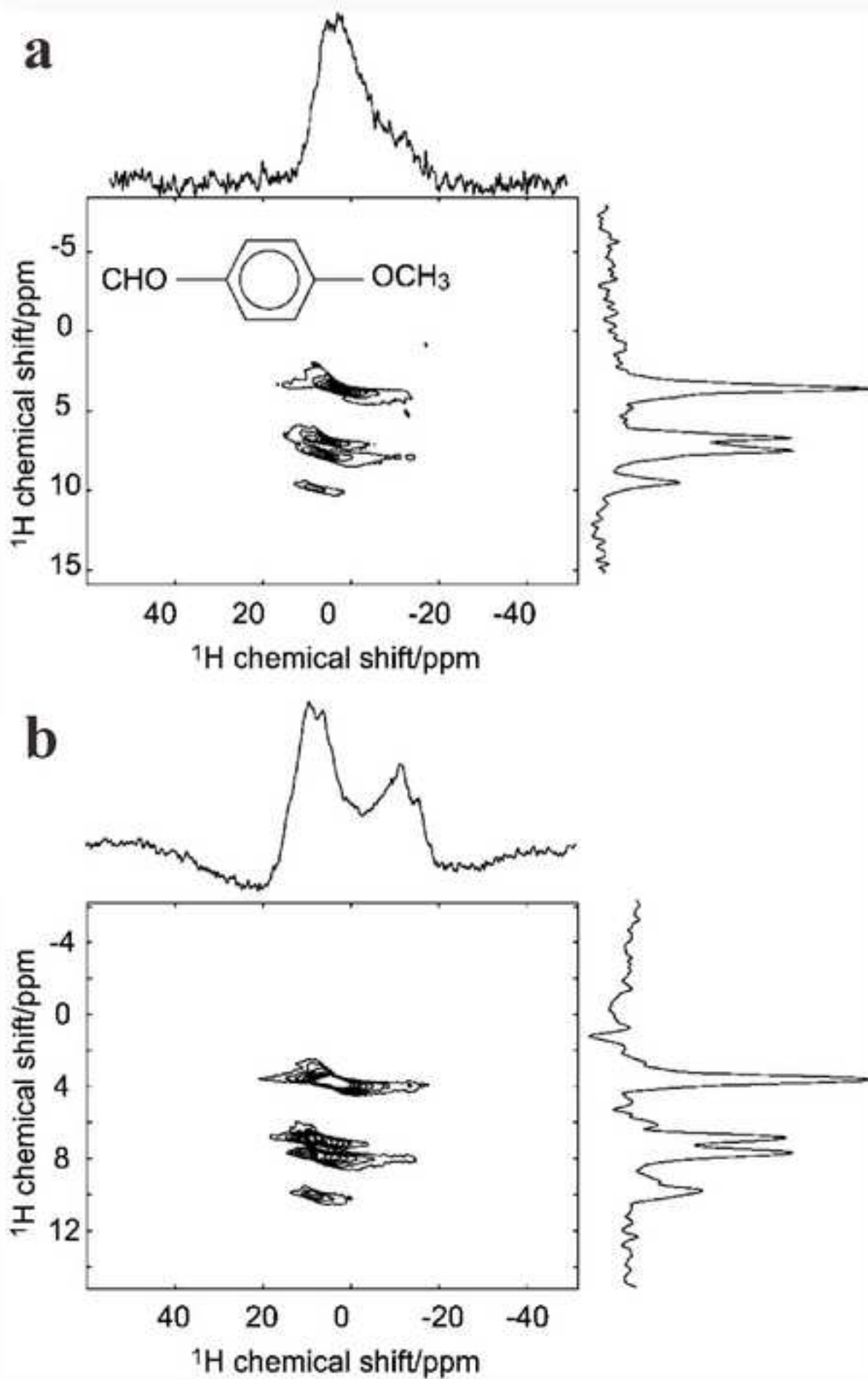
Fig.29

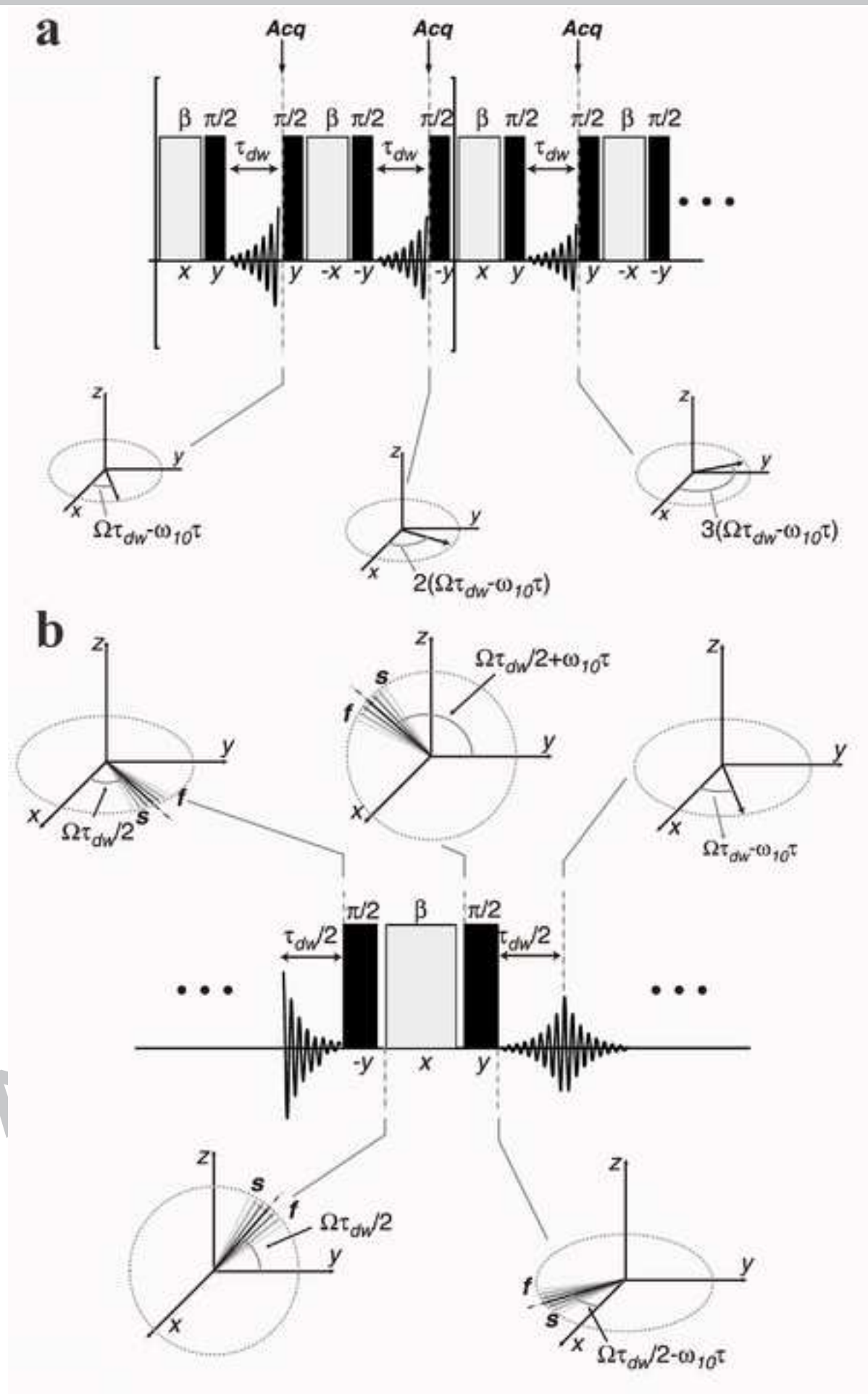
a



b







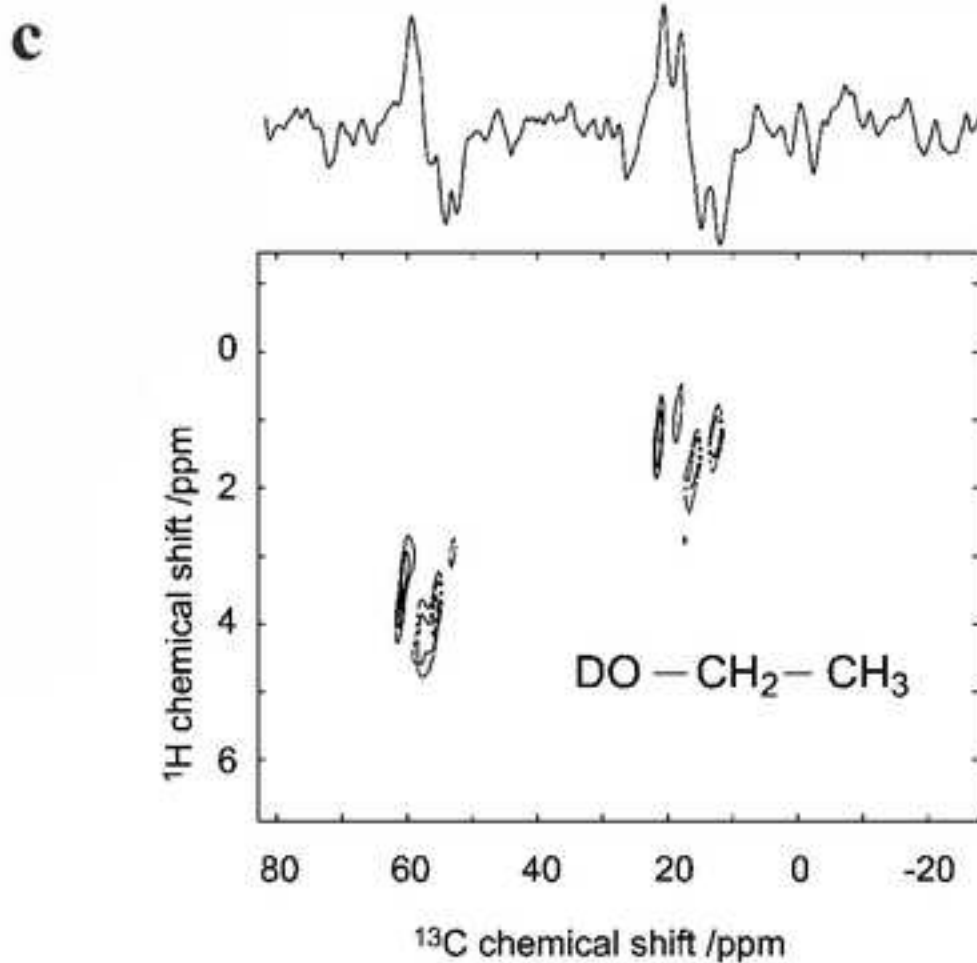
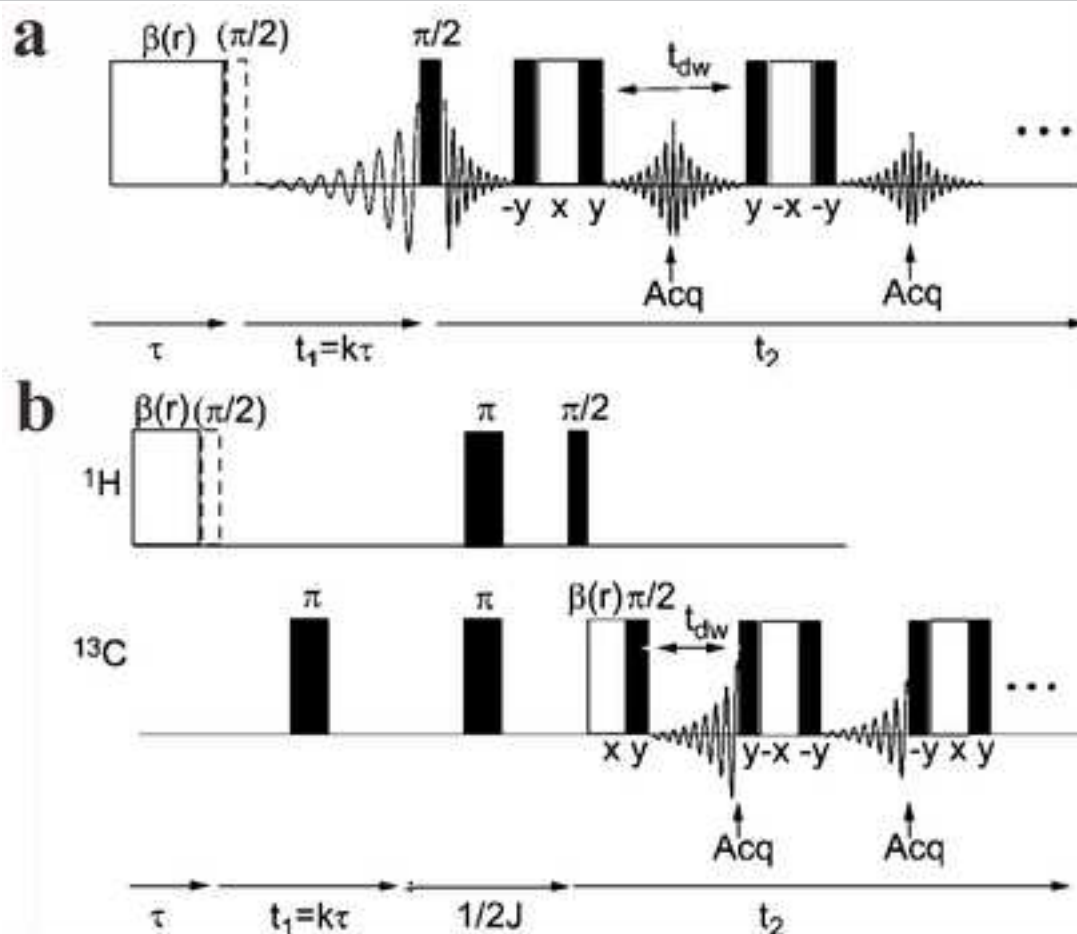
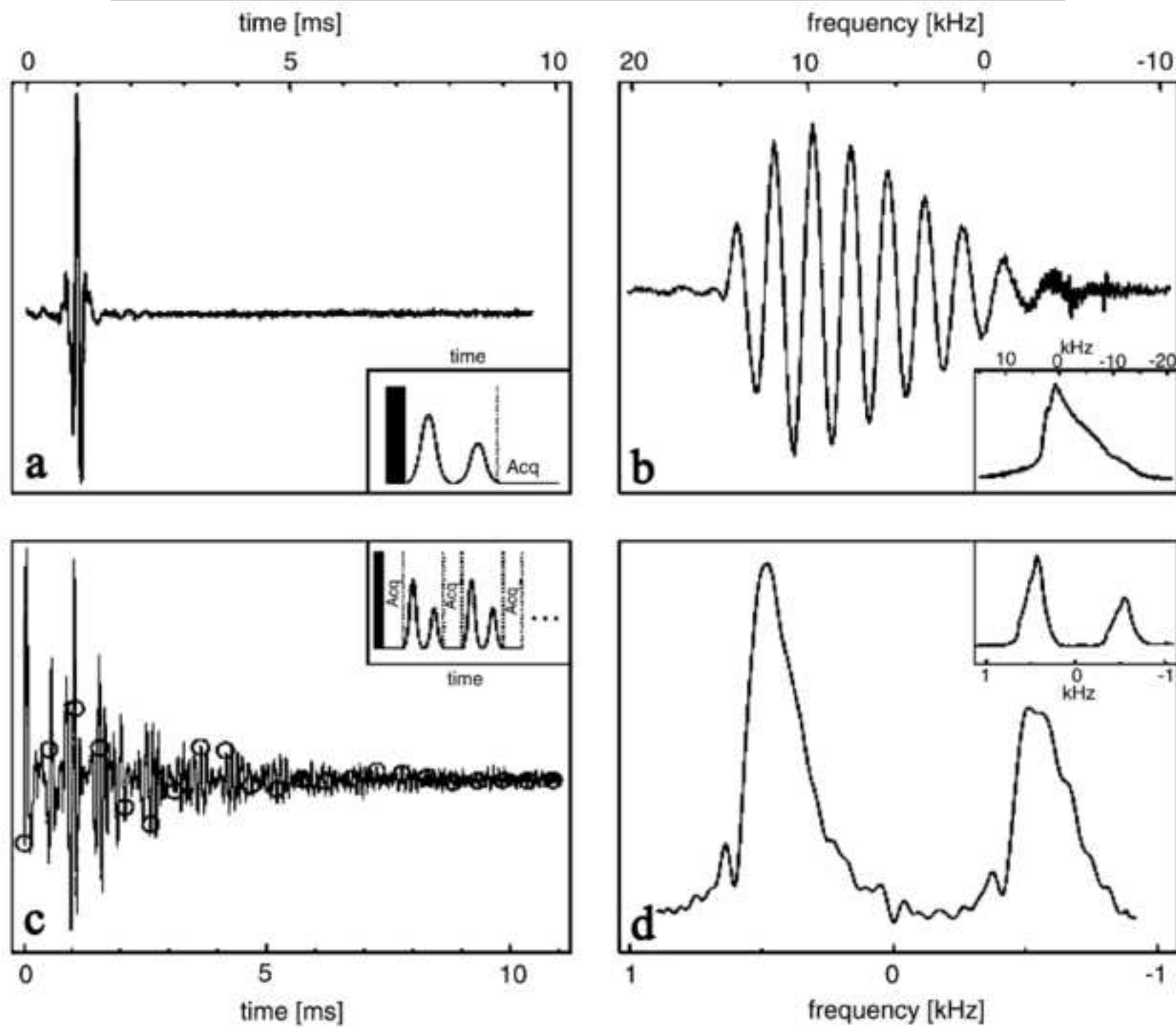
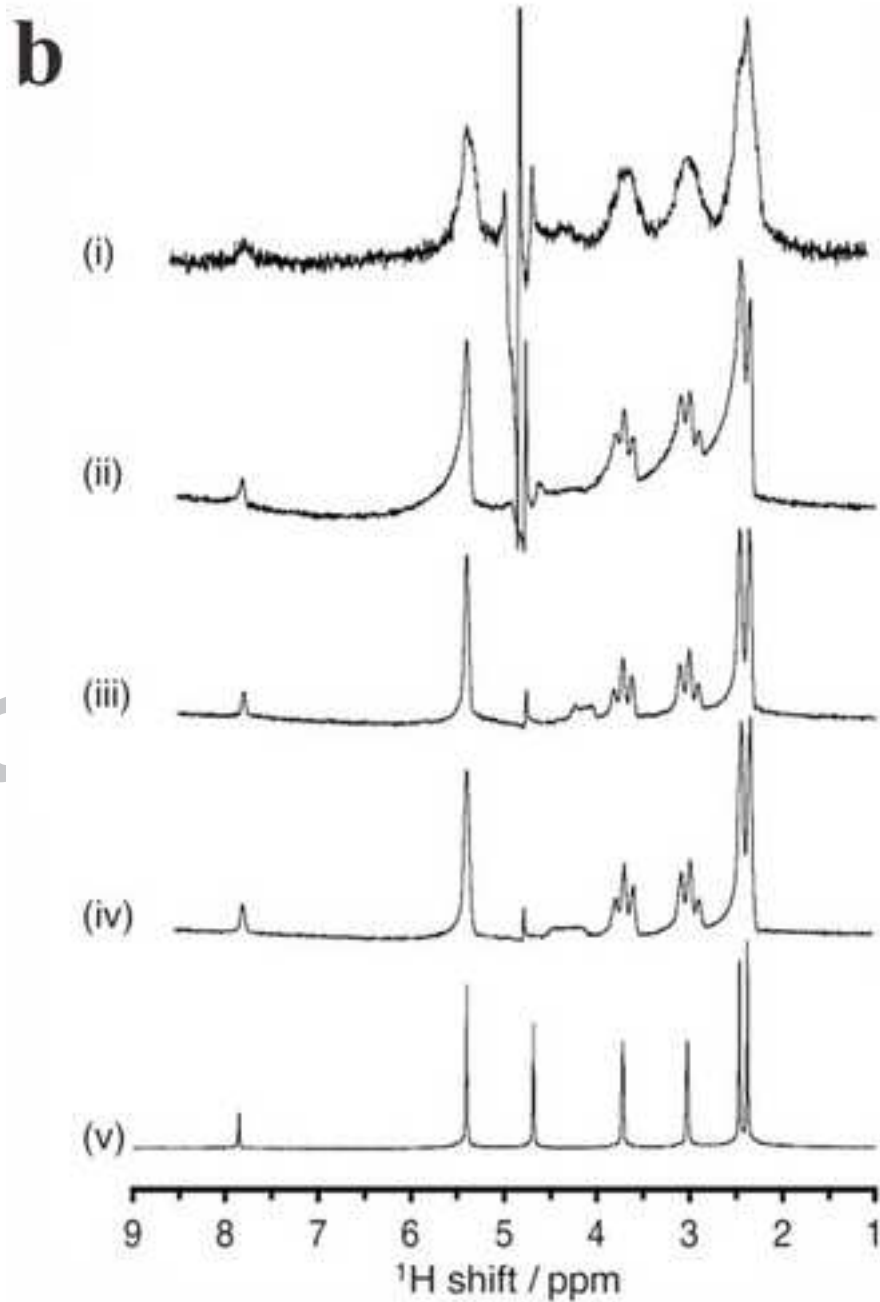
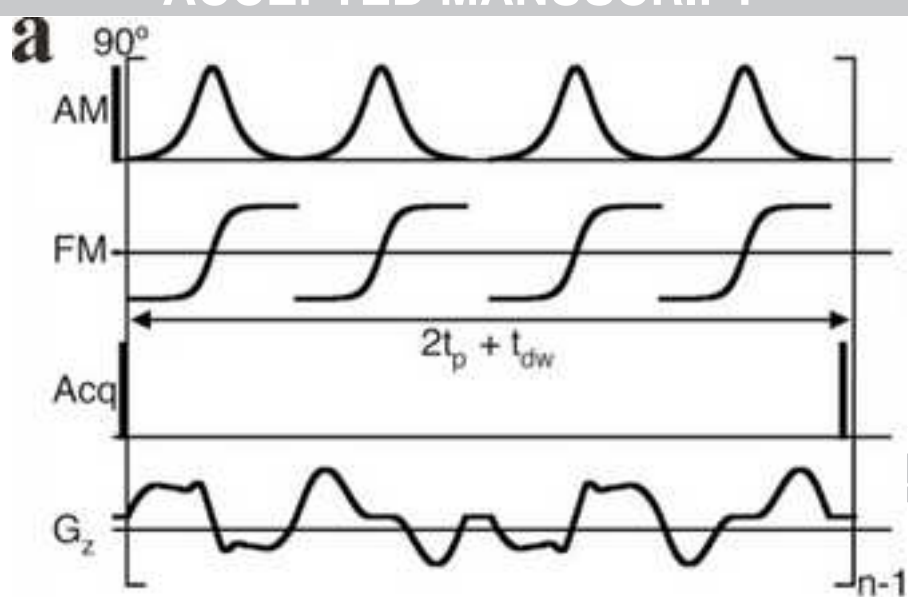
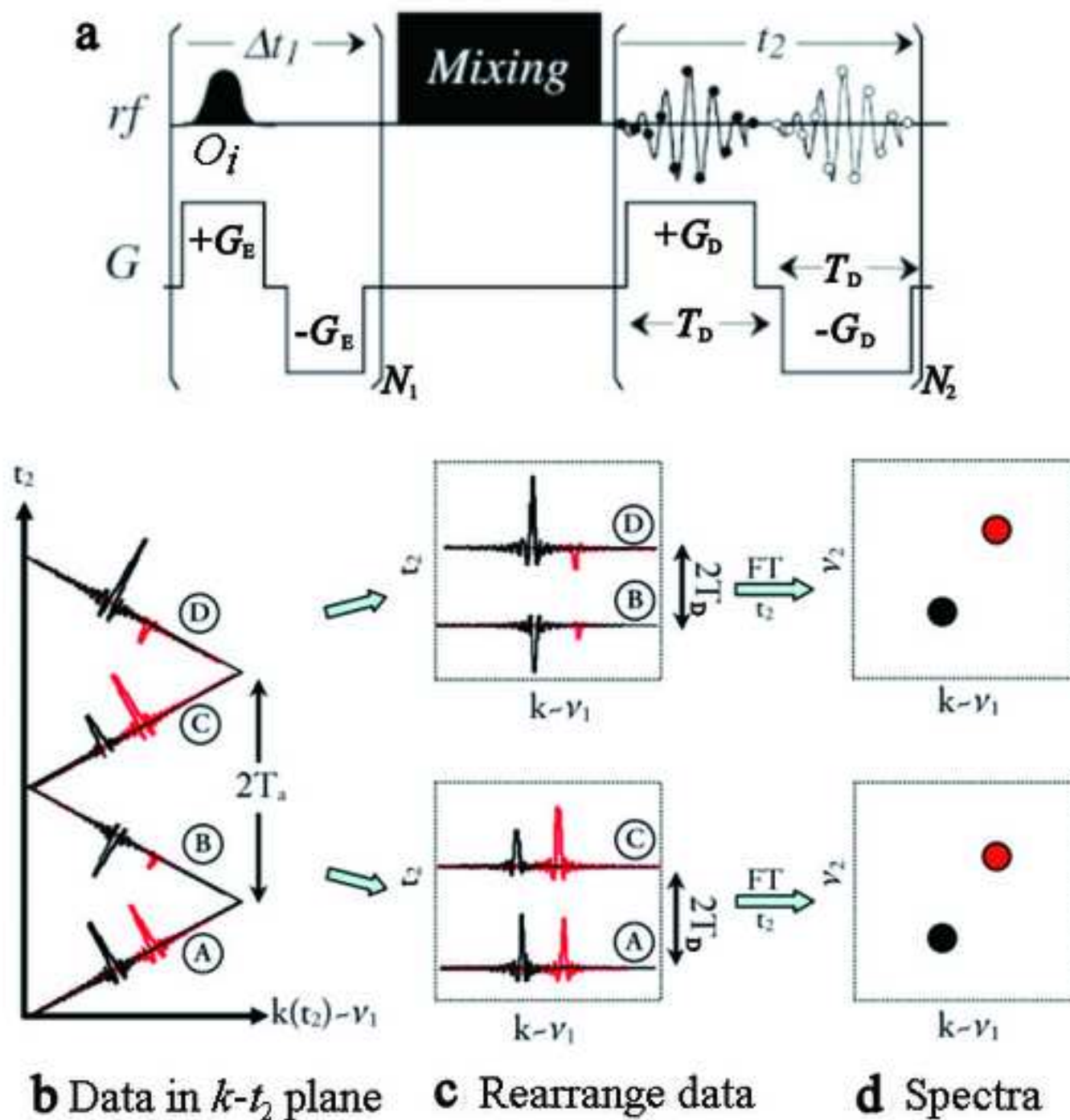


Fig.33







a Discrete longitudinal compensation

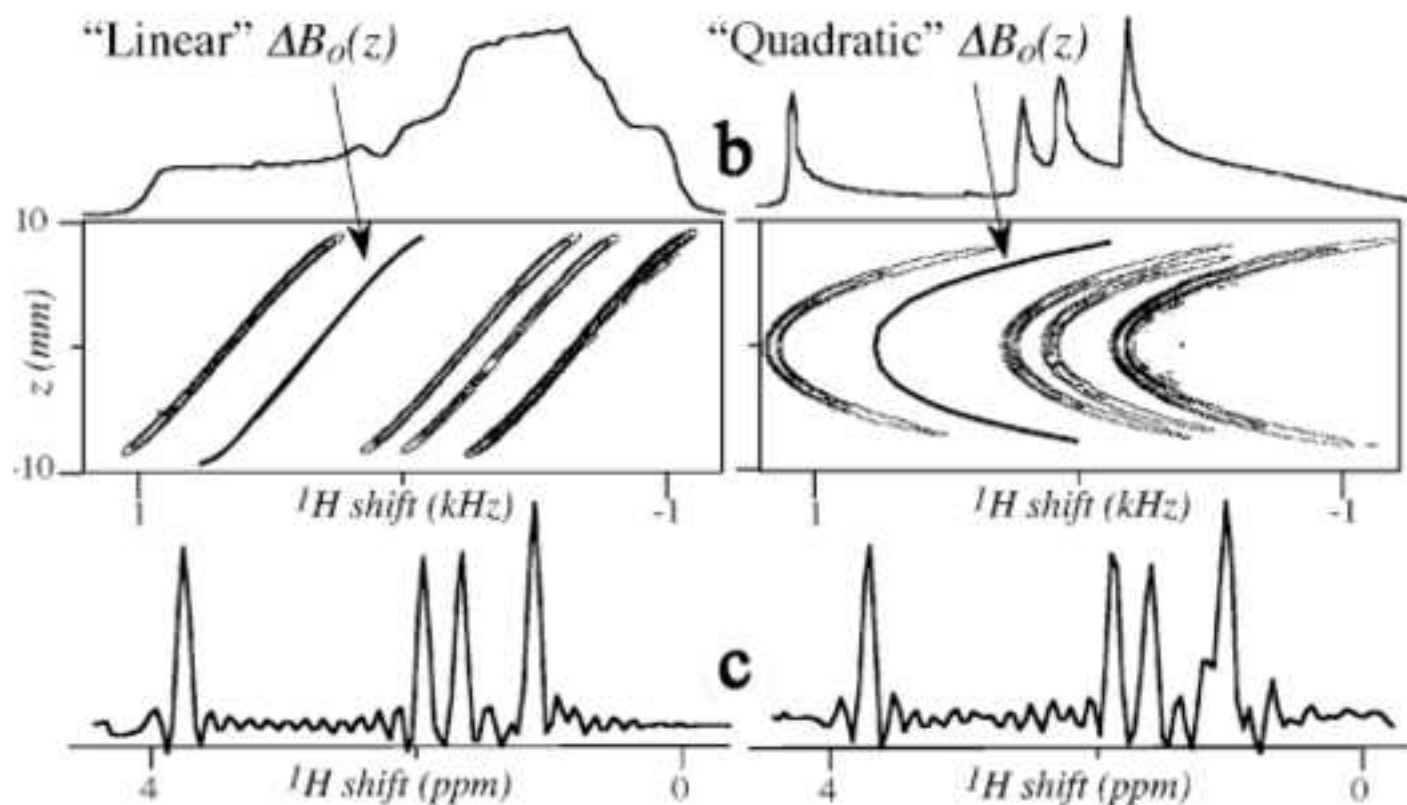
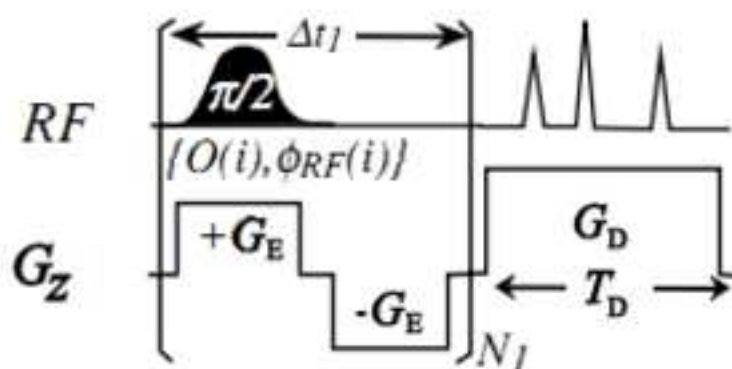


Fig.37

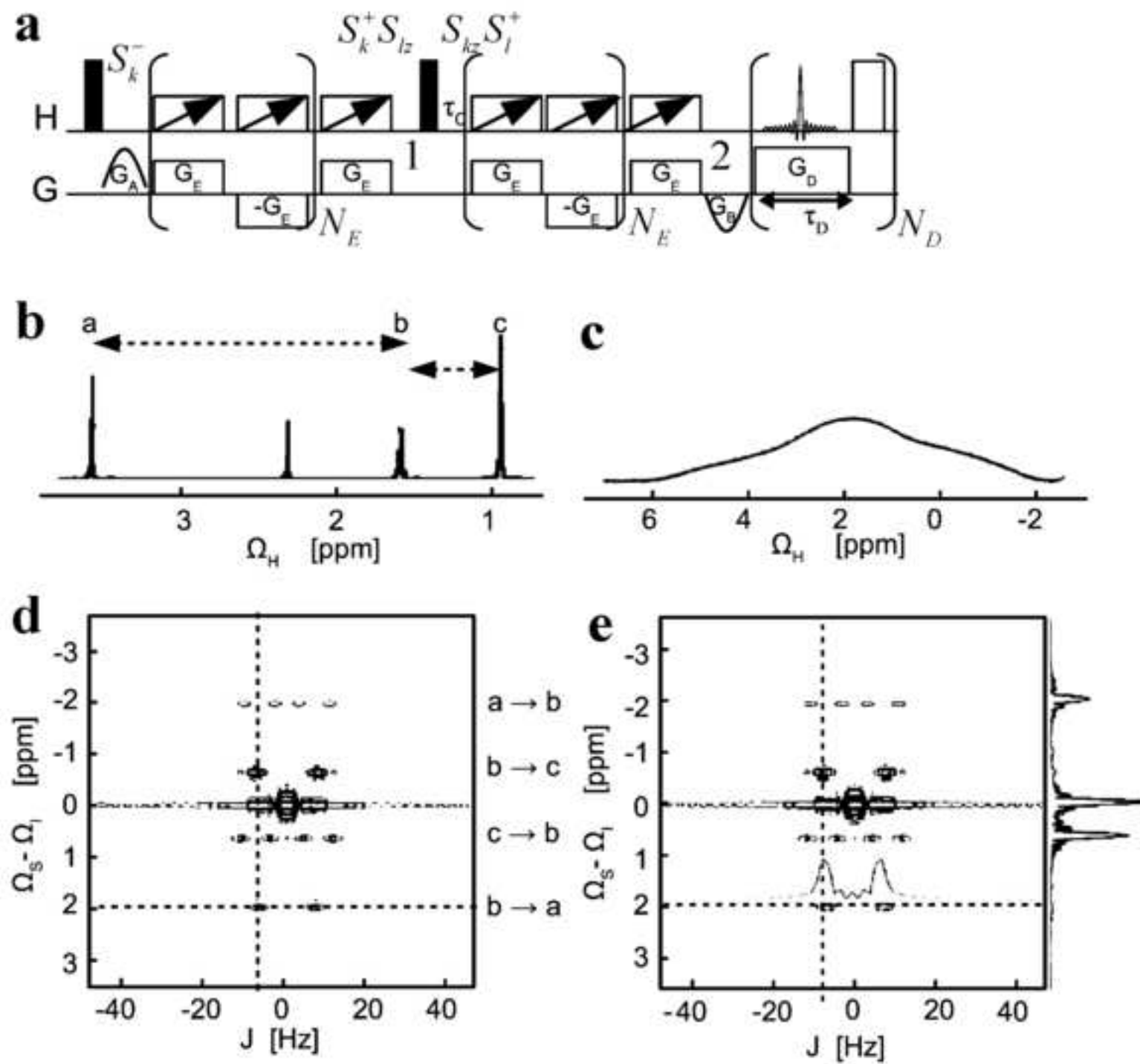
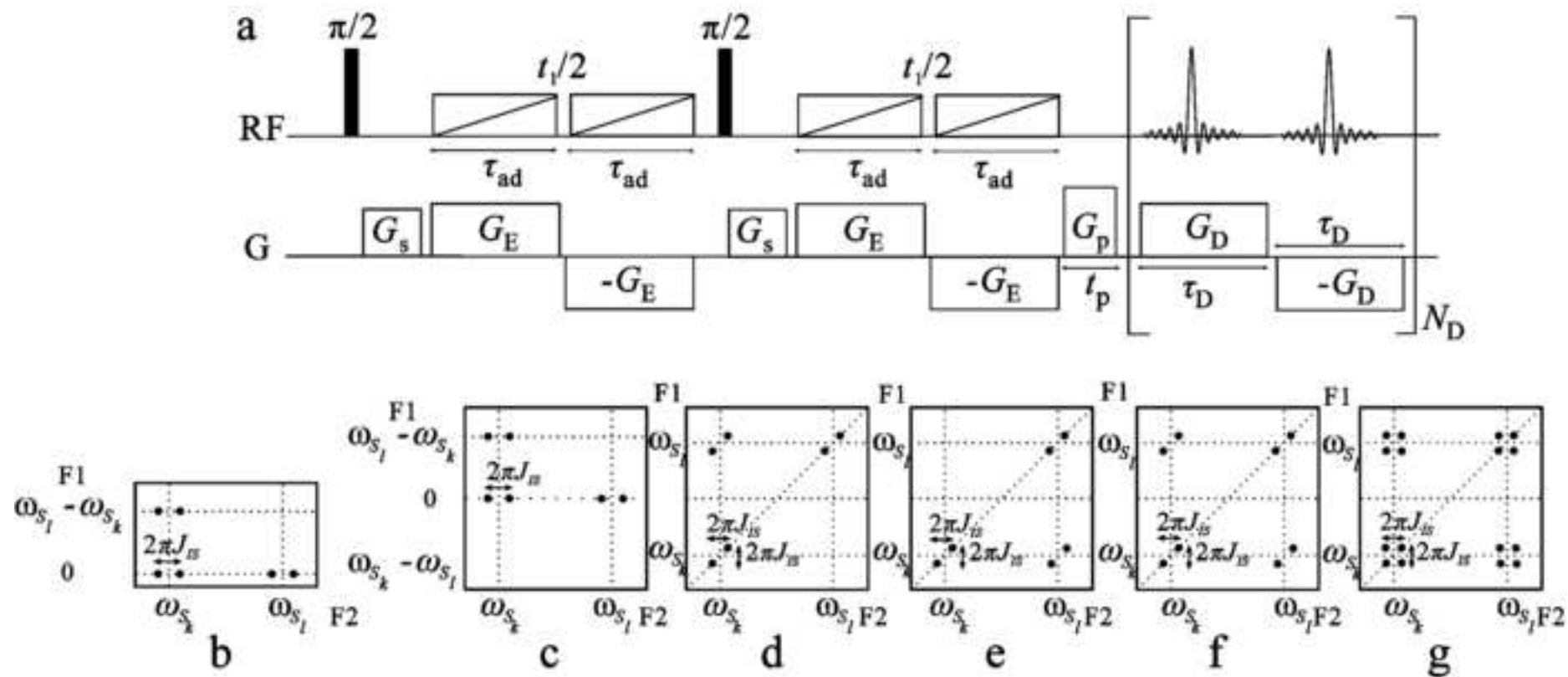
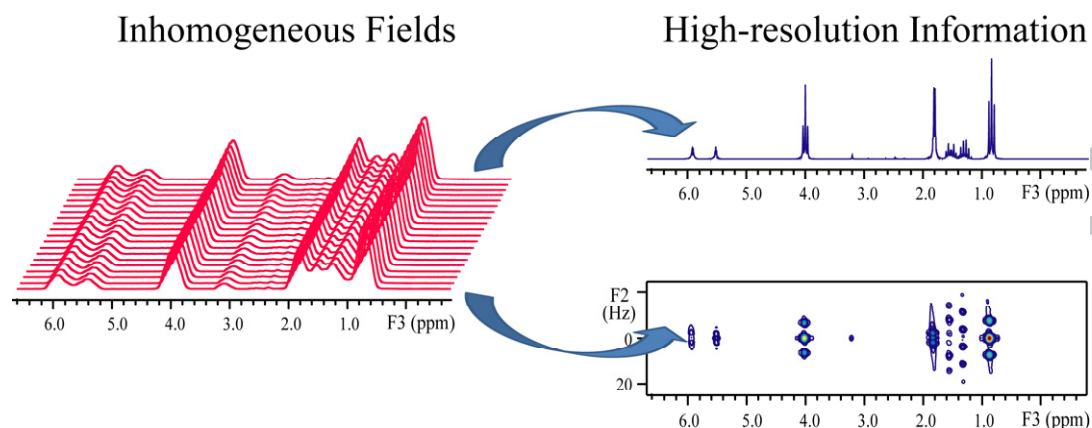


Fig.38



Graphical abstract



Highlights

- NMR spectroscopic approaches for high-resolution information in inhomogeneous fields are reviewed.
- Underlying mechanisms of these approaches are presented with analyses of their applicability and efficiency.
- Example applications for all these approaches are presented in various inhomogeneous field conditions.
- These approaches hold advantages and disadvantages in practical applications, and none of them is versatile and applicable to all practical circumstances.